

# 博士学位论文

曲面的几何、拓扑与组合结构研究

**ON THE GEOMETRIC, TOPOLOGICAL AND  
COMBINATORIAL STRUCTURES OF SURFACES**

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## 理学博士学位论文

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# **ON THE GEOMETRIC, TOPOLOGICAL AND COMBINATORIAL STRUCTURES OF SURFACES**

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## 摘 要

本文系统性地从几何、拓扑和组合的角度研究了一类由多面体乘积构造的实矩角复形曲面及其对偶曲面。对偶曲面的边结构自然给出一族最短闭测地线，这为研究表面上的闭测地线系统、同调群与双曲结构之间的关系提供了理想模型。同时，这些曲面具有高度对称性，这也为将相关研究拓展到正则地图理论，并进一步构造更多具体的对称曲面族，提供了自然动机。

我们首先考察了该系统中基本群与一维同调群之间的映射性质。通过将 Hurewicz 映射限制在由闭测地线生成的自由 Abel 群上，我们利用代数正规形式证明了该映射的核有一组标准基，其中的每个向量与圈图  $C_n$  的非空独立集一一对应。利用组合计数方法，我们确定了核与像的秩，得到了用 Lucas 数  $L(n)$  表达的显式公式。这一结果揭示了曲面几何与图论之间的联系。

随后，用同样的方法，我们研究了超立方体图  $Q_n$  正则嵌入的同调性质。研究由图的边构成的闭测地线系统，我们计算并证明了其中一系列嵌入当  $n$  为偶数时 Hurewicz 映射的核与像的秩公式，结果是用 Pell–Lucas 数  $Q(n)$  表达的显式公式。

从双曲几何的角度，为曲面赋予标准双曲度量，我们给出了计算最短测地线长度函数在 Teichmüller 空间上的连续 Morse 指标的新方法。该指标通过计算曲面的 systole 函数梯度向量张成的子空间的维数得出。利用 Walsh–Hadamard 变换分解方法，我们证明了猜想的指标公式，将曲面的几何结构与其组合结构联系起来。

在论文的最后部分，我们研究具有平坦度量的代数曲面与直纹面，特别关注在非 Hermite 量子力学中临界曲面的几何结构，分析了直纹面上的直线与圆锥曲线的性质，我们证明了在仿射参数条件下，临界曲面是切可展的直纹面。

**关键词：**多面体乘积; 实矩角复形; 闭测地线; Lucas 数; 正则嵌入; 双曲几何; Morse 指标; Teichmüller 空间; 直纹面

## ABSTRACT

In this paper, we systematically study a class of real moment-angle complex surfaces constructed via polyhedral products, together with their dual surfaces, from geometric, topological and combinatorial perspectives. The edge structure of each dual surface naturally gives rise to a collection of shortest closed geodesics, providing an ideal model for investigating the relationships among systems of closed geodesics, homology groups, and hyperbolic structures on surfaces. Meanwhile, the high degree of symmetry of these surfaces provides a natural motivation for extending the related study to the theory of regular maps and for further constructing more explicit families of symmetric surfaces.

We first examine the mapping property between the fundamental group and the first homology group in this system. By restricting the Hurewicz map to the free abelian group generated by closed geodesics and using algebraic normal forms, we prove that the kernel of this restricted map admits a standard basis whose basis vectors are in bijective correspondence with the nonempty independent sets of the cycle graph  $C_n$ . Using combinatorial counting, we determine the ranks of the kernel and the image, obtaining explicit formulas expressed in terms of the Lucas numbers  $L(n)$ . This result reveals a connection between surface geometry and graph theory.

Subsequently, using the same method, we study the homological properties of regular embeddings of the hypercube graph  $Q_n$ . By examining the system of closed geodesics formed by the edges of the graph, we compute and prove formulas for the ranks of the kernel and image of the Hurewicz map for a series of embeddings when  $n$  is even, resulting in explicit expressions in terms of Pell–Lucas numbers  $Q(n)$ .

From the perspective of hyperbolic geometry, we endow the surfaces with a canonical hyperbolic metric and present a new method for computing the continuous Morse index of the shortest geodesic length functional on Teichmüller space. This index is obtained by calculating the dimension of the subspace spanned by the gradient vectors of the systole function. Using the Walsh–Hadamard transform decomposition method, we prove a conjectured index formula, linking the geometric structure of the surface to its combinatorial structure.

In the final part of the paper, we study algebraic surfaces and ruled surfaces equipped with flat metrics, with particular attention to the geometric structure of degenerate surfaces

in non-Hermitian quantum mechanics. We analyze the properties of lines and conics on ruled surfaces and prove that, under affine parameter conditions, the degenerate surface is a tangent developable ruled surface.

**Keywords:** polyhedral products; real moment-angle complexes; closed geodesics; Lucas numbers; regular embeddings; hyperbolic geometry; Morse index; Teichmüller space; ruled surfaces

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## CHAPTER 1 INTRODUCTION

### 1.1 Research Background

A Riemann surface is a remarkably rich object that reveals different facets of mathematics depending on the perspective from which it is viewed. From the standpoint of geometry, a compact Riemann surface of genus at least two carries a canonical hyperbolic structure by the Uniformization Theorem. In this way, it appears as a closed hyperbolic manifold whose geometry is governed by Fuchsian groups, with its moduli controlled by Teichmüller space and mapping class groups. Hyperbolic geometry thus provides a natural language for understanding its global structure, deformations, and symmetries.

From a topological and combinatorial perspective, the same surface may be regarded simply as a smooth two-dimensional real manifold. Graph embeddings on the surface decompose it into vertices, edges, and faces, often producing highly symmetric tessellations. These embedded graphs encode the surface's combinatorial structure and link it to permutation representations and group actions. In particular, regular and highly symmetric maps highlight deep connections between graph theory, combinatorics, and group theory, allowing geometric symmetry to be studied through algebraic data.

Finally, a Riemann surface can be viewed as a complex algebraic curve defined over  $\mathbb{C}$ . In this algebro-geometric incarnation, it becomes an object of algebraic geometry and arithmetic: its function field, divisors, and morphisms connect to the theory of algebraic curves, while additional arithmetic structure relates it to number theory. These three viewpoints: hyperbolic-geometric, combinatorial-topological, and algebraic-arithmetic are not separate descriptions but complementary lenses through which the same profound object can be understood.

### 1.2 Statement of Main Results

First, we study the geometry and topology of a class of real moment-angle complexes  $X = Z(C_n, (D^1, S^0))$  given by polyhedral products, which is a surface, and its dual surface  $X^*$ . The edges of the dual surface exactly form a set of shortest closed geodesics, and we investigate the homological properties of this set.

Given the real moment-angle complex  $X$ , let  $S$  denote the set of systoles of  $X$ , namely

the set of shortest non-contractible closed geodesics. After choosing an orientation for each systole, every element of  $S$  determines a free homotopy class of loops, equivalently a conjugacy class in  $\pi_1(X)$ . Thus we obtain a natural map

$$\kappa : S \longrightarrow \text{Conj}(\pi_1(X)), \quad \gamma \mapsto [\gamma]_{\text{conj}}.$$

In general, this map need not be injective, since distinct systoles may represent the same conjugacy class.

The Hurewicz homomorphism

$$h : \pi_1(X) \longrightarrow H_1(X; \mathbb{Z})$$

coincides with the abelianization map. Since it is constant on conjugacy classes, it induces a well-defined map of sets

$$\bar{h} : \text{Conj}(\pi_1(X)) \longrightarrow H_1(X; \mathbb{Z}).$$

Therefore, the composition

$$\bar{h} \circ \kappa : S \longrightarrow H_1(X; \mathbb{Z})$$

assigns to each systole its corresponding homology class.

Let  $\tilde{S} := \mathbb{Z}^S$  be the free abelian group generated by  $S$ , and let  $i : S \hookrightarrow \tilde{S}$  be the canonical inclusion sending each  $\gamma \in S$  to the corresponding basis element of  $\tilde{S}$ . By the universal property of the free abelian group, the map  $\bar{h} \circ \kappa : S \rightarrow H_1(X; \mathbb{Z})$  extends uniquely to a group homomorphism

$$\tilde{h} : \tilde{S} \longrightarrow H_1(X; \mathbb{Z}),$$

defined on generators by

$$\tilde{h}(\gamma) = \bar{h}(\kappa(\gamma)), \quad \gamma \in S.$$

This construction is summarized by the following commutative diagram:

$$\begin{array}{ccccc} S & \xrightarrow{\kappa} & \text{Conj}(\pi_1(X)) & \xrightarrow{\bar{h}} & H_1(X; \mathbb{Z}) \\ \downarrow i & & & \nearrow \tilde{h} & \\ \tilde{S} & & & & \end{array}$$

**Theorem 1.1:** A standard basis for the kernel of the map  $\tilde{h}$  can be given, where each vector in this basis corresponds to an element of the set of nonempty independent sets of the cycle graph  $C_n$ .

Since the number of independent sets of a cycle graph can be obtained by standard

combinatorial methods, we know this number is  $L(n)$ , i.e., the Lucas numbers, which follow the same recurrence  $L(n) = L(n-1) + L(n-2)$  as the Fibonacci type numbers but with initial conditions  $L(1) = 1, L(2) = 3$ . Thus we can immediately determine the dimension of this space; similarly, we can determine the dimension of the subspace spanned by the image of this map in the homology group.

**Corollary 1.1:** By the enumeration formula, we can determine the ranks of the free abelian groups:

$$\text{rank Ker } \tilde{h} = L(n) - 1,$$

$$\text{rank Im } \tilde{h} = n \cdot 2^{n-3} - L(n) + 1.$$

We next investigate the embedding properties of this surface, considering both higher-dimensional sphere and the three-dimensional Euclidean spaces.

**Theorem 1.2:** One can embed  $X^* = Z(C_n, (D^1, S^0))^*$  into the  $(n-1)$ -dimensional sphere  $S^{n-1} \subset \mathbb{R}^n$  with the standard round metric, such that every edge is a geodesic arc, and also require that all edges have the same length  $\arccos\left(\frac{n-3}{n-2}\right)$ .

**Theorem 1.3:** For  $n \geq 5$ , the surface  $Z(C_n, (D^1, S^0))^*$  can be smoothly embedded into the 3-dimensional Euclidean space  $\mathbb{R}^3$ . The isometry group of this embedded surface in  $\mathbb{R}^3$  is isomorphic to the Klein four-group  $V_4$ .

As shown in Figure 1-1, the left figure shows the genus 5 surface (corresponding to  $n = 5$ ), and the right figure shows the genus 17 surface (corresponding to  $n = 6$ ). Both surfaces are embedded in  $\mathbb{R}^3$  together with their systoles. Vertices are represented by small blue circles, and edges connect pairs of vertices. In the top view, vertices that coincide are represented by larger circles, coincident edges are represented by black line segments, and non-coincident edges are represented by red line segments.

Using techniques of polyhedral decomposition, Al-Raisi studied the action of the cyclic group  $\mathbb{Z}/n$  on the cohomology groups of surfaces and obtained the following identity.

**Theorem 1.4:** [65]Proposition 3.5.10 For  $n \geq 4$ , we have

$$\frac{1}{n} \left( \sum_{d|n} \mu(d) 2^{\frac{n}{d}} \right) - (n-1) + \sum_{\substack{d|n \\ 1 < d < n}} \left( \frac{1}{d} \sum_{d_1|d} \mu(d_1) 2^{\frac{d}{d_1}} \right) = \frac{1}{n} \left( \sum_{d|n} \varphi(d) 2^{\frac{n}{d}} \right) - (n+1),$$

which relates the Möbius function and the Euler totient function by equating two different methods of orbit counting.

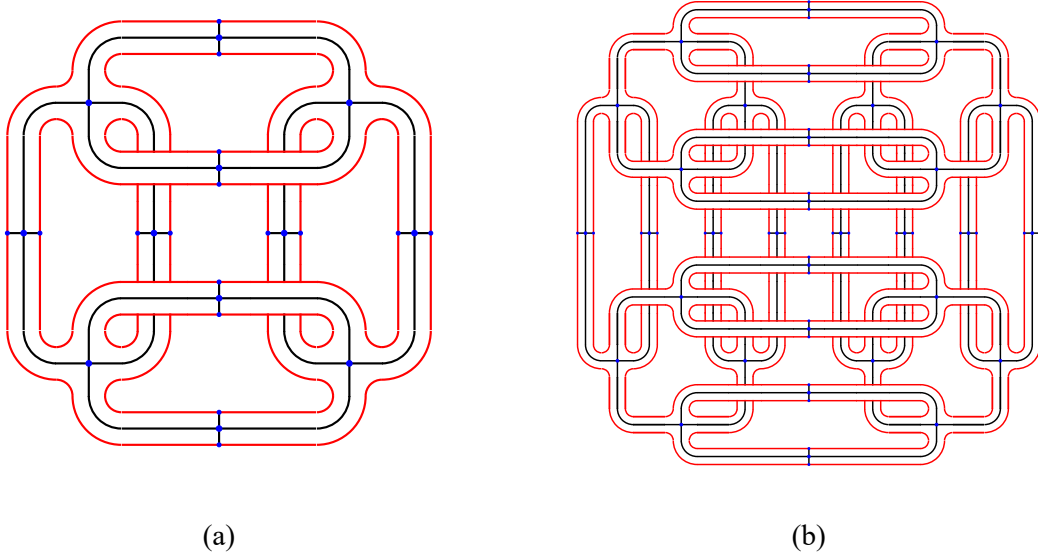


Figure 1-1 (a) The genus  $g = 5$  surface corresponding to  $n = 5$ ; (b) the genus  $g = 17$  surface corresponding to  $n = 6$ . Both surfaces are embedded in  $\mathbb{R}^3$  together with their systoles.

In a similar way, by counting orbits of the kernel of the Hurewicz map under the action of a cyclic group in two different ways, we obtain another identity.

**Theorem 1.5:** For  $n \geq 4$ , we have

$$\frac{1}{n} \left( \sum_{d|n} \mu(d) L\left(\frac{n}{d}\right) \right) + \sum_{\substack{d|n \\ 1 < d < n}} \left( \frac{1}{d} \sum_{d_1|d} \mu(d_1) L\left(\frac{d}{d_1}\right) \right) = \frac{1}{n} \left( \sum_{d|n} \phi(d) L\left(\frac{n}{d}\right) \right) - 1.$$

relating the Möbius inversion function, the Euler totient function and the Lucas numbers by equating the two different orbit counting methods of the systoles.

The homological properties of the closed geodesics formed by the edges of the dual surface  $X^*$  have been clarified by the preceding theorems. Next, we investigate the homological relations among the closed geodesics formed by edges arising from the cellular structure of the surface  $X$  itself. This case is more complex, as it requires distinguishing between odd and even cases. Only in the even case can all edges of the hypercube graph form closed cycles on the surface. In this case, edge cycles only form closed geodesics but not systoles, and we can determine the homological rank and kernel dimension of these closed geodesics.

**Theorem 1.6:** The homological rank and kernel dimension of the regular maps on surface  $X$  for even values of  $n$  can be captured by the Pell-Lucas numbers, which are defined by the recurrence relation  $Q(n) = 2Q(n-1) + Q(n-2)$  with initial conditions  $Q(0) = 2$ ,  $Q(1) = 2$ , and it has explicit formula  $Q(n) = (1 + \sqrt{2})^n + (1 - \sqrt{2})^n$ . The kernel di-

mension of Hurewicz map is given by:

$$\text{rank Ker } \tilde{h} = n2^{n-5} + (1 + \sqrt{2})^{n/2} + (1 - \sqrt{2})^{n/2} - 2.$$

The homological rank of closed geodesics formed by edges is given by:

$$\text{rank Im } \tilde{h} = 3n2^{n-5} - (1 + \sqrt{2})^{n/2} - (1 - \sqrt{2})^{n/2} + 2.$$

When  $n \geq 5$ , we can endow the surface with the standard hyperbolic metric and compute the Morse index of the systole function on Teichmüller space. In topological Morse theory, the *index* of a critical point is the dimension of the space of descending directions. For the systole function  $\text{sys} : \mathcal{T}_g \rightarrow \mathbb{R}$  at the critical point  $X$ , one has the important formula

$$\text{index}(X) = \dim \text{span}\{\nabla \ell_\gamma(X) : \gamma \in S(X)\},$$

where  $S(X)$  denotes the set of systoles on the surface  $X$ , and  $\nabla \ell_\gamma(X)$  denotes the gradient of the length function of the closed geodesic  $\gamma$ .

**Theorem 1.7:** Endowing each right-angled regular polygon of the surface  $X = Z(C_n, (D^1, S^0))^*$  with the standard hyperbolic metric,  $\text{index}(X)$  is given by the following close form formula for all  $n \geq 5$ :

$$\begin{aligned} \text{index}(X) &= \dim \text{span}\{\nabla \ell_\gamma(X) : \gamma \in S(X)\} \\ &= \#S(X) - (n + 3) \\ &= n2^{n-3} - (n + 3). \end{aligned}$$

We investigate discriminant surfaces arising from the characteristic polynomials of  $3 \times 3$  matrices possessing special symmetries, a study motivated by models in condensed matter physics involving Parity-Time and Minkowski-like pseudo-Hermitian Hamiltonians. In this part, we will demonstrate how these discriminant varieties naturally give rise to ruled and developable surfaces. Through explicit examples, we will explore their geometric properties and discuss some generalizations.

**Theorem 1.8 (Ruledness):** The generic part of discriminant surface of characteristic polynomial of matrix

$$H = \begin{bmatrix} g_1(f_1, f_2, f_3) & f_1 & f_2 \\ -f_1 & g_2(f_1, f_2, f_3) & f_3 \\ -f_2 & f_3 & g_3(f_1, f_2, f_3) \end{bmatrix}$$

is ruled, where  $g_1, g_2, g_3$  are linear functions of parameters  $f_1, f_2, f_3$ , i.e.  $g_i = a_{i,1}f_1 + a_{i,2}f_2 + a_{i,3}f_3 + b_i$ . Furthermore, the discriminant surface of  $H$  is tangent developable

and the associated directing curve is the curve of three multiplicity.

### 1.3 Outline of the Thesis

Chapter 1 of this thesis briefly introduces the research background and the main results of the dissertation.

Chapter 2 focuses on the theory of polyhedral products and its applications to the topology of surfaces. We investigate the topological properties of a special class of real moment-angle complexes, which are examples of polyhedral products in homotopy theory. The core result is a counting formula for geodesics in homology groups, proved using combinatorial methods. Finally, this result leads to a combinatorial identity. We also study the embedding properties of these real moment-angle complex surfaces.

Chapter 3 extends this problem to the realm of graph-theoretic regular embeddings. The surfaces and maps discussed in Chapter 1 arise as embeddings of hypercube graphs, which can be constructed via polyhedral products. For the remaining embeddings, we study analogous counting problems for closed geodesics, leading to intricate formulas and corresponding structures. In particular, these different embeddings themselves possess internal structures; we perform some calculations and propose conjectures.

Chapter 4 adopts the perspective of hyperbolic geometry on surfaces. The surfaces studied in Chapters 2 and 3 can each be endowed with a natural, highly symmetric hyperbolic structure. We investigate local features of the Teichmüller space near this point, its deformations, and the critical point indices of geodesic functions.

Chapter 5 focuses on algebraic surfaces and ruled surfaces that admit a flat metric. The rulings of these ruled surfaces are also geodesics; notably, they remain geodesics in the ambient space. These are geometric properties of ruled surfaces arising as critical surfaces in non-Hermitian quantum mechanics. We first analyze a special case and then prove a theorem for more general forms.

Appendix A provides an introduction to hyperbolic trigonometry, which is used in computations throughout Chapter 4. Appendix B contains the source code.

## CHAPTER 2 POLYHEDRAL PRODUCT HOMOTOPY THEORY

Polyhedral products form an important class of topological spaces constructed under the control of simplicial complexes. Originally conceived as a unification and generalization of several classical constructions, they have since evolved into a fundamental framework connecting combinatorial topology, homotopy theory, commutative algebra, and toric topology<sup>[7,9,17,63-64]</sup>.

Historically, although the terms “polyhedral product” and “moment-angle complex” appeared relatively late, their conceptual origins can be traced back to Porter’s work on higher products. Subsequently, with the rise of toric topology, the importance of moment-angle complexes became increasingly prominent. The work of Buchstaber and Panov showed that these spaces are not only natural objects for studying torus actions and simple polytopes but also have deep connections with Davis-Januszkiewicz spaces, Stanley-Reisner face rings, toric manifolds, and quasitoric manifolds, thereby serving as a crucial bridge between toric topology and toric geometry<sup>[17,28-30,62]</sup>.

On the other hand, moment-angle manifolds are also closely related to the theory of intersection quadrics. Studies by López de Medrano, López de Medrano-Verjovsky, and Bosio-Meersseman, among others, have shown that these spaces can also be understood from the perspectives of complex geometry and manifold topology, thus endowing moment-angle theory with a richer geometric background<sup>[16,24]</sup>.

At the algebraic and homological level, moment-angle complexes are intimately connected to Stanley-Reisner rings, Tor algebras, and the Hochster decomposition, allowing combinatorial properties of simplicial complexes to be directly reflected in topological and homological structures. The work of Baskakov, Buchstaber-Panov, Franz, and others has further advanced this direction, establishing moment-angle complexes as an important link between combinatorial commutative algebra and algebraic topology<sup>[10,17,28-30,37,62]</sup>.

The systematic development of polyhedral product theory owes much to the work of Bahri, Bendersky, Cohen, and Gitler. They established a unified framework for the polyhedral product functor and proved the famous stable decomposition theorem, providing a systematic method for studying the homotopy types and homological properties of moment-angle complexes and their real analogues<sup>[7-9]</sup>.

This paper focuses on real moment-angle complexes. Compared to their complex counterparts, real moment-angle complexes often exhibit more intuitive geometric realizations and are more suitable for studying group actions, homological structures, and stable decompositions. In recent years, a series of significant results concerning the surface structures, homological properties, and equivariant decompositions of real moment-angle complexes have emerged, indicating their increasingly important role within polyhedral product theory<sup>[23,25,65]</sup>.

Overall, polyhedral product theory has developed from a specific construction into a unifying perspective spanning combinatorics, topology, and algebra. The real moment-angle complex stands out as a central object possessing both geometric intuitiveness and algebraic richness. Within this context, this paper further discusses the homological properties, stable decompositions, and related issues of real moment-angle complexes<sup>[7,9,17,23,65]</sup>.

## 2.1 Introduction: Polyhedral Product Functors

Let  $[m] = \{1, 2, \dots, m\}$ . We first define a category  $\mathcal{C}([m])$  whose objects are pairs  $(\underline{X}, \underline{A})$ , where

$$(\underline{X}, \underline{A}) = \{(X_1, A_1), (X_2, A_2), \dots, (X_m, A_m)\}$$

is a family of based CW-pairs. A morphism in  $\mathcal{C}([m])$

$$\underline{f} : (\underline{X}, \underline{A}) \rightarrow (\underline{Y}, \underline{B})$$

consists of a collection of continuous maps  $f_i : X_i \rightarrow Y_i$  for  $1 \leq i \leq m$ , such that  $f_i(A_i) \subseteq B_i$  for each  $i$ .

Now let  $K$  be a simplicial complex with vertex set  $[m]$ . We regard  $K$  as a category whose objects are the simplices of  $K$  and whose morphisms are given by inclusions; that is, there is a morphism  $\sigma \rightarrow \tau$  if and only if  $\sigma \subseteq \tau$ .

**Definition 2.1:** The *polyhedral product* is the functor

$$Z(K; -) : \mathcal{C}([m]) \rightarrow \mathbf{Top}$$

defined by

$$Z(K; (\underline{X}, \underline{A})) := \operatorname{colim} D \subseteq \prod_{i=1}^m X_i,$$

where  $D : K \rightarrow CW_*$  is the diagram given by

$$D(\sigma) := \prod_{i=1}^m W_i, \quad \text{with} \quad W_i = \begin{cases} X_i, & \text{if } i \in \sigma, \\ A_i, & \text{if } i \in [m] \setminus \sigma, \end{cases}$$

for each simplex  $\sigma \in K$ .

Equivalently, the colimit may be described as the union

$$Z(K; (\underline{X}, \underline{A})) = \bigcup_{\sigma \in K} D(\sigma).$$

Similarly, one defines the *smash polyhedral product*

$$\widehat{Z}(K; (\underline{X}, \underline{A}))$$

by replacing Cartesian products with smash products throughout. Namely,

$$\widehat{D}(\sigma) := \bigwedge_{i=1}^m W_i, \quad \text{and} \quad \widehat{Z}(K; (\underline{X}, \underline{A})) = \bigcup_{\sigma \in K} \widehat{D}(\sigma).$$

**Remark 2.1:** If all pairs  $(X_i, A_i)$  are equal to a single pair  $(X, A)$ , then the family  $(\underline{X}, \underline{A})$  is denoted simply by  $(X, A)$ . In this case, the automorphism group  $\text{Aut}(K)$  acts naturally on  $Z(K; (X, A))$ .

**Remark 2.2:** Polyhedral products arise in a remarkably wide range of mathematical settings. For instance, the choice  $(D^2, S^1)$  leads to spaces central in toric geometry and topology, while  $(D^1, S^0)$  gives real moment-angle complexes, which are closely related to surfaces, number theory, representation theory, and linear programming. Other choices recover spaces connected with right-angled Artin groups and robotics through  $(S^1, *)$ , right-angled Coxeter groups through  $(\mathbb{RP}^\infty, *)$ , and complements of coordinate or more general subspace arrangements through  $(\mathbb{C}, \mathbb{C}^*)$  and  $(\mathbb{R}^n, (\mathbb{R}^n)^*)$ . Polyhedral products also appear in the study of monomial ideal rings, free groups and monodromy, combinatorics and aspherical spaces, Whitehead products in homotopy theory, and graph products and quadratic algebras, depending on the choice of the pair  $(X, A)$ <sup>[7]</sup>. This breadth of applications is one of the main reasons why polyhedral products have become a useful organizing framework across several areas of topology, algebra, and geometry.

## 2.2 Stable Decompositions of a Polyhedral Product

It is well known that the product space  $X^m$  of a connected CW-complex with base point admits a stable decomposition as a wedge of smash products. More precisely, there

is a homotopy equivalence<sup>[40,54]</sup>

$$H : \Sigma(X^m) \rightarrow \Sigma\left(\bigvee_{I \subset [m]} \widehat{X}^I\right),$$

where the index set  $I$  ranges over all non-empty subsequences of  $[m] := \{1, 2, \dots, m\}$ .

This may be viewed as a special case of the polyhedral product construction, namely when all pairs  $(X_i, A_i)$  are equal to  $(X, *)$  and  $K = \Delta^m$ , the  $m$ -simplex. A natural question is whether an analogous decomposition holds for the polyhedral product

$$Z(K; (\underline{X}, \underline{A})) \subset X_1 \times \dots \times X_m$$

when  $K \neq \Delta^m$ . In other words, does there exist a stable splitting for polyhedral products associated to an arbitrary simplicial complex  $K$ ? For general pairs  $(X, A)$  and simplicial complexes  $K$ , such splittings were established in<sup>[8]</sup> and<sup>[9]</sup>Theorem 2.10.

**Theorem 2.1:** Let  $K$  be an abstract simplicial complex on the vertex set  $[m]$ . Let

$$(\underline{X}, \underline{A}) = \{(X_i, A_i)\}_{i=1}^m$$

be a family of pointed CW-pairs  $(X_i, A_i, p_i)$ . Then there is a natural pointed homotopy equivalence

$$H : \Sigma(Z(K; (\underline{X}, \underline{A}))) \rightarrow \Sigma\left(\bigvee_{I \subset [m]} \widehat{Z}(K_I; (\underline{X}, \underline{A})_I)\right).$$

Here  $(\underline{X}, \underline{A})_I$  denotes the restricted family of pointed CW-pairs  $\{(X_i, A_i)\}_{i \in I}$ , and  $K_I$  denotes the full subcomplex (subcategory) of  $K$  on the vertex set  $I$ .

## The James Construction

Let  $J(X)$  denote the *James construction* or *James reduced product* on a based space  $(X, e)$ . Introduced by James<sup>[40]</sup>,  $J(X)$  is defined as the free topological monoid generated by  $X$ , with the basepoint  $e$  serving as the identity element. This construction proceeds explicitly as follows:

Consider the disjoint union  $\coprod_{k \geq 1} X^k$ , where  $X^k$  is the  $k$ -fold Cartesian product of  $X$ . The space  $J(X)$  is obtained by taking the quotient of this disjoint union under the identifications  $(x_1, \dots, x_i, \dots, x_k) \sim (x_1, \dots, \hat{x}_i, \dots, x_k)$  whenever  $x_i = e$ . Consequently, points of  $J(X)$  correspond to equivalence classes of  $k$ -tuples  $(x_1, \dots, x_k)$  for  $k \geq 0$  with no entry equal to the basepoint  $e$ .

$J(X)$  admits a natural filtration  $\{J_n(X)\}_{n \geq 0}$  by word length, where  $J_n(X)$  is the sub-

space consisting of points represented by tuples of length at most  $n$ . This subspace  $J_n(X)$  can be identified with the quotient of  $X^n$  under the identifications

$$(x_1, \dots, x_{i-1}, e, x_{i+1}, \dots, x_n) \sim (x_1, \dots, x_{j-1}, e, x_{j+1}, \dots, x_n)$$

for any positions  $i, j$  containing  $e$ . For instance:

- $J_0(X) = e$
- $J_1(X) = X$ .
- $J_2(X) = (X \times X) / \{(x, e) \sim (e, x) \mid \forall x \in X\}$ .

If  $X$  is a CW complex with  $e$  a 0-cell, the quotient map  $q_n : X^n \rightarrow J_n(X)$  identifies the  $n$  subcomplexes of  $X^n$  where one coordinate is fixed at  $e$ . These identifications are performed via homeomorphisms that map cells onto cells. Consequently,  $J_n(X)$  inherits a CW complex structure from  $X^n$ . The natural inclusions  $J_n(X) \subset J_{n+1}(X)$  are cellular, making  $J(X) = \bigcup_{n \geq 0} J_n(X)$  a CW complex.

The successive quotients in the filtration are homeomorphic to smash products:

$$J_n(X) / J_{n-1}(X) \cong X^{\wedge n},$$

where  $X^{\wedge n}$  denotes the  $n$ -fold self-smash product of  $X$ .

James proved<sup>[40]</sup> that  $J(X)$  is weak homotopy equivalent to  $\Omega\Sigma X$ , the based loop space of the reduced suspension of  $X$ , whenever  $X$  is a connected CW complex. The equivalence arises as follows:

A natural map  $\lambda : X \rightarrow \Omega\Sigma X$  is defined by the adjunction between suspension and loop spaces, for  $x \in X, t \in I$ :

$$\begin{aligned} \lambda : X &\longrightarrow \Omega\Sigma X \\ x &\longmapsto (t \mapsto (t, x)) \end{aligned}$$

Since  $J(X)$  is the free topological monoid on  $X$  (with  $e$  as identity), the universal property of free monoids allows us to extend  $\lambda$  multiplicatively to a map  $\bar{\lambda} : J(X) \rightarrow \Omega\Sigma X$ . Explicitly, for a point with word length  $n$  in  $J(X)$  can be represented by a formal product  $x_1 x_2 \cdots x_n$ :

$$\begin{aligned} \bar{\lambda} : J(X) &\longrightarrow \Omega\Sigma X \\ x_1 x_2 \cdots x_n &\longmapsto \lambda(x_1) \circ \lambda(x_2) \circ \cdots \circ \lambda(x_n) \end{aligned}$$

where the product on the right is the loop composition in  $\Omega\Sigma X$ . This map  $\bar{\lambda}$  induces the weak homotopy equivalence.

## Equivariant Properties of Stable Splitting Maps

Now suppose that all pairs  $(X_i, A_i)$  are equal. In this case, the automorphism group  $\text{Aut}(K)$  may be identified with a subgroup of the symmetric group  $\Sigma_m$ , and it acts naturally on both the domain and codomain of the stable splitting map  $H$ . Using James–Hopf invariants, A. Al-Raisi<sup>[65]</sup> showed that the stable splitting map  $H$  is in fact equivariant with respect to the  $\text{Aut}(K)$ -action.

**Theorem 2.2:** <sup>[65]Section 2.2</sup> The adjoint of the stable decomposition,

$$\theta_H : Z(K; (X, A)) \rightarrow J\left(\bigvee_{I \subseteq [m]} \widehat{Z}(K_I; (X, A)_I)\right) \rightarrow \Omega\Sigma\left(\bigvee_{I \subseteq [m]} \widehat{Z}(K_I; (X, A)_I)\right),$$

is  $\text{Aut}(K)$ -equivariant.

## 2.3 Real Moment-angle Complexes and Relation to Surface Topology

Let  $K = C_n$  denote the cycle graph, i.e. the one-dimensional simplicial complex given by the boundary of an  $n$ -gon or a ring with  $n$  vertices and  $n$  edges. We also use  $C_n$  to denote the cyclic group of order  $n$ ; there will be no ambiguity, as the meaning is determined by whether  $C_n$  appears in a topological-theoretic or group-theoretic context.

The real moment-angle manifold  $Z(C_n; (D^1, S^0))$  was studied very early on; as a closed surface tiled by quadrilaterals, the first to give its genus formula was perhaps Coxeter<sup>[22]</sup>.

**Theorem 2.3:** The real moment-angle manifold  $Z(C_n; (D^1, S^0))$  corresponding to a cycle graph  $C_n$ , for  $n \geq 3$ , is an oriented surface of genus  $g = 1 + (n - 4) \cdot 2^{n-3}$ .

**Remark 2.3:** Das<sup>[23]</sup> directly computed the genus of the *hypercube graph*, i.e., the 1-skeleton of the  $n$ -cube, which can be identified with the real moment-angle complex  $Z(N_n; (D^1, S^0))$ , and obtained the formula  $1 + (n - 4)2^{n-3}$ . The computation proceeds by equivariantly embedding the hypercube graph into the real moment-angle complex  $Z(C_n; (D^1, S^0))$ . The polyhedral product  $Z(N_n; (D^1, S^0))$  corresponding to the edgeless null graph  $N_n$  on  $n$  vertices is isomorphic to the hypercube graph  $Q_n$ . By functoriality of the inclusion  $N_n \hookrightarrow C_n$ , we obtain a regular embedding  $Q_n \hookrightarrow X_n := Z(C_n; (D^1, S^0))$ . This genus fact was originally established by Ringel<sup>[67]</sup> and independently by Beineke and Harary<sup>[11]</sup>. Subsequently, Catalano et al.<sup>[19]</sup> classified all regular embeddings of  $Q_n$  into surfaces and showed that the smallest possible genus equals  $1 + (n - 4)2^{n-3}$ , while the

largest genus equals  $1 + (n - 3)2^{n-2}$ . The real moment-angle complex  $Z(C_n; (D^1, S^0))$  itself provides the unique regular embedding of  $Q_n$  into a surface of this smallest genus.

### An invitation to the basic research objects

In the special case of the pair solid spheres and their sphere boundary  $(D^{m+1}, S^m, *)$  the stable decomposition of  $\Sigma(Z(K; (D^{m+1}, S^m)))$  can be simplified as the following form:

**Corollary 2.1 (Bahri et al.<sup>[9]</sup> Corollary 2.24):** The stable decomposition theorem states that for all  $i, m \geq 0$  and any simplicial complex  $K$ , there exists homotopy equivalence maps for the suspension of polyhedral product  $Z(K; (D^{m+1}, S^m))$ :

$$\Sigma(Z(K; (D^{m+1}, S^m))) \rightarrow \Sigma\left(\bigvee_{I \notin K} |K_I| * S^{m|I|}\right) \rightarrow \bigvee_{I \notin K} \Sigma^{2+m|I|} |K_I|,$$

where  $|I|$  stands for the size of  $I$ ,  $K_I$  stands for the full subcomplexes of  $K$  which have all of their vertices in  $I$ , so  $K_I = K \cap \Delta^I$  and  $|K_I|$  stands for the geometric realization of  $K_I$ .

In the more special case where  $K = K_n$  (the boundary of the  $n$ -gon), and  $m = 0$ , there is a homotopy equivalence:

$$\Sigma(Z(C_n; (D^1, S^0))) \rightarrow \bigvee_{I \notin K_n} \Sigma^2 |K_I|.$$

**Example 2.1:** When  $n = 4$ ,  $K_4$  represents a square. The simplices of  $K_4$  consist of the vertices  $\{1\}, \{2\}, \{3\}, \{4\}$ , and the edges  $\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}$ .

The simplices represented by the sets  $\{1, 3\}, \{2, 4\}$ , all 3-element subsets  $\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}$ , and the full set  $\{1, 2, 3, 4\}$  are *not* included in  $K_4$ .

For the 2-element subsets, by the definition of  $K_I$ , we have  $|K_{\{1,3\}}| \cong |K_{\{2,4\}}|$ , and both are two discrete points, which have the homotopy type of  $S^0$ . For each of the three 3-element subsets,  $|K_I|$  forms a “V”-shaped complex, which is contractible. Moreover,  $|K_{\{1,2,3,4\}}|$  is equal to  $K_4$  itself and has the homotopy type of a circle.

Therefore, by the preceding corollary, we obtain:

$$\Sigma(Z(C_4; (D^1, S^0))) \simeq \bigvee_{i=1}^2 \Sigma^2(S^0) \bigvee \Sigma^2(S^1) = \bigvee_{i=1}^2 S^2 \bigvee S^3.$$

Hence,  $H_1(Z(C_4; (D^1, S^0))) = \mathbb{Z}^2$ , and the rank of this group agrees with the genus formula  $g = 1$ .

**Example 2.2:** When  $n = 5$ ,  $K_5$  represents a pentagon. Its simplices are the five vertices  $\{1\}, \{2\}, \{3\}, \{4\}, \{5\}$  and the five edges  $\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 5\}, \{5, 1\}$ .

The simplices represented by the 2-element sets  $\{1, 3\}$ ,  $\{2, 4\}$ ,  $\{3, 5\}$ ,  $\{1, 4\}$ ,  $\{2, 5\}$ ; all 3-element subsets  $\{1, 2, 4\}$ ,  $\{2, 3, 5\}$ ,  $\{1, 3, 4\}$ ,  $\{2, 4, 5\}$ ,  $\{1, 3, 5\}$ ; all 4-element subsets  $\{1, 2, 3, 4\}$ ,  $\{1, 2, 3, 5\}$ ,  $\{1, 3, 4, 5\}$ ,  $\{2, 3, 4, 5\}$ ; and the full set  $\{1, 2, 3, 4, 5\}$  are *not* included in  $K_5$ .

By the definition of  $K_I$ , for each of the listed 2-element and 3-element subsets  $I$ , the complex  $|K_I|$  has the homotopy type of two discrete points,  $S^0$ . For a 3-element subset, such as  $\{1, 2, 4\}$ , this corresponds to two vertices connected by an edge and one isolated vertex. For any 4-element subset  $I$ ,  $|K_I|$  is a path of 3 edges, hence contractible, and contributes no wedge summand. Finally,  $|K_{\{1,2,3,4,5\}}|$  is the pentagon itself and has the homotopy type of a circle,  $S^1$ .

Therefore, by the preceding corollary, we obtain:

$$\Sigma(Z(C_5; (D^1, S^0))) \simeq \bigvee_{i=1}^{10} \Sigma^2(S^0) \bigvee \Sigma^2(S^1) = \bigvee_{i=1}^{10} S^2 \bigvee S^3.$$

Hence,  $H_1(Z(C_5; (D^1, S^0))) = \mathbb{Z}^{10}$ , and the rank of this group agrees with the genus formula  $g = 5$ .

**Example 2.3:** When  $n = 6$ ,  $K_6$  is the hexagon. Its simplices are the six vertices  $\{1\}$ ,  $\{2\}$ ,  $\{3\}$ ,  $\{4\}$ ,  $\{5\}$ ,  $\{6\}$  and the six edges  $\{1, 2\}$ ,  $\{2, 3\}$ ,  $\{3, 4\}$ ,  $\{4, 5\}$ ,  $\{5, 6\}$ ,  $\{6, 1\}$ .

The following simplices, represented by their vertex subsets, are *not* included in  $K_6$ . According to the definition of  $K_I$ , we analyze the homotopy types they contribute to the final wedge sum. Note that although all simplices of  $\Delta^6$  with more than two vertices are absent from  $K_6$ , we focus only on those that contribute non-contractible homotopy types.

**Some 2-element subsets (diagonals):**  $\{1, 3\}$ ,  $\{1, 4\}$ ,  $\{1, 5\}$ ,  $\{2, 4\}$ ,  $\{2, 5\}$ ,  $\{2, 6\}$ ,  $\{3, 5\}$ ,  $\{3, 6\}$ ,  $\{4, 6\}$ . For each of them,  $|K_I|$  consists of two discrete points, so  $\Sigma^2|K_I|$  contributes a copy of  $S^2$ . These give 9 copies of  $S^2$ .

**All 3-element subsets:** Among these, two types contribute non-contractible spaces.

- The first type, the configuration such as  $\{1, 2, 4\}$  and  $\{1, 2, 5\}$  and their images under the  $C_6$  rotation action, yield  $|K_I|$  with the homotopy type of two discrete points. There are 12 such subsets, each contributing an  $S^2$ .

- The second type, only 2 subsets  $\{1, 3, 5\}$  and  $\{2, 4, 6\}$ , have  $|K_I|$  with three discrete points, which has the homotopy type of  $V^2 S^1$  after once suspension. Each contributes a homotopy type  $V^2 S^2$  after twice suspension.

**All 4-element subsets:** Non-contractible contributions come from the  $C_6$ -orbits of two types of 4-element subsets. The first type is represented by  $\{1, 2, 4, 5\}$ , whose orbit

consists of the three subsets:  $\{1, 2, 4, 5\}$ ,  $\{2, 3, 5, 6\}$ , and  $\{1, 3, 4, 6\}$ . The second type is represented by  $\{1, 2, 3, 5\}$ , whose orbit under the cyclic group action contains six distinct subsets. In all these nine cases,  $|K_I|$  has the homotopy type of  $S^0$ , so  $\Sigma^2|K_I| \simeq S^2$ , with each subset contributing one copy of  $S^2$ .

**All 5-element subsets:** All such subsets yield contractible spaces ( $K_I$  is always a path of 4 edges), and thus contribute no wedge summands.

**The full set [6]:** Here,  $|K_{[6]}|$  is the boundary of the hexagon, which has the homotopy type of  $S^1$ . Therefore, it contributes  $\Sigma^2 S^1 \simeq S^3$ .

Therefore, by the preceding corollary, we obtain:

$$\Sigma(Z(C_6; (D^1, S^0))) \simeq \bigvee_{i=1}^{30} \Sigma^2(S^0) \bigvee_{i=1}^2 \Sigma^2(\coprod_{i=1}^3 *) \bigvee \Sigma^2(S^1) = \bigvee_{i=1}^{34} S^2 \bigvee S^3.$$

Hence,  $H_1(Z(C_6; (D^1, S^0))) = \mathbb{Z}^{34}$ , and the rank of this group agrees with the genus formula  $g = 17$ .

## Application of equivalent stable splitting theorem

In the case  $K = C_n$ ,  $\text{Aut}(C_n) = D_{2n}$ , the dihedral group with subgroup cyclic group  $C_n$ , the equivalent stable splitting theorem has interesting consequences. The real moment-angle manifold is the conduit through which Al-Raisi<sup>[65]</sup> uses Theorem 2.2 to link combinatorics of  $C_n$  and the rotation action of  $C_n$  on a choice of basis for the cohomology of a surface of genus  $g = 1 + (n - 4) \cdot 2^{n-3}$ .

In this setting, one may choose generators indexed, up to sign, by subsequences

$$I = (i_1, \dots, i_r) \subset \{1, \dots, n\},$$

where  $I$  is required to have gap number greater than 1. Here the gap number refers to the number of connected components of  $I$  when regarded as a subset of the cyclically ordered set  $\{1, \dots, n\}$ . In other words, we view  $\{1, \dots, n\}$  as arranged on a circle, and  $I$  decomposes into maximal consecutive blocks; the gap number counts how many such blocks occur. The condition that the gap number is greater than 1 means that  $I$  does not form a single contiguous block around the circle.

Equivalently, these generators may be encoded by binary colorings of an  $n$ -bead necklace: a black bead records membership in  $I$ , while a white bead records its complement. Under this correspondence, the action of  $C_n$  becomes the usual rotation action on

necklaces. Thus, the problem of understanding the  $C_n$ -orbits in

$$H^2(\Sigma Z(K_n; (D^1, S^0)))$$

reduces to a combinatorial orbit-counting problem.

There are two natural ways to count these orbits.

**Method 1: Global orbit counting via Burnside's lemma**

The first counts all admissible orbits at once. By Burnside's lemma, the number of rotational equivalence classes of binary colorings of an  $n$ -bead necklace is

$$\frac{1}{n} \sum_{d|n} \phi(d) 2^{n/d},$$

where  $\phi(d)$  is the Euler totient function which counts the number of integers up to  $n$  that are relatively prime to  $n$ . However, not every binary coloring contributes to the cohomology basis considered above: the excluded cases are exactly those colorings in which all beads of the same color occur consecutively around the necklace. There are precisely  $n + 1$  such colorings, namely the two constant colorings together with the  $n - 1$  colorings having one contiguous black block of length  $i$ , for  $1 \leq i \leq n - 1$ . Hence the total number of relevant  $C_n$ -orbits is

$$\frac{1}{n} \sum_{d|n} \phi(d) 2^{n/d} - (n + 1).$$

**Method 2: Stratified orbit counting via primitive necklaces decomposition**

The second method counts the same orbits by their exact size. A word of length  $n$  is called an *aperiodic word* if it has  $n$  distinct rotations or equivalently the stabilizer group of this coloring is trivial group under the action of cyclic group  $C_n$ . An equivalence class of an aperiodic word under rotation is called a *primitive necklace* (also called *Lyndon word* or *primitive strings*). If  $d \mid n$  with  $1 < d < n$ , then an orbit of size  $d$  corresponds to a coloring obtained by repeating an aperiodic word of length  $d$ , or equivalently to a primitive  $d$ -necklace on a two-letter alphabet. By Moreau's formula, the number of primitive  $d$ -necklaces is

$$M(2, d) = \frac{1}{d} \sum_{d_1|d} \mu(d_1) 2^{d/d_1},$$

where  $\mu$  is the Möbius inversion function:

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n \text{ is square-free with } k \text{ distinct prime factors,} \\ 0 & \text{if } n \text{ has a squared prime factor.} \end{cases}$$

This number is the number of  $C_n$ -orbits of size  $d$  arising from basis elements up to sign. Therefore, for every proper divisor  $d$  of  $n$  with  $1 < d < n$ , the number of such size- $d$  orbits is

$$\frac{1}{d} \sum_{d_1|d} \mu(d_1) 2^{d/d_1}.$$

For orbits of full size  $n$ , one again starts from primitive necklaces, now of length  $n$ . In this case one must remove the  $n - 1$  primitive necklaces in which all black beads occur consecutively, since these correspond to excluded colorings and do not contribute to the cohomology basis. Thus the number of orbits of size  $n$  is

$$M(2, n) - (n - 1) = \frac{1}{n} \sum_{d|n} \mu(d) 2^{n/d} - (n - 1).$$

Since every relevant orbit has size  $d$  for some divisor  $d$  of  $n$ , the total number of orbits is the sum of the numbers of orbits of each possible size. Equating this decomposition by orbit size with the Burnside count above yields the following identity.

**Theorem 2.4:** [65]Proposition 3.5.10 For  $n \geq 4$ , there is an identity

$$\frac{1}{n} \left( \sum_{d|n} \mu(d) 2^{\frac{n}{d}} \right) - (n - 1) + \sum_{\substack{d|n \\ 1 < d < n}} \left( \frac{1}{d} \sum_{d_1|d} \mu(d_1) 2^{\frac{d}{d_1}} \right) = \frac{1}{n} \left( \sum_{d|n} \phi(d) 2^{\frac{n}{d}} \right) - (n + 1)$$

relating the Möbius inversion function and the Euler totient function by equating the two different orbit counting methods.

## 2.4 Enumeration Formulas for Homological Ranks of Closed Geodesics on Hyperbolic Surfaces and Combinatorial Identities

We have constructed an embedding of the hypercube graph in the real moment-angle complex and computed its genus. This reveals an interesting geometric relationship between the hypercube graph and real moment-angle complexes. We now turn our attention to the dual complex of the real moment-angle complex  $Z(C_n, (D^1, S^0))$ .

## Dual complex and hyperbolic geometric structure

The definition of tessellation is adopt from<sup>[66]Sec 6.8</sup>.

**Definition 2.2:** A *tessellation* (or tiling) of an  $n$ -dimensional topological manifold  $X$  is a way of covering  $X$  with a collection  $\mathcal{P}$  of  $n$ -dimensional convex pieces (like polyhedra) such that:

- (1) the pieces do not overlap (their interiors are disjoint);
- (2) together, they cover all of  $X$ ;
- (3) every point in  $X$  has a neighborhood that meets only finitely many pieces.

**Definition 2.3:** A tessellation  $\mathcal{P}$  of  $X$  is called *exact* if every face (or side) of every piece is shared by exactly two pieces. In other words, there are no gaps or hanging edges.

**Definition 2.4:** A *regular tessellation* of  $X$  is an exact tessellation where all the pieces are identical regular polytopes (like perfect squares, cubes, or their higher-dimensional analogues).

**Definition 2.5:** A tessellation of a Riemann surface is called *platonic* if its symmetry group can move any flag (a triple consisting of a vertex, an edge, and a face, all incident to each other) to any other flag. Such a tessellation is also known as a regular map. A Riemann surface is called *platonic* if it admits at least one platonic tessellation.

A map  $M$  on a surface  $S$  can be understood as a graph  $G$  embedded in  $S$  such that each connected component of the complement  $S \setminus G$  is a 2-cell (a face), i.e., homeomorphic to an open disk.

**Definition 2.6 (Dual Map):** Let  $M$  be a map on a surface  $S$ . Its **dual map**  $M^*$  (or dual embedding  $G^*$ ) is defined as follows:

- **Vertices (dual vertices):** For each face  $f$  of  $M$  (i.e., a connected component of  $S \setminus G$ ), choose a point  $v_f$  in the interior of  $f$ . These points serve as the vertices of  $M^*$ . Consequently,  $V(M^*) \leftrightarrow F(M)$ .

- **Edges (dual edges):** For each edge  $e$  of  $M$ , it is adjacent to two faces  $f_1, f_2$  in the embedding (allowing the case  $f_1 = f_2$ , where both sides of the edge belong to the same face). Draw an arc  $e^*$  on the surface connecting  $v_{f_1}$  and  $v_{f_2}$ , satisfying that it intersects the original edge  $e$  transversely in exactly one point and  $e^*$  does not intersect any other edge of  $G$  (except at that single transverse intersection point with  $e$ ). The homotopy classes of these arcs (with endpoints fixed within their faces) determine the dual edges. Therefore,  $E(M^*) \leftrightarrow E(M)$ .

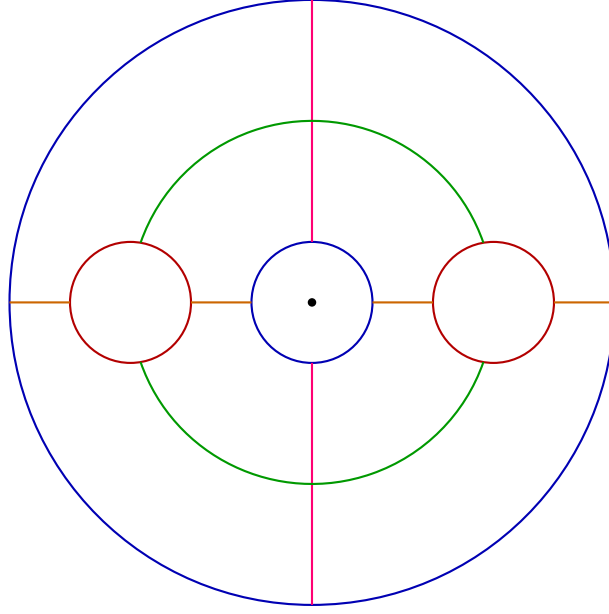
- **Faces (dual faces):** After drawing all dual edges, we obtain an embedded graph

$G^* \subset S$ . Its faces correspond bijectively to the vertices of the original graph: around each primal vertex  $u \in V(M)$ , a dual face  $f_u^*$  is formed. Thus,  $F(M^*) \leftrightarrow V(M)$ .

Under the definition given above, the dual map satisfies the following properties: The double dual  $(M^*)^*$  is naturally isomorphic to  $M$  (isomorphism in the topological sense, i.e., as graphs up to homeomorphism).

Our main object of study next is the dual map  $Z(C_n, (D^1, S^0))^*$ , which is composed of  $2^n$   $n$ -gons glued together, with four edges meeting at each vertex. A schematic of one-quarter of  $Z(C_5, (D^1, S^0))$  unfolded onto the plane.

Structure of one piece of  $n=5$



## Symmetric embedding of surfaces

We study the embedding properties of closed surfaces and their 1-skeletons into higher dimensional spaces. Although the ultimate goal is embedding into three dimensional Euclidean space for its advantages in visualization, and because every orientable closed surface of any genus can be embedded there, it is natural to first consider embeddings into higher dimensional Euclidean spaces and spheres, as this corresponds more directly with the intrinsic construction of the surfaces.

Specifically, by definition, the complex  $Z(C_n, (D^1, S^0))$  is a subcomplex of the standard high-dimensional cube  $I^n$ . Therefore, it can be naturally embedded into Euclidean space  $\mathbb{R}^n$  by taking the cube as  $[-1, 1]^n$ , so that its vertices lie exactly at the coordinates  $(\epsilon_1, \dots, \epsilon_n)$ , where each  $\epsilon_i = \pm 1$ .

Based on this embedding, we consider the geometric dual subdivision of the complex

on the surface: each vertex of the dual is placed at the center of the corresponding face of the primal complex. Then, on the original surface, we connect these dual vertices sequentially with straight line segments to form the edges of the dual complex; the regions enclosed by these edges on the original surface constitute the faces of the dual complex. Finally, by projecting these regions as a whole onto the unit sphere  $S^{n-1}$ , we complete the geometric construction of the new surface.

**Theorem 2.5:** One can embed  $Z(C_n, (D^1, S^0))^*$  into the  $(n - 1)$ -dimensional sphere  $S^{n-1} \subset \mathbb{R}^n$  with the standard round metric, such that every edge is a geodesic arc, and also require that all edges have the same length  $\arccos\left(\frac{n-3}{n-2}\right)$ .

**Proof:** In  $n$ -dimensional Euclidean space  $\mathbb{R}^n$ , the geodesics on the  $(n - 1)$ -dimensional unit sphere  $S^{n-1}$  with the standard round metric are precisely the great circles obtained by intersecting the sphere with a two-dimensional plane passing through the center (origin). We can write the parametric equation as follows: given two linearly independent unit vectors  $\mathbf{u}, \mathbf{v} \in S^{n-1}$  (satisfying  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ , i.e., orthogonal), the intersection of the plane spanned by them with  $S^{n-1}$  (i.e., the great circle) has the parametric equation:

$$\mathbf{r}(t) = \cos(t) \cdot \mathbf{u} + \sin(t) \cdot \mathbf{v}, \quad t \in [0, 2\pi)$$

Here, taking  $t = 0$  corresponds to the point  $\mathbf{u}$ , and  $t = \pi/2$  corresponds to the point  $\mathbf{v}$ . The parameter  $t$  is the arc-length parameter starting from  $\mathbf{u}$ , because  $\|\mathbf{r}'(t)\| = 1$ . Therefore, if we take the parameter range  $t \in [0, \pi/2]$ , the length of this arc connecting  $\mathbf{u}$  and  $\mathbf{v}$  is  $\pi/2$ .

Even if  $\mathbf{u}$  and  $\mathbf{v}$  are not orthogonal ( $\langle \mathbf{u}, \mathbf{v} \rangle \neq 0$ ), the geodesic (great circular arc) connecting them still exists and is unique, but the basis in its parametric equation needs to be reconstructed from  $\mathbf{u}$  and  $\mathbf{v}$ .

Project  $\mathbf{v}$  onto  $\mathbf{u}$  to obtain the component parallel to  $\mathbf{u}$ , and then find the component perpendicular to  $\mathbf{u}$ .

$$\mathbf{v}_{\parallel} = \langle \mathbf{v}, \mathbf{u} \rangle \mathbf{u}, \quad \mathbf{v}_{\perp} = \mathbf{v} - \mathbf{v}_{\parallel}$$

Normalize the perpendicular component to obtain a unit vector  $\mathbf{w}$  orthogonal to  $\mathbf{u}$ .

$$\mathbf{w} = \frac{\mathbf{v}_{\perp}}{\|\mathbf{v}_{\perp}\|}$$

Here,  $\|\mathbf{v}_{\perp}\| = \sqrt{1 - \langle \mathbf{u}, \mathbf{v} \rangle^2}$ . The parametric equation of the geodesic (great circular arc) connecting  $\mathbf{u}$  and  $\mathbf{v}$  is:

$$\mathbf{r}(t) = \cos(t) \cdot \mathbf{u} + \sin(t) \cdot \mathbf{w}, \quad t \in [0, \theta]$$

where  $\theta = \arccos(\langle \mathbf{u}, \mathbf{v} \rangle)$  is the spherical angular distance (i.e., the great circular arc length) between  $\mathbf{u}$  and  $\mathbf{v}$ . When  $t = 0$ ,  $\mathbf{r}(0) = \mathbf{u}$ . When  $t = \theta$ , because  $\cos \theta = \langle \mathbf{u}, \mathbf{v} \rangle$  and  $\sin \theta = \|\mathbf{v}_\perp\|$ , substituting gives  $\mathbf{r}(\theta) = \mathbf{v}$ .

The parametric equation of the unique geodesic (shortest path) connecting two points  $\mathbf{u}$  and  $\mathbf{v}$  on the unit sphere ( $\langle \mathbf{u}, \mathbf{v} \rangle \neq -1$ ) is:

$$\mathbf{r}(t) = \cos(t) \mathbf{u} + \sin(t) \frac{\mathbf{v} - \langle \mathbf{v}, \mathbf{u} \rangle \mathbf{u}}{\sqrt{1 - \langle \mathbf{u}, \mathbf{v} \rangle^2}}, \quad t \in [0, \arccos(\langle \mathbf{u}, \mathbf{v} \rangle)]$$

In  $\mathbb{R}^n$ , take the standard basis  $\mathbf{e}_1, \dots, \mathbf{e}_n$ . For each vertex in the dual graph,

$$\mathbf{v}^* = (a_1, \dots, *, *, \dots, a_n),$$

which corresponds to a face in the original graph, with coordinates where the  $i$ -th and  $(i + 1)$ -th positions are  $*$  and the remaining coordinates are arbitrarily chosen as 0 or 1, with indices taken modulo  $n$ . The coordinate vector is:

$$((-1)^{a_1}, \dots, 0, 0, \dots, (-1)^{a_n}),$$

where the  $i$ -th and  $(i + 1)$ -th components are 0, and the  $j$ -th component in the remaining positions is  $\epsilon_j := (-1)^{a_j} \in \{\pm 1\}$ . Project this onto a point  $\mathbf{u}_{i,i+1}$  on the sphere such that  $\|\mathbf{u}_{i,i+1}\| = 1$ , so its coordinates are

$$\frac{1}{\sqrt{n-2}}((-1)^{a_1}, \dots, 0, 0, \dots, (-1)^{a_n}).$$

After embedding all  $n \cdot 2^{n-2}$  points in this way, we consider how to embed the edges. Each edge  $e^* = (\dots, a_{i-1}, *, a_{i+1}, \dots)$  in the dual graph connects two vertices:  $(\dots, a_{i-2}, *, *, a_{i+1}, \dots)$  and  $(\dots, a_{i-1}, *, *, a_{i+2}, \dots)$ . On the original surface complex, this edge consists of two straight line segments, i.e.

$$\begin{aligned} & (\dots, \epsilon_{i-2}, 0, 0, \epsilon_{i+1}, \epsilon_{i+2}, \dots) - \\ & - (\dots, \epsilon_{i-2}, \epsilon_{i-1}, 0, \epsilon_{i+1}, \epsilon_{i+2}, \dots) - \\ & - (\dots, \epsilon_{i-2}, \epsilon_{i-1}, 0, 0, \epsilon_{i+2}, \dots) \end{aligned}$$

These two line segments can be parameterized as

$$(\dots, \epsilon_{i-2}, t_1 \epsilon_{i-1}, 0, \epsilon_{i+1}, \epsilon_{i+2}, \dots), \quad t_1 \in [0, 1]$$

and

$$(\dots, \epsilon_{i-2}, \epsilon_{i-1}, 0, \epsilon_{i+1}(1 - t_2), \epsilon_{i+2}, \dots), \quad t_2 \in [0, 1].$$

The intermediate point lies in the two-dimensional plane spanned by the endpoints, so its projection onto the unit sphere is a great circular arc. We only need to consider the

endpoints of the great circular arc.

Assume the two vertices, embedded as defined above, are  $\mathbf{u}_{i-1,i}$  and  $\mathbf{u}_{i,i+1}$ . The inner product of these two points is

$$\langle \mathbf{u}_{i-1,i}, \mathbf{u}_{i,i+1} \rangle = \frac{n-3}{n-2}.$$

Therefore, the spherical angle (i.e., the shortest geodesic arc length) between the two points is

$$\theta = \arccos\left(\frac{n-3}{n-2}\right).$$

Using the general parameterization formula given earlier, this edge can be embedded as the unit-speed geodesic arc connecting  $\mathbf{u}_{i-1,i}$  and  $\mathbf{u}_{i,i+1}$ :

$$\mathbf{r}(t) = \cos(t) \cdot \mathbf{u}_{i-1,i} + \sin(t) \frac{\mathbf{u}_{i,i+1} - \langle \mathbf{u}_{i,i+1}, \mathbf{u}_{i-1,i} \rangle \mathbf{u}_{i-1,i}}{\sqrt{1 - \langle \mathbf{u}_{i-1,i}, \mathbf{u}_{i,i+1} \rangle^2}}, \quad t \in \left[0, \arccos\left(\frac{n-3}{n-2}\right)\right]$$

This parameterization is unit-speed, so the length of each edge is  $\arccos\left(\frac{n-3}{n-2}\right)$ . In this way, all  $n \cdot 2^{n-1}$  edges can be embedded.

Next, we consider the embedding of faces. It suffices to prove that for a high-dimensional cube  $[-1, 1]^n$ , the projections of each two-dimensional face onto the unit sphere are disjoint except at their boundaries.

The interiors of the projections of different two-dimensional faces do not intersect because radial projection is injective. Since the cube is convex and the origin lies in its interior, each ray from the origin intersects the boundary of the cube at exactly one point. Therefore, radial projection establishes a bijection between the boundary of the cube and the unit sphere. Each two-dimensional face projects onto the unit sphere such that, except for possible overlaps at boundaries, the interior regions are mutually disjoint. ■

**Remark 2.4:** In Theorem 2.5, the embedded manifold is not smooth except in the case  $n = 3$ . Specifically, when  $n = 3$ , the embedded manifold is exactly the entire space  $S^2$ , and hence is smooth. For  $n \geq 4$ , the manifold is only piecewise smooth, with the dihedral angle between each pair of faces given by  $\arccos\left(-\frac{1}{2n-5}\right)$ . For detailed proof, see the Theorem 2.6 below.

**Theorem 2.6 (Spherical angles after radial projection):** Let  $n \geq 4$  and work in the Euclidean space  $\mathbb{R}^n$  with the standard inner product  $\langle \cdot, \cdot \rangle$ . Let  $\epsilon_j \in \{-1, 1\}$  for all  $j$ . To investigate the local geometry around a point  $P$  and its four adjacent points, we define points  $P, A, B, C, D \in \mathbb{R}^n$  as follows:

For all coordinates  $j \notin \{i-1, i, i+1, i+2\}$ , set them to be equal to the given sign

$$P_j = A_j = B_j = C_j = D_j = \epsilon_j.$$

On the coordinates  $(i-1, i, i+1, i+2)$ , set

$$P = (\epsilon_{i-1}, 0, 0, \epsilon_{i+2}), \quad A = (0, 0, -1, \epsilon_{i+2}), \quad B = (0, 0, 1, \epsilon_{i+2}),$$

$$C = (\epsilon_{i-1}, 1, 0, 0), \quad D = (\epsilon_{i-1}, -1, 0, 0),$$

where the displayed 4-tuples occupy the index positions  $(i-1, i, i+1, i+2)$ . A schematic diagram of the geometry is depicted in Figure 2-1.

Let the radial projection onto the unit sphere  $S^{n-1}$  be

$$\hat{X} := \frac{X}{\|X\|} \in S^{n-1}.$$

Then the spherical angles at  $\hat{P}$  satisfy

$$\angle \hat{A} \hat{P} \hat{B} = \angle \hat{A} \hat{P} \hat{C} = \arccos\left(-\frac{1}{2n-5}\right),$$

independent of the choices of  $\{\epsilon_j\}$ .

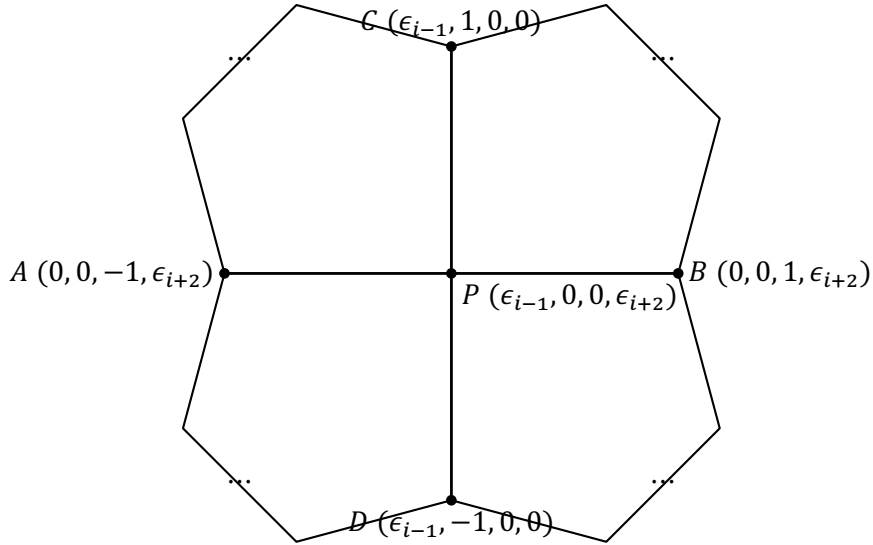


Figure 2-1 Local geometry around a point  $P$  and its four adjacent points  $A, B, C, D$

**Proof:** First compute the relevant dot products. The common  $(n-4)$  coordinates contribute  $n-4$  to every dot product. Using the specified 4-tuples:

$$\langle P, A \rangle = (n-4) + (\epsilon_{i+2} \cdot \epsilon_{i+2}) = (n-4) + 1 = n-3,$$

$$\langle P, B \rangle = n-3, \quad \langle P, C \rangle = (n-4) + (\epsilon_{i-1} \cdot \epsilon_{i-1}) = (n-4) + 1 = n-3,$$

$$\langle A, B \rangle = (n-4) + ((-1) \cdot 1) + (\epsilon_{i+2} \cdot \epsilon_{i+2}) = (n-4) - 1 + 1 = n-4,$$

$$\langle A, C \rangle = (n - 4) + 0 = n - 4.$$

Dividing by  $n - 2$  gives

$$\langle \hat{P}, \hat{A} \rangle = \langle \hat{P}, \hat{B} \rangle = \langle \hat{P}, \hat{C} \rangle = \frac{n-3}{n-2}, \quad \langle \hat{A}, \hat{B} \rangle = \langle \hat{A}, \hat{C} \rangle = \frac{n-4}{n-2}.$$

Then we study the spherical angle formula at  $\hat{P}$ . Let  $p = \hat{P}$  and  $x = \hat{A}$ ,  $y = \hat{B}$  (or  $y = \hat{C}$ ). The tangent space at  $p$  is  $T_p S^{n-1} = \{v : \langle v, p \rangle = 0\}$ . Project  $x, y$  to  $T_p S^{n-1}$  by

$$u := x - \langle x, p \rangle p, \quad v := y - \langle y, p \rangle p.$$

Then  $\angle xpy$  equals the Euclidean angle between  $u$  and  $v$ :

$$\cos \angle xpy = \frac{\langle u, v \rangle}{\|u\| \cdot \|v\|}.$$

A direct expansion using  $\|p\| = 1$  yields

$$\langle u, v \rangle = \langle x, y \rangle - \langle x, p \rangle \langle y, p \rangle,$$

and

$$\|u\|^2 = 1 - \langle x, p \rangle^2, \quad \|v\|^2 = 1 - \langle y, p \rangle^2.$$

In our case  $\langle x, p \rangle = \langle y, p \rangle = \frac{n-3}{n-2}$ , so

$$\cos \angle xpy = \frac{\langle x, y \rangle - \langle x, p \rangle^2}{1 - \langle x, p \rangle^2}.$$

Using  $\langle x, y \rangle = \frac{n-4}{n-2}$  and  $\langle x, p \rangle = \frac{n-3}{n-2}$ ,

$$\cos \angle xpy = \frac{\frac{n-4}{n-2} - \left(\frac{n-3}{n-2}\right)^2}{1 - \left(\frac{n-3}{n-2}\right)^2} = -\frac{1}{2n-5}.$$

Taking  $y = \hat{B}$  gives  $\angle \hat{A} \hat{P} \hat{B} = \arccos\left(-\frac{1}{2n-5}\right)$ , and taking  $y = \hat{C}$  gives the same value because the dot products used above satisfy  $\langle \hat{A}, \hat{C} \rangle = \langle \hat{A}, \hat{B} \rangle$  and  $\langle \hat{P}, \hat{C} \rangle = \langle \hat{P}, \hat{B} \rangle$ .

Therefore

$$\angle \hat{A} \hat{P} \hat{B} = \angle \hat{A} \hat{P} \hat{C} = \arccos\left(-\frac{1}{2n-5}\right).$$

■

**Theorem 2.7:** For  $n = 4$ , there exists a smooth embedding  $Z(C_4, (D^1, S^0))^* \hookrightarrow S^3$ , which can be realized by the Clifford torus and a subcomplex on it.

**Proof:** The Clifford torus is a special, flat two-dimensional torus embedded in four-dimensional Euclidean space  $\mathbb{R}^4$  (or in its submanifold, the three-dimensional sphere  $S^3$ ).

It can be simply defined as the surface in  $\mathbb{R}^4$  given by the equations

$$\left\{ (x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1^2 + x_2^2 = \frac{1}{2}, x_3^2 + x_4^2 = \frac{1}{2} \right\}.$$

Clearly, it lies on the unit sphere  $S^3$ , since  $x_1^2 + x_2^2 + x_3^2 + x_4^2 = \frac{1}{2} + \frac{1}{2} = 1$ .

It admits a parametrization given by

$$x_1 = \frac{1}{\sqrt{2}} \cos u, \quad x_2 = \frac{1}{\sqrt{2}} \sin u, \quad x_3 = \frac{1}{\sqrt{2}} \cos v, \quad x_4 = \frac{1}{\sqrt{2}} \sin v$$

with  $u, v \in [0, 2\pi)$ . One can see that it is essentially a two-dimensional torus  $T^2 = S^1 \times S^1$ , where both circles have radius  $\frac{1}{\sqrt{2}}$ , corresponding respectively to the two orthogonal planes  $(x_1, x_2)$  and  $(x_3, x_4)$  in  $\mathbb{R}^4$ .

Let  $C = \{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$ . Taking the parameters  $u, v$  to be values in  $C$ , we obtain the coordinates of 16 points:  $(0, 0, \pm 1, \pm 1)$ ,  $(0, \pm 1, \pm 1, 0)$ ,  $(\pm 1, \pm 1, 0, 0)$ ,  $(\pm 1, 0, 0, \pm 1)$ . These can serve as the embedding points of the surface for  $n = 4$ . The edges are obtained by varying only one of the parameters  $u$  or  $v$  by  $\pi/2$ . ■

In the following part, we will consider methods for embedding closed surfaces into three-dimensional Euclidean space.

To efficiently visualize the topological structure of large-scale closed surfaces, we adopt a standard simplification method: tubular regions of the form  $S^1 \times D^1$  are reduced to line segments, and cross-shaped junctions are simplified into nodes where line segments intersect. Thus, the diagram becomes a one-dimensional CW complex that fully encodes the homeomorphism type of the surface. Consequently, the two-dimensional closed manifold is simplified into a one-dimensional CW complex without endpoints.

There is a canonical way to reconstruct the original surface from this diagram: treat the diagram  $D$  as the core of a three-dimensional solid handlebody  $H$ , thicken it into solid handles by taking a regular neighborhood of the embedding of this graph in three-dimensional Euclidean space, and finally take the boundary  $\partial H$  of this solid handlebody to obtain a closed surface homeomorphic to the original one.

In schematic diagrams, nodes at line intersections are typically represented as solid dots. To further express the specific embedding of the surface in three-dimensional space, when two lines representing the surface cross in projection, we follow the drawing convention of knot diagrams: the line being covered is indicated with a break to clarify the over-and-under relationship of the embedding.

**Construction 2.7:** Starting with the initial case  $n = 4$ , draw a square centered at the origin with side length 2 and sides parallel to the coordinate axes. For any  $n \geq 5$ , first

duplicate the diagram for size  $n - 1$  twice, shifting one copy left and the other right by a distance of  $2^{n-4}$ ; then multiply all  $y$ -coordinates of the resulting diagram by 2, effectively stretching its vertical segments to twice their original length. Next, rotate a copy of the current diagram by  $90^\circ$  about the origin (applying the coordinate transformation  $(x, y) \mapsto (-y, x)$ ) and superimpose it onto the original to obtain the final pattern for size  $n$ . When handling all intersections except corner points, they are classified into two types: if an intersection lies at the midpoint of any segment, it is marked as a “node” with a solid dot in the same color as that segment; all other intersections receive a visual 3D layering treatment where horizontal lines always lie on top of vertical lines, i.e., a short portion of the vertical line is obscured near the intersection to achieve depth separation.

Schematic diagrams for  $n = 5, 6, 7, 8$  are shown in Figure 2-3, Figure 2-4, Figure 2-5, and Figure 2-6, respectively.

We now discuss why the embedding construction described above suits our needs, along with some of its properties.

**Theorem 2.8:** Construction 2.7 above provides an embedding of surfaces into  $\mathbb{R}^3$ ; the resulting surface is exactly the object of our study, i.e.  $Z(C_n, (D^1, S^0))$ .

**Proof:** We prove this by induction. First, for the base case  $n = 4$ , the line arrangement is a square, which represents a torus. We know that this corresponds to the polyhedral complex  $Z(C_n, (D^1, S^0))$ . Subsequently, the larger-scale surface for  $n + 1$  is obtained by taking two surfaces of scale  $n$ , removing two small disks (holes) from each, and then connecting these holes with a handle. The resulting surface is exactly the surface of scale  $n + 1$ , which is consistent with the definition given by the polyhedral product. ■

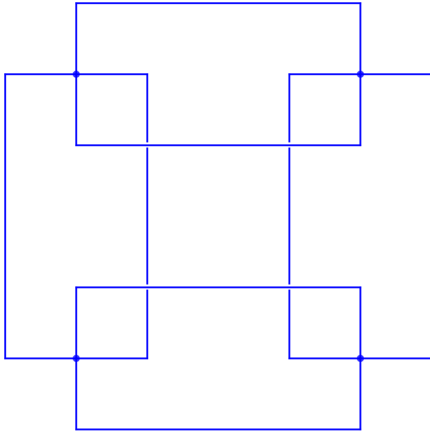
**Theorem 2.9:** For  $n \geq 5$ , the isometry group of the surface embedded in  $\mathbb{R}^3$  by Construction 2.7 is isomorphic to the Klein four-group  $V_4$ .

**Proof:** Since each surface embedding is obtained by thickening its corresponding line graph, it suffices to show that the line graph itself has the same symmetry. The line graph of order  $n + 1$  is constructed from that of order  $n$  by a doubling and superposition procedure; we therefore proceed by induction on  $n$ .

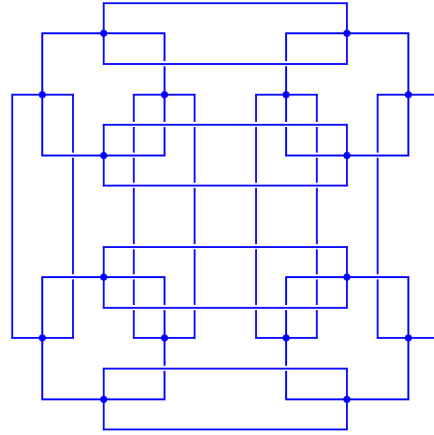
Base case  $n = 5$ . We denote the four points in the space as follows, corresponding to the four rectangles in the embedding:

$$\begin{aligned} A &= (a, 0, -b), & B &= (-a, 0, -b), \\ C &= (0, a, b), & D &= (0, -a, b), \end{aligned}$$

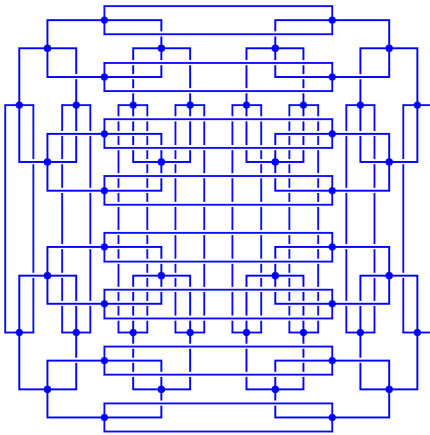
with  $a > 0$ ,  $b > 0$ , and  $a \neq \sqrt{2}b$  (so that the tetrahedron is not regular). The orientation-



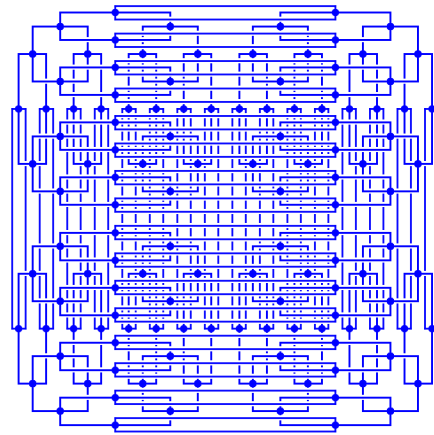
(a)  $n = 5, g = 5$



(b)  $n = 6, g = 17$



(c)  $n = 7, g = 49$



(d)  $n = 8, g = 129$

Figure 2-2 The line graph corresponding to different values of  $n = 5, 6, 7, 8$ , and genus  $g = (n - 4) \cdot 2^{n-3} + 1$

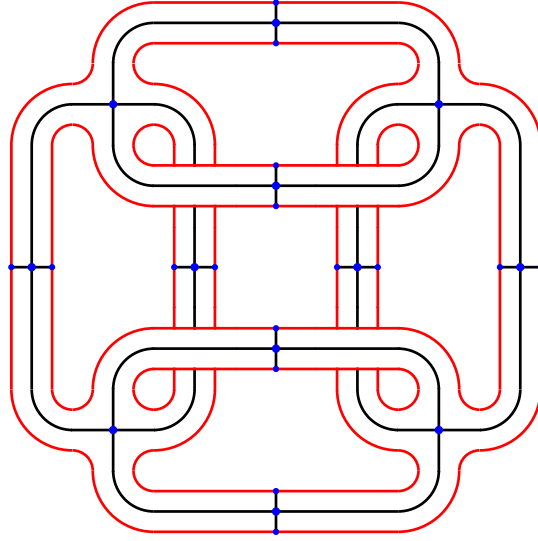


Figure 2-3 The genus  $g = 5$  surface corresponding to  $n = 5$ , together with its systoles, embedded in  $\mathbb{R}^3$

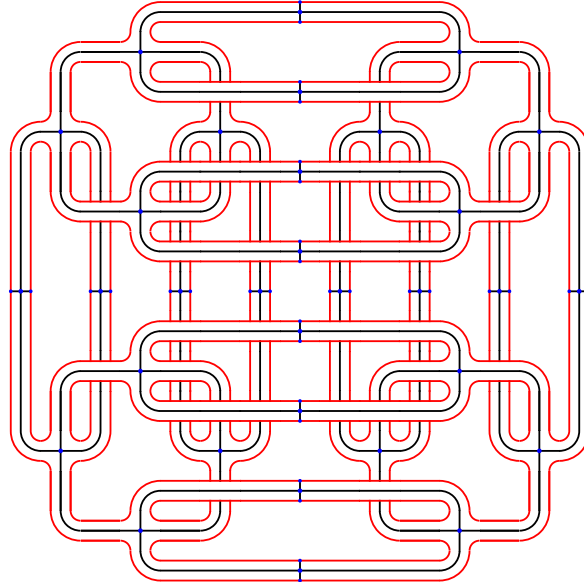


Figure 2-4 The genus  $g = 17$  surface corresponding to  $n = 6$ , together with its systoles, embedded in  $\mathbb{R}^3$

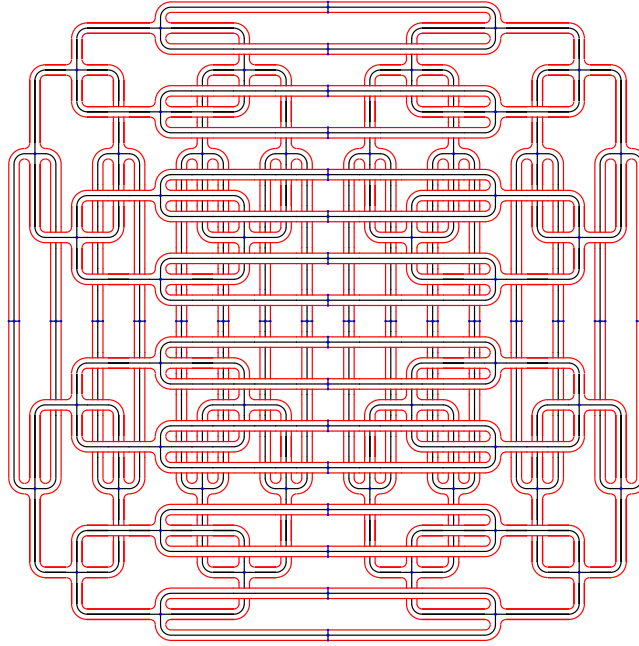


Figure 2-5 The genus  $g = 49$  surface corresponding to  $n = 7$ , together with its systoles, embedded in  $\mathbb{R}^3$

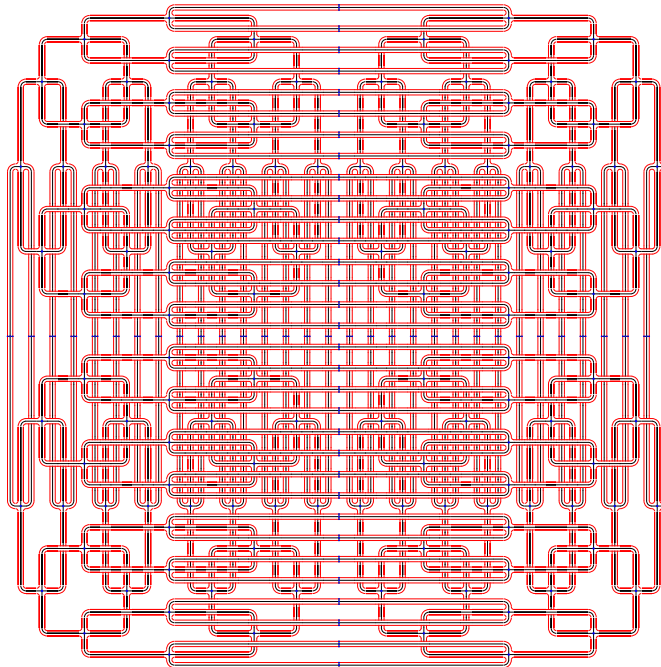


Figure 2-6 The genus  $g = 129$  surface corresponding to  $n = 8$ , together with its systoles, embedded in  $\mathbb{R}^3$

preserving rigid symmetry group of this configuration consists of four rotations:

- (1) the identity;
- (2) a  $180^\circ$  rotation about the  $z$ -axis, swapping  $A \leftrightarrow B$  and  $C \leftrightarrow D$ ;
- (3) a  $180^\circ$  rotation about the axis  $(1, -1, 0)$ , swapping  $A \leftrightarrow D$  and  $B \leftrightarrow C$ ;
- (4) and a  $180^\circ$  rotation about the axis  $(1, 1, 0)$ , swapping  $A \leftrightarrow C$  and  $B \leftrightarrow D$ .

These three nontrivial rotations are pairwise perpendicular, each of order two, and the set is closed under composition, forming an abelian group of order four; hence it is isomorphic to  $V_4$ . (If  $a = \sqrt{2}b$ , the configuration becomes a regular tetrahedron and the symmetry group enlarges; in that degenerate case the symmetry group is isomorphic to  $A_4$ .)

Inductive step: Assume the claim holds for some  $n \geq 5$ . To construct the line graph of size  $n + 1$ , we first take two copies of the scale  $n$  graph, shifting one left and the other right; we can regard these two parts to be new points  $C'$  and  $D'$ . We then scale all  $y$ -coordinates by a factor of 2 and stretch every vertical segment to twice its original length, this operation does not affect the symmetry. Finally, we rotate a copy of the current configuration by  $90^\circ$  about the origin (i.e., apply the transformation  $(x, y) \mapsto (-y, x)$ ) and superimpose it onto the original. This yields the configuration of size  $n + 1$ ; the new copies of the stretched graph can be denote the new points  $A'$  and  $B'$ . Hence the resulting line graph still admits the same four isometries, completing the induction. ■

**Remark 2.5:** With suitable local fine-tuning of the shape, it is conjectured that this global configuration may yield a good embedding of the surface, possibly providing an example of a Willmore surface.

Let  $\Sigma$  be a smooth surface in  $\mathbb{R}^3$ . Its Willmore energy is defined as

$$\mathcal{W}(\Sigma) = \int_{\Sigma} H^2 dA,$$

where  $H$  denotes the mean curvature of the surface and  $dA$  is the area element. A surface  $\Sigma$  is called a *Willmore surface* if it is a critical point of the Willmore energy, the first variation of the energy vanishes for all infinitesimal variations preserving the conformal structure.

Equivalently, Willmore surfaces satisfy the corresponding Euler-Lagrange equation:

$$\Delta H + 2H(H^2 - K) = 0,$$

where  $\Delta$  is the Laplace–Beltrami operator on the surface and  $K$  denotes the Gaussian curvature.

**Remark 2.6:** We note that in our constructed embedding, the edge lengths within each block are not uniform when measured with the standard Euclidean metric. Another idea is to adjust the metric on the three-dimensional space to make every edge length equal, which would also introduce some interesting geometry and metric considerations.

We observe that in regions higher along the  $z$ -axis, the horizontal edges become longer while the vertical edges become shorter. Conversely, in lower regions along the  $z$ -axis, the folding pattern behaves oppositely. Moreover, the magnitude of this variation grows exponentially with changes in the  $z$ -coordinate. Therefore, if we adjust the metric according to the  $z$ -coordinate, we can ensure that each edge within every block has roughly the same order of magnitude, that is, they become approximately uniform. Following this approach, we can equip the three-dimensional space with the following metric:

$$ds^2 = e^{2z}dx^2 + e^{-2z}dy^2 + dz^2$$

This metric appears in many other areas of mathematics.

*Thurston geometries* provide a unified framework for the classification of three-dimensional manifolds. Thurston conjectured, and Perelman later completed the proof using techniques from the Ricci flow in geometric analysis, that any closed 3-manifold can be decomposed into pieces, each admitting one of *eight model geometries*. One of these is **Sol** geometry. The metric  $ds^2 = e^{2z}dx^2 + e^{-2z}dy^2 + dz^2$  is the standard model metric for Sol on  $\mathbb{R}^3$ . Unlike spherical or hyperbolic geometry, Sol is anisotropic: it expands exponentially in the  $x$ -direction and contracts exponentially in the  $y$ -direction. Consequently, its curvature is neither constant nor isotropic. A canonical example arises from the mapping torus of a two-dimensional torus  $T^2$ , where the gluing map is given by a hyperbolic matrix in  $SL(2, \mathbb{Z})$ ; the resulting 3-manifold naturally admits a Sol geometric structure.

In dynamical systems, this metric is intimately related to Anosov systems. More specifically, a hyperbolic automorphism of the torus, given by a matrix  $A \in SL(2, \mathbb{Z})$  with  $|\text{tr}(A)| > 2$ , admits a splitting into stable and unstable directions: one direction contracts exponentially, while the other expands exponentially. When this discrete dynamical system is suspended to a continuous-time flow via its mapping torus, the corresponding geometric model is precisely Sol space. In this formulation, the factors  $e^z$  and  $e^{-z}$  in the metric encode the exponential growth and decay of the unstable and stable directions, respectively. In this sense, the metric can be interpreted as a geometric realization of Anosov dynamics: it encodes exponential stretching and contracting behavior directly

into the geometry of the space, thereby serving as an important bridge between geometric topology and hyperbolic dynamical systems.

## Cellular homology and notion for closed geodesics

The dual complex

$$X^* = Z(C_n, (D^1, S^0))^*$$

is a cell complex whose 2-cells are  $2^n$  many  $n$ -gons, whose 1-cells are  $n \cdot 2^{n-1}$  edges, and whose 0-cells are  $n \cdot 2^{n-2}$  vertices.

The 2-cells of  $X^*$  correspond to the vertices of  $X$ , that is, to the vertices of  $Q_n$ . Hence they can be labeled by elements of the vector space  $\mathbb{F}_2^n$ , namely by binary sequences

$$(a_1, a_2, \dots, a_n), \quad a_i \in \mathbb{F}_2,$$

where the indices are understood cyclically modulo  $n$ .

We use the same notation for corresponding cells in  $X$  and  $X^*$ : vertices correspond to 2-cells, and edges correspond to edges. More precisely:

- A 2-cell of  $X^*$  corresponds to a vertex of  $X$ , and is labeled by its coordinate

$$(a_1, a_2, \dots, a_n).$$

- A 1-cell of  $X^*$  corresponds to an edge of  $X$ , obtained by changing the  $i$ -th coordinate. We introduce the symbol  $*$  to indicate that the corresponding entry takes both values 0 and 1. Thus such an edge is denoted by

$$(a_1, \dots, *, \dots, a_n),$$

where the  $i$ -th entry is  $*$ , and all other entries are fixed in  $\{0, 1\}$ .

- A 0-cell of  $X^*$  corresponds to a 2-cell of  $X$ . This 2-cell is determined by allowing two adjacent coordinates, say the  $i$ -th and  $(i + 1)$ -st, to vary over  $\{0, 1\}$ , producing four vertices. Accordingly, such a vertex of  $X^*$  is denoted by

$$(a_1, \dots, *, *, \dots, a_n),$$

where the  $i$ -th and  $(i + 1)$ -st entries are  $*$ , and all other entries are fixed in  $\{0, 1\}$ .

The cellular homology of  $X^*$  can be computed from its cellular chain complex. For convenience, we work throughout with coefficients in  $\mathbb{F}_2$ . According to the incidence relations among faces, edges, and vertices, the boundary operators are described as follows.

First, each 2-cell is adjacent to  $n$  edges, obtained by replacing one coordinate with

\*. Therefore,

$$\begin{aligned} d_2(a_1, a_2, \dots, a_n) &= (*, a_2, \dots, a_n) + (a_1, *, a_3, \dots, a_n) + \dots \\ &\quad + (a_1, a_2, \dots, *, a_n) + (a_1, a_2, \dots, a_{n-1}, *). \end{aligned}$$

Second, the boundary of each edge consists of the two vertices obtained by replacing the  $*$  together with one of its two adjacent positions by  $*$ . For example,

$$d_1(a_1, *, a_3, \dots, a_n) = (*, *, a_3, \dots, a_n) + (a_1, *, *, \dots, a_n).$$

Hence the cellular chain complex of  $X^*$  over  $\mathbb{F}_2$  is

$$0 \rightarrow C_2 \xrightarrow{d_2} C_1 \xrightarrow{d_1} C_0 \rightarrow 0,$$

that is,

$$0 \rightarrow \mathbb{F}_2^{2^n} \rightarrow \mathbb{F}_2^{n \cdot 2^{n-1}} \rightarrow \mathbb{F}_2^{n \cdot 2^{n-2}} \rightarrow 0.$$

Moreover, the first homology group is

$$H_1(X^*) \cong \mathbb{Z}^{(n-4) \cdot 2^{n-2} + 2}.$$

It is generated by the 1-cells of  $X^*$ , and the relations among these generators are exactly those coming from the boundaries of the 2-cells, namely the  $n$ -gons.

Next we describe a labeling of the shortest closed geodesics, namely the systoles. Each systole is composed of four edges, which form a square of the following form:

$$(\dots, 0, *, 0, \dots) \rightarrow (\dots, 0, *, 1, \dots) \rightarrow (\dots, 1, *, 1, \dots) \rightarrow (\dots, 1, *, 0, \dots) \rightarrow (\dots, 0, *, 0, \dots).$$

After choosing an orientation, equivalently, after fixing the position of the symbol  $*$  to be the  $i$ -th coordinate, the systole can be labeled by this cyclic sequence of four edges. In this description, the  $(i-1)$ -st and  $(i+1)$ -st coordinates vary through the pattern

$$(0, 0) \rightarrow (0, 1) \rightarrow (1, 1) \rightarrow (1, 0),$$

while all remaining coordinates are fixed arbitrarily in  $\{0, 1\}$ .

Accordingly, a systole may be denoted by

$$(a_1, \dots, *, *, *, \dots, a_n),$$

where the symbol  $*$  indicates the fixed direction of the systole, and the two symbols  $*$  indicate the two adjacent coordinates involved in the above cyclic pattern. The other coordinates  $a_j$  are arbitrary elements of  $\mathbb{F}_2$ . Indeed, there are  $n$  possible choices for the position of the distinguished coordinate  $*$ , and for each such choice, the remaining  $n-3$  coordinates can be chosen freely in  $\mathbb{F}_2$ . Therefore, the total number of distinct systoles is

$n 2^{n-3}$ .

## Standard Kernel Bases Construction

We briefly recall the construction introduced in the introduction. Let  $S$  be the set of oriented systoles of  $X$ , and let  $\tilde{S} := \mathbb{Z}^S$  be the free abelian group generated by  $S$ . Each  $\gamma \in S$  determines a conjugacy class  $\kappa(\gamma) \in \text{Conj}(\pi_1(X))$ . Since the Hurewicz homomorphism  $h : \pi_1(X) \rightarrow H_1(X; \mathbb{Z})$  is constant on conjugacy classes, it induces a well-defined map  $\tilde{h} : \text{Conj}(\pi_1(X)) \rightarrow H_1(X; \mathbb{Z})$ . Consequently, the assignment  $\gamma \mapsto \tilde{h}(\kappa(\gamma))$  from  $S$  to  $H_1(X; \mathbb{Z})$  extends uniquely to a group homomorphism

$$\tilde{h} : \tilde{S} \rightarrow H_1(X; \mathbb{Z}).$$

This homomorphism records the homology classes represented by systoles. Associated to  $\tilde{h}$  we have the canonical exact sequence

$$0 \rightarrow \text{Ker } \tilde{h} \rightarrow \tilde{S} \xrightarrow{\tilde{h}} H_1(X; \mathbb{Z}) \rightarrow \text{Coker } \tilde{h} \rightarrow 0.$$

In what follows, we study the algebraic information carried by systoles through this sequence, with particular emphasis on the kernel  $\text{Ker } \tilde{h}$  and the image  $\text{Im } \tilde{h} \subseteq H_1(X; \mathbb{Z})$ .

**Construction 2.8:** We now describe the general construction for an arbitrary independent set. Fix a vector  $v = (v_1, \dots, v_n) \in \mathbb{F}_2^n$ . The particular choice of  $v$  is not essential; it only specifies which value is imposed on each fixed coordinate. Thus the choice  $v = (1, \dots, 1)$  is a convenient normalization, but replacing it by any other vector yields the same construction and a different basis of the same intersection space.

Let  $A \subset [n]$  be a non-empty independent set. Define

$$\psi_f(A) := \sum_{\substack{(a_1, \dots, a_n) \in \mathbb{F}_2^n \\ a_i = v_i \text{ for all } i \in A}} (a_1, \dots, a_n).$$

In other words,  $\psi_f(A)$  is the sum of all 2-cells obtained by fixing the coordinates in  $A$  to the corresponding values of  $v$ , while allowing all coordinates outside  $A$  to vary freely.

Applying the boundary map  $d_2$ , all contributions coming from directions not in  $A$  cancel in pairs over  $\mathbb{F}_2$ . Hence only the directions indexed by elements of  $A$  remain, and we obtain

$$d_2(\psi_f(A)) = \sum_{i \in A} \sum_{\substack{a_j \in \mathbb{F}_2, j \notin A \\ a_j = v_j, j \in A \setminus \{i\}}} (a_1, \dots, a_{i-1}, *, a_{i+1}, \dots, a_n).$$

We define

$$\psi_s(A) := d_2(\psi_f(A)).$$

Since  $A$  is independent, for every  $i \in A$  neither  $i - 1$  nor  $i + 1$  belongs to  $A$ . Therefore, in each summand with the symbol  $(*)$  in the  $i$ -th coordinate, the two adjacent coordinates  $i - 1$  and  $i + 1$  are free. For each fixed choice of the remaining free coordinates, these four possibilities form exactly one systole. Thus  $\psi_s(A)$  is a linear combination of systoles.

Moreover, since it is the boundary of the 2-chain  $\psi_f(A)$ , it also lies in the boundary space. Therefore

$$\psi_s(A) \in \tilde{S} \cap B.$$

The construction is summarized by the following commutative diagram:

$$\begin{array}{ccccc} & & C_2 & & \\ & \nearrow \psi_f & & \searrow d_2 & \\ I(C_n) \setminus \{\emptyset\} & \xrightarrow{\psi_s} & \tilde{S} \cap B & \hookrightarrow & Z_1 \subset C_1. \end{array}$$

Equivalently, for every non-empty independent set  $A$ ,

$$\psi_s(A) = d_2(\psi_f(A)).$$

As  $A$  ranges over the non-empty independent sets of  $C_n$ , these elements give the desired basis of  $\tilde{S} \cap B$ . Different choices of the vector  $v$  give different representatives, but the same construction and the same conclusion.

The reason independent sets appear in this construction is precisely that, when  $i \in A$ , the two neighboring coordinates  $i - 1$  and  $i + 1$  must remain free in order to produce systoles. The following algebraic normal form argument proves that this condition is not only sufficient but also necessary: every 2-chain whose boundary is a linear combination of systoles is generated by subcubes whose fixed coordinate sets are independent.

**Example 2.4:** We illustrate the construction in the case  $n = 6$ . In both examples we take  $v = (1, 1, 1, 1, 1, 1)$ .

The symbol  $\star$  is used only as a shorthand: in an expression of the form  $(\dots, \star, \star, \star, \dots)$ , the symbol  $\star$  indicates the direction of the corresponding edges, while the two symbols  $\star$  indicate the two adjacent coordinates which vary freely. Thus such an expression denotes the sum of the four edges forming one systole, with all other coordinates fixed or summed over as indicated.

First consider the independent set

$$A = \{1, 3, 5\} = (\bullet, \circ, \bullet, \circ, \bullet, \circ).$$

Then

$$\psi_f(A) = \sum_{a_2, a_4, a_6 \in \mathbb{F}_2} (1, a_2, 1, a_4, 1, a_6).$$

Applying  $d_2$ , only the directions 1, 3, 5 survive, and hence

$$\psi_s(A) = d_2(\psi_f(A)) = \sum_{a_4 \in \mathbb{F}_2} (*, *, 1, a_4, 1, *) + \sum_{a_6 \in \mathbb{F}_2} (1, *, *, *, 1, a_6) + \sum_{a_2 \in \mathbb{F}_2} (1, a_2, 1, *, *, *).$$

Equivalently, this is represented by the commutative diagram

$$\begin{array}{ccc}
 & \sum_{a_2, a_4, a_6 \in \mathbb{F}_2} (1, a_2, 1, a_4, 1, a_6) & \\
 \psi_f \nearrow & & \downarrow d_2 \\
 \{1, 3, 5\} & & \\
 \psi_s \searrow & & \\
 \sum_{a_4 \in \mathbb{F}_2} (*, *, 1, a_4, 1, *) + \sum_{a_6 \in \mathbb{F}_2} (1, *, *, *, 1, a_6) + \sum_{a_2 \in \mathbb{F}_2} (1, a_2, 1, *, *, *) & & 
 \end{array}$$

Second consider the independent set

$$A = \{1, 4\} = (\bullet, \circ, \circ, \bullet, \circ, \circ).$$

Then

$$\psi_f(A) = \sum_{a_2, a_3, a_5, a_6 \in \mathbb{F}_2} (1, a_2, a_3, 1, a_5, a_6).$$

Therefore

$$\psi_s(A) = d_2(\psi_f(A)) = \sum_{a_3, a_5 \in \mathbb{F}_2} (*, *, a_3, 1, a_5, *) + \sum_{a_2, a_6 \in \mathbb{F}_2} (1, a_2, *, *, *, a_6).$$

Equivalently, we have

$$\begin{array}{ccc}
 & \sum_{a_2, a_3, a_5, a_6 \in \mathbb{F}_2} (1, a_2, a_3, 1, a_5, a_6) & \\
 \psi_f \nearrow & & \downarrow d_2 \\
 \{1, 4\} & & \\
 \psi_s \searrow & \sum_{a_3, a_5 \in \mathbb{F}_2} (*, *, a_3, 1, a_5, *) + \sum_{a_2, a_6 \in \mathbb{F}_2} (1, a_2, *, *, *, a_6) &
 \end{array}$$

The general construction can be summarized by the following commutative diagram:

$$\begin{array}{ccc}
 & \psi_f(A) = \sum_{a_j \in \mathbb{F}_2, j \notin A} (a_1, \dots, a_n), \quad a_i = v_i \text{ for } i \in A & \\
 \psi_f \nearrow & & \downarrow d_2 \\
 A \in I(C_n) \setminus \{\emptyset\} & & \\
 \psi_s \searrow & \psi_s(A) = \sum_{i \in A} \sum_{\substack{a_j \in \mathbb{F}_2, j \notin A \\ a_j = v_j, j \in A \setminus \{i\}}} (a_1, \dots, a_{i-1}, *, a_{i+1}, \dots, a_n) &
 \end{array}$$

This diagram shows that every independent set produces an element of  $\tilde{S} \cap B$ . We now use the algebraic normal form argument to prove the converse: if a 2-chain has boundary lying in the span of systoles, then its fixed coordinate set must be independent. Thus the independent-set construction captures exactly the chains whose boundaries are generated by systoles.

**Theorem 2.10:** The image of  $\psi_s$  is a basis of  $\text{Ker } \tilde{h}$ .

**Corollary 2.2:** By the enumeration formula, we can determine the ranks of the free abelian groups:

$$\text{rank Ker } \tilde{h} = L(n) - 1,$$

$$\text{rank Im } \tilde{h} = n \cdot 2^{n-3} - L(n) + 1.$$

**Proof:** The exact sequence and its rank of each terms is displays as following:

$$\begin{array}{ccccccc} 0 & \rightarrow & \text{Ker } \tilde{h} & \rightarrow & \tilde{S}(X) & \xrightarrow{\tilde{h}} & H_1(X) & \rightarrow & \text{Coker } \tilde{h} & \rightarrow & 0 \\ 0 & \rightarrow & \mathbb{F}_2^{L_{n-1}} & \rightarrow & \mathbb{F}_2^{n \cdot 2^{n-3}} & \rightarrow & \mathbb{F}_2^{2((n-4)2^{n-3}+1)} & \rightarrow & \mathbb{F}_2^{(n-8)2^{n-3}+L_n+1} & \rightarrow & 0 \end{array}$$

■

## ANF and Boolean function

Throughout the paper, all indices are understood cyclically in  $\mathbb{Z}/n\mathbb{Z}$ . In particular,  $m-1$  and  $m+1$  are taken modulo  $n$ . We assume  $n \geq 3$ .

Under the identification of  $C_2$  with Boolean functions on  $\mathbb{F}_2^n$ , a 2-chain is viewed as the indicator function of a subset of vertices of the cube. The boundary map  $d_2$  becomes the Boolean derivative in each coordinate. The condition that  $d_2\beta$  is a linear combination of systoles is therefore translated into an independence condition on the neighboring variables.

**Definition 2.9 (Boolean functions and chain spaces):** Let  $\mathbb{F}_2 = \{0, 1\}$  be the field with two elements. Let  $F = \mathbb{F}_2^n$ . Define  $C_2 = \mathbb{F}_2\langle F \rangle$ , the  $\mathbb{F}_2$ -vector space with basis indexed by the elements of  $F$ . We may identify every element  $\beta \in C_2$  with a Boolean function  $\beta : F \rightarrow \mathbb{F}_2$ .

**Definition 2.10 (The space of one-dimensional faces):** Let  $E = \mathbb{Z}/n\mathbb{Z} \times \mathbb{F}_2^{n-1}$ . An element of  $E$  is interpreted as a vector with one distinguished coordinate replaced by a symbol  $*$ :  $(x_1, \dots, x_{m-1}, *, x_{m+1}, \dots, x_n)$ . The symbol  $*$  appears in the  $m$ -th coordinate.

Define  $C_1 = \mathbb{F}_2\langle E \rangle$ . Equivalently,  $C_1$  decomposes as a direct sum

$$C_1 = \bigoplus_{m \in \mathbb{Z}/n\mathbb{Z}} C_{1,m},$$

where  $C_{1,m}$  is the subspace spanned by those basis elements whose  $*$  appears in the  $m$ -th coordinate.

**Definition 2.11 (The differential):** For the algebraic reformulation below, we write  $d$  for the same boundary map  $d_2$ . Define a linear map  $d : C_2 \rightarrow C_1$  on basis elements by

$$d(x_1, \dots, x_n) = \sum_{m \in \mathbb{Z}/n\mathbb{Z}} (x_1, \dots, x_{m-1}, *, x_{m+1}, \dots, x_n).$$

Extending linearly gives the differential  $d : C_2 \rightarrow C_1$ .

If  $\beta \in C_2$  is viewed as a Boolean function, then the  $m$ -th component of  $d\beta$  is denoted by  $(d\beta)_m$ . It is a Boolean function of all variables except  $x_m$ .

**Definition 2.12 (Boolean algebra):** Let  $R_n = \mathbb{F}_2[x_1, \dots, x_n]/(x_1^2 - x_1, \dots, x_n^2 - x_n)$ .

Since each variable takes values in  $\mathbb{F}_2$ , one has  $x_i^2 = x_i$  as functions on  $\mathbb{F}_2^n$ . Therefore every polynomial function on  $\mathbb{F}_2^n$  can be represented by a squarefree polynomial.

For every subset  $I \subseteq \mathbb{Z}/n\mathbb{Z}$ , write  $x_I = \prod_{i \in I} x_i$ , with the convention  $x_\emptyset = 1$ .

**Theorem 2.11 (Algebraic normal form):** Every Boolean function  $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$  has a unique expression of the form

$$f(x) = \sum_{I \subseteq \mathbb{Z}/n\mathbb{Z}} c_I x_I, \quad c_I \in \mathbb{F}_2.$$

This expression is called the algebraic normal form (ANF) of  $f$ , also called the Zhegalkin polynomial.

**Proof:** For every subset  $J \subseteq \mathbb{Z}/n\mathbb{Z}$ , let  $1_J \in \mathbb{F}_2^n$  be the point defined by

$$(1_J)_i = \begin{cases} 1, & i \in J, \\ 0, & i \notin J. \end{cases}$$

Suppose first that  $f(x) = \sum_{I \subseteq \mathbb{Z}/n\mathbb{Z}} c_I x_I$ . Evaluating at  $1_J$ , we get  $f(1_J) = \sum_{I \subseteq J} c_I$ , because  $x_I(1_J) = 1$  if and only if  $I \subseteq J$ .

Conversely, define  $c_I = \sum_{J \subseteq I} f(1_J)$ . Then for any  $K \subseteq \mathbb{Z}/n\mathbb{Z}$ ,  $\sum_{I \subseteq K} c_I = \sum_{I \subseteq K} \sum_{J \subseteq I} f(1_J)$ . Changing the order of summation gives  $\sum_{I \subseteq K} c_I = \sum_{J \subseteq K} \#\{I : J \subseteq I \subseteq K\} f(1_J)$ . But  $\#\{I : J \subseteq I \subseteq K\} = 2^{|K|-|J|}$ . In the field  $\mathbb{F}_2$ , this number is zero unless  $J = K$ . Hence  $\sum_{I \subseteq K} c_I = f(1_K)$ . Therefore the polynomial  $\sum_{I \subseteq \mathbb{Z}/n\mathbb{Z}} c_I x_I$  agrees with  $f$  on every point of  $\mathbb{F}_2^n$ . Hence it represents  $f$ .

The same formula also proves uniqueness. Indeed, if  $\sum_{I \subseteq \mathbb{Z}/n\mathbb{Z}} c_I x_I$  is the zero function, then all values  $f(1_J)$  are zero, and therefore every coefficient  $c_I$  is zero. ■

**Definition 2.13 (Subcubes and their indicator functions):** Let  $I \subseteq \mathbb{Z}/n\mathbb{Z}$  and let  $a = (a_i)_{i \in I} \in \mathbb{F}_2^I$ . Define the subcube  $Q(I, a) = \{x \in \mathbb{F}_2^n : x_i = a_i \text{ for all } i \in I\}$ . The set  $I$  is called the set of fixed coordinates of the subcube.

The indicator function of  $Q(I, a)$  is  $\chi_{I,a}(x) = \prod_{i \in I} (x_i + a_i + 1)$ . Indeed, if  $a_i = 1$ , then  $x_i + a_i + 1 = x_i$ , which is the indicator of the condition  $x_i = 1$ . If  $a_i = 0$ , then  $x_i + a_i + 1 = x_i + 1$ , which is the indicator of the condition  $x_i = 0$ .

**Definition 2.14 (Independent sets in the cycle graph):** We regard  $\mathbb{Z}/n\mathbb{Z}$  as the vertex set of the cycle graph  $C_n$ , where  $i$  is adjacent to  $i + 1$ .

A subset  $I \subseteq \mathbb{Z}/n\mathbb{Z}$  is called an independent set if it contains no two adjacent vertices. Equivalently, there is no  $i \in \mathbb{Z}/n\mathbb{Z}$  such that  $i \in I$  and  $i + 1 \in I$ .

**Definition 2.15 (Special generators):** Fix  $m \in \mathbb{Z}/n\mathbb{Z}$ . Fix all coordinates except  $m - 1, m, m + 1$ . Let the fixed values be  $b_i \in \mathbb{F}_2$ ,  $i \notin \{m - 1, m, m + 1\}$ . Define an

element  $s_{m,b} \in C_1$  by

$$s_{m,b} = \sum_{a_{m-1}, a_{m+1} \in \mathbb{F}_2} (\dots, a_{m-1}, *, a_{m+1}, \dots),$$

where the  $m$ -th coordinate is  $*$ , the coordinates  $m-1$  and  $m+1$  vary freely over  $\mathbb{F}_2$ , and all remaining coordinates are fixed by  $b$ .

Let  $S = \{s_{m,b}\}$  be the set of all such special generators, and let  $\langle S \rangle \subseteq C_1$  denote the  $\mathbb{F}_2$ -linear span of  $S$ .

**Lemma 2.1:** For fixed  $m$ , the subspace of  $C_{1,m}$  spanned by the generators  $s_{m,b}$  is exactly the space of functions which do not depend on the two variables  $x_{m-1}, x_{m+1}$ .

**Proof:** Fix  $m$ . Elements of  $C_{1,m}$  may be regarded as Boolean functions in all variables except  $x_m$ .

Now fix all variables except  $x_{m-1}, x_{m+1}$ . The basis elements of  $C_{1,m}$  are partitioned into four-point blocks:  $\{(a_{m-1}, a_{m+1}) : a_{m-1}, a_{m+1} \in \mathbb{F}_2\}$ . Each generator  $s_{m,b}$  is precisely the indicator function of one such four-point block.

Therefore any linear combination of the  $s_{m,b}$ 's is constant on each such block. Hence it does not depend on  $x_{m-1}, x_{m+1}$ .

Conversely, if a function in  $C_{1,m}$  does not depend on  $x_{m-1}, x_{m+1}$ , then it is constant on every such four-point block. Therefore it can be written as the sum of precisely those block indicators on which it takes the value 1. Hence it lies in the span of the  $s_{m,b}$ 's. ■

**Lemma 2.2 (Boolean derivative in algebraic normal form):** Let  $\beta(x) = \sum_{I \subseteq \mathbb{Z}/n\mathbb{Z}} c_I x_I$  be the algebraic normal form of  $\beta$ . Then  $(d\beta)_m = \sum_{I \ni m} c_I x_{I \setminus \{m\}}$ .

**Proof:** It suffices to compute the effect of  $d$  on a monomial  $x_I$ .

If  $m \notin I$ , then  $x_I$  does not involve  $x_m$ . Therefore  $x_I|_{x_m=0} + x_I|_{x_m=1} = x_I + x_I = 0$ .

If  $m \in I$ , then  $x_I = x_m x_{I \setminus \{m\}}$ . Hence  $x_I|_{x_m=0} + x_I|_{x_m=1} = 0 \cdot x_{I \setminus \{m\}} + 1 \cdot x_{I \setminus \{m\}} = x_{I \setminus \{m\}}$ . By linearity, we obtain  $(d\beta)_m = \sum_{I \ni m} c_I x_{I \setminus \{m\}}$ . ■

**Lemma 2.3:** For  $\beta \in C_2$ , the condition  $d\beta \in \langle S \rangle$  holds if and only if, for every  $m \in \mathbb{Z}/n\mathbb{Z}$ , the component  $(d\beta)_m$  does not depend on the variables  $x_{m-1}, x_{m+1}$ .

**Proof:** Since  $C_1 = \bigoplus_{m \in \mathbb{Z}/n\mathbb{Z}} C_{1,m}$ , the condition  $d\beta \in \langle S \rangle$  is equivalent to the condition that each component  $(d\beta)_m$  lies in the span of the special generators of type  $m$ .

By the previous lemma on special generators, that span is precisely the space of functions independent of  $x_{m-1}, x_{m+1}$ . Therefore  $d\beta \in \langle S \rangle$  if and only if every  $(d\beta)_m$  is independent of those two neighboring variables. ■

**Theorem 2.12 (Spanning theorem):** The set of all  $\beta \in C_2$  satisfying  $d\beta \in \langle S \rangle$  is ex-

actly the  $\mathbb{F}_2$ -linear span of the indicator functions of subcubes whose fixed coordinate sets are independent sets in the cycle graph  $C_n$ . That is,  $\{\beta \in C_2 : d\beta \in \langle S \rangle\} = \text{span}_{\mathbb{F}_2} \{\chi_{I,a} : I \subseteq \mathbb{Z}/n\mathbb{Z} \text{ is independent, } a \in \mathbb{F}_2^I\}$ .

**Proof:** First suppose  $d\beta \in \langle S \rangle$ . Write the algebraic normal form of  $\beta$  as  $\beta(x) = \sum_{I \subseteq \mathbb{Z}/n\mathbb{Z}} c_I x_I$ . By the formula for the Boolean derivative,  $(d\beta)_m = \sum_{I \ni m} c_I x_{I \setminus \{m\}}$ .

By the preceding lemma, for every  $m$ , the function  $(d\beta)_m$  does not depend on  $x_{m-1}, x_{m+1}$ . Hence the algebraic normal form of  $(d\beta)_m$  cannot contain either of the variables  $x_{m-1}, x_{m+1}$ .

Now suppose  $c_I \neq 0$  and  $m \in I$ . Then the monomial  $x_{I \setminus \{m\}}$  appears in  $(d\beta)_m$ . Since algebraic normal form is unique, this monomial cannot be cancelled by any other monomial. Therefore it cannot contain either  $x_{m-1}$  or  $x_{m+1}$ . Thus  $m-1 \notin I$  and  $m+1 \notin I$ .

Since this holds for every  $m \in I$ , the set  $I$  contains no two adjacent indices. Therefore, whenever  $c_I \neq 0$ , the set  $I$  must be an independent set in  $C_n$ .

It follows that

$$\beta(x) = \sum_{\substack{I \subseteq \mathbb{Z}/n\mathbb{Z} \\ I \text{ independent}}} c_I x_I.$$

But  $x_I$  is the indicator function of the subcube  $\{x \in \mathbb{F}_2^n : x_i = 1 \text{ for all } i \in I\}$ , whose fixed coordinate set is  $I$ . Hence  $\beta$  lies in the span of subcube indicators whose fixed coordinate sets are independent.

Conversely, let  $I \subseteq \mathbb{Z}/n\mathbb{Z}$  be an independent set. Consider the monomial  $x_I$ . For any  $m \in \mathbb{Z}/n\mathbb{Z}$ , if  $m \notin I$ , then  $(dx_I)_m = 0$ . If  $m \in I$ , then  $(dx_I)_m = x_{I \setminus \{m\}}$ . Since  $I$  is independent, neither  $m-1$  nor  $m+1$  belongs to  $I$ . Hence  $x_{I \setminus \{m\}}$  does not contain the variables  $x_{m-1}, x_{m+1}$ . Therefore each component  $(dx_I)_m$  is independent of  $x_{m-1}, x_{m+1}$ . By the preceding lemma,  $dx_I \in \langle S \rangle$ . By linearity, every linear combination of such monomials also satisfies the condition.

It remains to relate these monomials to arbitrary subcube indicators. If  $I$  is independent and  $a \in \mathbb{F}_2^I$ , then  $\chi_{I,a}(x) = \prod_{i \in I} (x_i + a_i + 1)$ . Expanding this product gives  $\chi_{I,a}(x) = \sum_{J \subseteq I} d_J x_J$  for some coefficients  $d_J \in \mathbb{F}_2$ . Since every subset of an independent set is again independent, each monomial  $x_J$  appearing in this expansion has independent support. Thus every such  $\chi_{I,a}$  lies in the span of independent monomials.

Conversely, the monomial  $x_I$  itself is the indicator function of the subcube defined by  $x_i = 1$  for all  $i \in I$ . Therefore the span of independent monomials is the same as the

span of all subcube indicators whose fixed coordinate sets are independent.

Hence  $\{\beta \in C_2 : d\beta \in \langle S \rangle\} = \text{span}_{\mathbb{F}_2} \{\chi_{I,a} : I \subseteq \mathbb{Z}/n\mathbb{Z} \text{ is independent, } a \in \mathbb{F}_2^I\}$ .

The theorem is proved.  $\blacksquare$

**Remark 2.7:** The main idea is that the differential  $d$  acts as the Boolean derivative in each coordinate. The condition  $d\beta \in \langle S \rangle$  forces the derivative in the  $m$ -th direction to be independent of the two neighboring variables  $x_{m-1}, x_{m+1}$ . In algebraic normal form, this is possible only if no monomial of  $\beta$  contains two adjacent variables. Thus the allowed monomials are precisely those whose supports are independent sets in the cycle graph  $C_n$ .

**Theorem 2.13 (Basis theorem):** Fix  $v = (v_1, \dots, v_n) \in \mathbb{F}_2^n$ . For every independent set  $A \subseteq \mathbb{Z}/n\mathbb{Z}$ , let  $\psi_f(A) \in C_2$  be the indicator function of the subcube defined by  $x_i = v_i$  for all  $i \in A$ .

Equivalently,

$$\psi_f(A) = \prod_{i \in A} (x_i + v_i + 1),$$

with the convention that  $\psi_f(\emptyset) = 1$ .

Then the set  $\{\psi_f(A) : A \subseteq I(C_n)\}$  is a basis of the vector space  $\{\beta \in C_2 : d\beta \in \langle S \rangle\}$ . Consequently, the set  $\{\psi_s(A) = d\psi_f(A) : A \subseteq I(C_n) \setminus \emptyset\}$  is a basis of  $\langle S \rangle \cap \text{im } d$ . In other words, it is a basis of  $\tilde{S} \cap B$ .

**Proof:** By the preceding theorem, the vector space

$$\{\beta \in C_2 : d\beta \in \langle S \rangle\}$$

is spanned by the monomials  $x_A$  whose supports  $A$  are independent sets in  $C_n$ .

We first show that, after fixing  $v$ , the elements  $\psi_f(A)$  form a basis of this same space. For every independent set  $A$ , one has

$$\psi_f(A) = \prod_{i \in A} (x_i + v_i + 1).$$

Expanding this product gives

$$\psi_f(A) = x_A + \sum_{J \subsetneq A} c_{A,J} x_J, \quad c_{A,J} \in \mathbb{F}_2.$$

Every subset  $J \subseteq A$  is again independent. Therefore, if the independent sets are ordered by cardinality, the change-of-basis matrix from the family  $\{\psi_f(A) : A \subseteq I(C_n)\}$  to the monomial family  $\{x_A : A \subseteq I(C_n)\}$  is triangular with diagonal entries equal to 1. Hence it is invertible over  $\mathbb{F}_2$ . Thus the elements  $\psi_f(A)$ , with  $A$  independent, form a basis of  $\{\beta \in C_2 : d\beta \in \langle S \rangle\}$ .

It remains to pass from 2-chains to their boundaries. We first identify the kernel of  $d$ . Let

$$\beta = \sum_{I \subseteq \mathbb{Z}/n\mathbb{Z}} c_I x_I$$

be the algebraic normal form of  $\beta$ . If  $d\beta = 0$ , then for every  $m \in \mathbb{Z}/n\mathbb{Z}$ ,

$$(d\beta)_m = \sum_{I \ni m} c_I x_{I \setminus \{m\}} = 0.$$

By uniqueness of algebraic normal form, this implies that  $c_I = 0$  for every non-empty subset  $I$ . Hence  $\beta$  is constant. Conversely, every constant function has zero boundary. Therefore

$$\ker d = \mathbb{F}_2 \langle 1 \rangle = \mathbb{F}_2 \langle \psi_f(\emptyset) \rangle.$$

Now suppose that there is a linear relation

$$\sum_{\substack{A \subseteq I(C_n) \\ A \neq \emptyset}} \lambda_A \psi_s(A) = 0.$$

Since  $\psi_s(A) = d\psi_f(A)$ , this means

$$d \left( \sum_{\substack{A \subseteq I(C_n) \\ A \neq \emptyset}} \lambda_A \psi_f(A) \right) = 0.$$

Hence

$$\sum_{\substack{A \subseteq I(C_n) \\ A \neq \emptyset}} \lambda_A \psi_f(A)$$

is constant. But the set  $\{\psi_f(A) : A \subseteq I(C_n)\}$  is a basis, and the constant function is  $\psi_f(\emptyset)$ . Since the above linear combination contains no  $\psi_f(\emptyset)$ -term, all coefficients  $\lambda_A$  must be zero. Therefore the elements  $\{\psi_s(A) : A \neq \emptyset\}$ , are linearly independent.

The spanning property is already a consequence of the preceding theorem. Indeed, if  $\gamma \in \langle S \rangle \cap \text{im } d$ , then  $\gamma = d\beta$  for some  $\beta \in C_2$ , and  $d\beta \in \langle S \rangle$ . Hence, by the preceding theorem,

$$\beta = \sum_{A \text{ independent}} \mu_A \psi_f(A).$$

Applying  $d$ , we obtain

$$\gamma = \sum_{A \text{ independent}} \mu_A d\psi_f(A) = \sum_{\substack{A \text{ independent} \\ A \neq \emptyset}} \mu_A \psi_s(A).$$

Thus the  $\psi_s(A)$ 's span  $\langle S \rangle \cap \text{im } d$ . Since they are linearly independent by the argument above, they form a basis. ■

## Combinatorial identities

Similar to the previous Theorem 2.4 (Proposition 3.5.10 in Al-Raisi<sup>[65]</sup>), by counting the number of orbits of a given set under the action of a cyclic group using different methods, we obtain another identity relating the Möbius function and the Euler totient function.

The cyclic group  $C_n$  acts naturally on both the set of systoles  $\tilde{S}$  and the first homology group  $H_1(X_n)$ . Moreover, the map  $\tilde{h} : \tilde{S} \rightarrow H_1(X_n)$  is equivariant with respect to this action. Consequently,  $C_n$  also acts on the kernel of  $\tilde{h}$ , giving rise to the following short exact sequence of  $\mathbb{Z}[\mathbb{Z}/n]$ -modules:

$$0 \rightarrow \text{Ker } \tilde{h} \rightarrow \tilde{S} \rightarrow H_1(X_n).$$

**Theorem 2.14:** For  $n \geq 4$ , there is an identity

$$\frac{1}{n} \left( \sum_{d|n} \mu(d) L\left(\frac{n}{d}\right) \right) + \sum_{\substack{d|n \\ 1 < d < n}} \left( \frac{1}{d} \sum_{d_1|d} \mu(d_1) L\left(\frac{d}{d_1}\right) \right) = \frac{1}{n} \left( \sum_{d|n} \phi(d) L\left(\frac{n}{d}\right) \right) - 1,$$

relating the Möbius inversion function, the Euler totient function and the Lucas numbers by equating the two different orbit counting methods of the systoles.

**Proof:** The action of  $\mathbb{Z}/n$  acts equivariantly on the free abelian group generated by the mapping systole set, mapping to the Hurewicz map of the cohomology class, so it must act on the kernel of this map. The way it acts is still by rotating the cycle graph  $C_n$ , and thus it acts on the non-empty independent sets. That is to say, it acts naturally on the standard basis we have given. We then count the orbits of this group action in two different ways to obtain such an identity. One method is to use Burnside's lemma for the calculation, which gives the right-hand side of the identity. The other method is to directly count the aperiodic independent sets, which gives the left-hand side of the identity.

By Theorem 2.15 and Theorem 2.16 we can get identity:

$$\frac{1}{n} \left( \sum_{d|n} \mu(d) L\left(\frac{n}{d}\right) \right) + \sum_{\substack{d|n \\ 1 < d < n}} \left( \frac{1}{d} \sum_{d_1|d} \mu(d_1) L\left(\frac{d}{d_1}\right) \right) = \frac{1}{n} \left( \sum_{d|n} \phi(d) \left( L\left(\frac{n}{d}\right) - 1 \right) \right).$$

The last  $-1$  term can be collected using the formula  $\sum_{d|n} \phi(d) = n$ . ■

**Theorem 2.15:** Let  $C_n$  be the cycle graph with  $n$  vertices ( $n \geq 3$ ). The number of orbits of all non-empty independent sets of  $C_n$  under the action of the cyclic group  $\mathbb{Z}/n$  (by rotations) is given by

$$N = \frac{1}{n} \sum_{d|n} \phi(d) \left( L\left(\frac{n}{d}\right) - 1 \right),$$

where  $L(m) = L_m$  is the Lucas number, defined as  $L_m = F_{m-1} + F_{m+1}$  (with  $F_m$  the Fibonacci numbers,  $F_0 = 0, F_1 = 1$ ), and  $\phi$  is Euler's totient function.

**Proof:** By Burnside's lemma, the number of orbits equals the average number of fixed points under the group action, i.e.,

$$N = \frac{1}{n} \sum_{k=0}^{n-1} \text{Fix}(k),$$

where  $\text{Fix}(k)$  denotes the number of non-empty independent sets that are invariant under a rotation by  $k$  steps.

For a given rotation by  $k$ , let  $d = \gcd(n, k)$ . This rotation partitions the vertex set into  $d$  orbits, each of length  $n/d$ . An independent set  $S$  is invariant under this rotation if and only if  $S$  consists of a union of whole orbits, and these orbits (viewed as vertices of the induced cycle graph  $C_d$ ) form an independent set in  $C_d$  (since vertices from adjacent orbits are adjacent in the original graph). Therefore, the number of such  $S$  (including the empty set) equals the number of all independent sets of the cycle graph  $C_d$ . It is well known that the number of independent sets of the cycle graph  $C_m$  is the Lucas number  $L_m = F_{m-1} + F_{m+1}$ . Hence, the number of independent sets (including empty) invariant under this rotation is  $L_d$ , and consequently the number of non-empty invariant sets is  $L_d - 1$ , i.e.,  $\text{Fix}(k) = L_d - 1$ .

Now we group the values of  $k$  by their greatest common divisor with  $n$ : for each divisor  $d$  of  $n$ , the number of  $k$  such that  $\gcd(n, k) = d$  is  $\phi(n/d)$  (with the convention that  $\phi(1) = 1$  corresponds to  $k = 0$  when  $d = n$ ). Thus,

$$\sum_{k=0}^{n-1} (L_{\gcd(n,k)} - 1) = \sum_{d|n} \phi\left(\frac{n}{d}\right) (L_d - 1).$$

Substituting into Burnside's formula yields

$$N = \frac{1}{n} \sum_{d|n} \varphi\left(\frac{n}{d}\right) (L_d - 1) = \frac{1}{n} \sum_{d|n} \varphi(d) (L_{n/d} - 1).$$

This completes the proof. ■

**Theorem 2.16:** Let  $C_n$  be the cycle graph on  $n \geq 3$  vertices. Consider binary necklaces of length  $n$ , up to cyclic rotation, where the vertices colored 1 form an independent set of  $C_n$ . Equivalently, these are cyclic binary words of length  $n$  with no two consecutive 1's, including the first and last positions.

Then the number of aperiodic (equivalently, primitive) such necklaces with the 1-set nonempty is

$$N = \frac{1}{n} \sum_{d|n} \mu(d) L_{n/d}$$

where  $\mu$  is the Möbius function and  $L_m = L(m)$  is the  $m$ -th Lucas number.

**Proof:** Let  $A(n)$  be the number of cyclic binary words of length  $n$  in which no two 1's are adjacent (cyclically). This is exactly the number of independent sets of the cycle graph  $C_n$ , and it is well known that

$$A(n) = L_n.$$

Now let  $P(n)$  denote the number of such cyclic words that are primitive, i.e., whose least period is exactly  $n$ . Every admissible cyclic word has a unique least period  $d \mid n$ , and it is obtained by repeating a primitive admissible cyclic word of length  $d$  exactly  $n/d$  times. Hence

$$A(n) = \sum_{d|n} P(d).$$

Applying Möbius inversion gives

$$P(n) = \sum_{d|n} \mu(d) A(n/d) = \sum_{d|n} \mu(d) L_{n/d}.$$

Each primitive necklace has exactly  $n$  distinct rotations, so the number  $N_n$  of primitive necklaces is

$$N = \frac{P(n)}{n} = \frac{1}{n} \sum_{d|n} \mu(d) L_{n/d}.$$

Finally, the condition that the 1-set be nonempty does not change the count: the only admissible necklace with empty 1-set is  $00 \cdots 0$ , which has period 1 and is therefore not primitive for  $n \geq 3$ . Thus the same formula counts primitive necklaces whose 1-set is a

nonempty independent set. ■

## 2.5 Generalization and Future Work

The following questions and directions remain open for further investigation.

We intend to write down the explicit edges corresponding to each homology class (using the stable decomposition theorem) and then compare them directly with the edges representing geodesics. This comparison may reveal deeper relationships between combinatorial cycles and minimal length representatives.

The maximum independent set cuts the surface into certain basic pieces, suggesting the possibility of a natural Fenchel-Nielsen coordinate system. Currently, there is no canonical pants decomposition that respects the cell structure—any decomposition into a union of pants would necessarily destroy the existing cell decomposition. We hope to find a more suitable coordinate system for Teichmüller space that uses fewer parameters and is more natural than the current approach based on edge lengths and angles.

Our current work focuses on the systole function (the shortest closed geodesic). In the future, we can study geodesics of arbitrary length. Due to the nature of hyperbolic geometry, as the length grows, the multiplicity of geodesics becomes unbounded. This allows us to introduce a filtration by geodesic length and investigate how the size of the kernel of the Hurewicz map varies as the length increases. This would provide a more refined understanding of the relationship between topology and geometry.

For infinite-type surfaces (those with infinite genus or infinitely many ends), the mapping class group often called a big mapping class group, it has become a distinct area of study. Research has shifted from the discrete, finitely generated groups of the finite-type case to non-locally compact, typically uncountable Polish groups, together with their actions on arc graphs, Teichmüller spaces, and newly developed “marked” moduli spaces. Recent progress focuses on large-scale geometry, continuity and rigidity phenomena, classification up to isomorphism, and topological-dynamical properties such as Rokhlin-type behavior and amenability, as well as on constructing appropriate moduli space frameworks for infinite-type surfaces. For instance, recent surveys have established the field as an independent direction; Mann-Rafi study their coarse geometry; and Tappu introduces moduli spaces of marked Nielsen-convex hyperbolic structures adapted to the infinite-type setting, proving continuity of the mapping class group action. This line of research is important because infinite-type surfaces form a natural boundary of classical Teichmüller

and moduli space theory, yet exhibit fundamentally new group-theoretic and dynamical phenomena; at the same time, they connect low-dimensional topology, geometric group theory, complex analysis, and moduli theory, and challenge the classical principle that a surface is determined by its symmetries in a non-compact, infinitely complex setting.

The family  $Z(C_n, (D^1, S^0))^*$ , in the limit  $n \rightarrow \infty$ , suggests a new class of infinite-type surfaces with ends whose topology is more complicated than the standard examples such as the Loch Ness monster surface, Jacob's ladder surface, and the blooming Cantor tree surface. Since its automorphism group contains an infinite cyclic subgroup, the surface carries a nontrivial recurring symmetry that may influence its mapping class group, its end space, and the corresponding moduli-theoretic structure. This makes it a natural object to incorporate into the broader program of studying big mapping class groups and moduli spaces of infinite-type surfaces, especially as a candidate example where classical intuition from known infinite-type surfaces may no longer suffice.

## CHAPTER 3 REGULAR MAP THEORY

### 3.1 Topological Graph Theory

The inquiry of graphs on surfaces is a relatively young branch of mathematics, commonly known as topological graph theory<sup>[73]</sup>. Although this discipline truly took shape in modern times, its intellectual origins can be traced back to the 19th century, or even earlier<sup>[33,56]</sup>. It is generally acknowledged that Heawood's research on Map Coloring Problems laid the modern foundation for this field<sup>[35,68]</sup>. However, from a broader historical perspective, its roots extend even further back: for instance, Kepler<sup>[48]</sup> studied highly symmetric polyhedra and attempted to use these "perfect solids" to explain the hierarchical relationships between planetary orbits; similarly, Plato and his school showed profound interest in the five regular polyhedra later known as the Platonic solids<sup>[21]</sup>. In fact, regular maps, as one of the central objects in topological graph theory, can be viewed as a natural generalization of the Platonic solids to arbitrary surfaces<sup>[45]</sup>.

Topological graph theory as we now understand it took shape largely from the study of map coloring problems in the 19th century<sup>[35,47]</sup>. Its landmark beginning is often identified with the Four Color Conjecture posed in 1852: Can every planar map be colored using only four colors so that adjacent regions differ in color? This deceptively simple combinatorial question profoundly revealed the deep connections between graph structures and the topological properties of surfaces. Mathematicians soon realized that map coloring could be reformulated in terms of vertex coloring of planar graphs<sup>[51]</sup>, thereby integrating it into the broader framework of graph theory.

As research progressed, it became clear that core questions, such as planarity testing and the embeddability of graphs on surfaces, were tied not only to combinatorial graph structure but also to the topological classification of surfaces (such as sphere, torus, double torus, etc.)<sup>[33,56]</sup>. The field matured rapidly in the mid-20th century alongside advances in topology and discrete mathematics. A major milestone came in 1976 with the computer-assisted proof of the Four Color Theorem<sup>[3-4,69]</sup>, which served as a powerful catalyst for the development of topological graph theory. The graph minor theory pioneered by Robertson and Seymour in the late 20th century provided a profound finite characterization for surface embeddings, elevating topological graph theory to a central position in

modern graph theory<sup>[70]</sup>. These achievements helped systematically establish important subfields such as graph embedding theory, surface coloring formulas, and graph minor theory, shaping the discipline into its modern form.

The background of topological graph theory is inseparable from the fusion of two major branches of mathematics: on the one hand, graph theory provides discrete, combinatorial models; on the other hand, topology offers continuous, spatial theoretical tools. The core motivation for this intersection stems from the mathematical abstraction of the fundamental act of “drawing map”—how to depict network structures on surfaces without crossings. From circuit board wiring in engineering to network embedding in theoretical computer science, and further to knot theory in low-dimensional topology, this fundamental problem has given rise to a rich research system.

## Regular Maps and Symmetry of surfaces

Regular maps are the most symmetric objects in classical map theory: a map is an embedding of a graph on a closed surface such that faces are topological discs, and it is regular when its automorphism group acts transitively on flags, which are incident vertex-edge-face triples. Equivalently, a regular map can be encoded by a highly symmetric permutation description (often via triangle groups), making the subject a meeting point of combinatorics, group theory, and surface topology. Standard foundational treatments develop the flag-transitivity viewpoint, the group-theoretic presentations, and the classification/programs for families of regular maps<sup>[45,75]</sup>. For a broader perspective emphasizing highly symmetric and related map classes and modern developments, see<sup>[73]</sup>.

Dessins d’Enfants (“children’s drawings”) are combinatorial objects—typically connected bipartite graphs embedded on an oriented surface—that encode rich geometric and arithmetic data. Concretely, a dessin can be viewed as a map (a graph embedded so that faces are simply connected) with vertices colored black/white, and it corresponds to a holomorphic branched covering of the Riemann sphere ramified only over  $\{0, 1, \infty\}$ , i.e. a Belyi map<sup>[42-44]</sup>. By Belyi’s theorem, a compact Riemann surface admits such a map iff it can be defined over  $\overline{\mathbb{Q}}$ ; thus dessins form a bridge between Riemann surfaces and number fields, and they admit a natural action of  $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ , an idea emphasized by Grothendieck following Belyi’s insights<sup>[42,46]</sup>.

Because dessins sit simultaneously in topology as embedded graphs, complex analysis (branched covers), and arithmetic geometry (fields of definition and Galois action), they connect to areas far beyond group theory—such as hyperbolic geometry and the uni-

formization of surfaces, the theory of Riemann surfaces, and surprisingly directly to Galois theory and arithmetic invariants<sup>[31,46]</sup>. For approachable treatments and useful viewpoints, see elementary introductions and surveys<sup>[34]</sup> as well as broader accounts focusing on Belyi maps, bipartite structures, and map-theoretic techniques<sup>[42-44]</sup>. Recent work also explores extensions beyond the classical compact setting, including non-compact surface generalizations<sup>[6]</sup>.

## 3.2 Fixed Graph Embedding

### Hypercube Graphs

#### A bird's-eye view

Catalano, Conder, Du, Kwon, Nedela, and Wilson<sup>[19]</sup> provided a complete classification of orientably regular embeddings of the  $n$ -dimensional binary hypercubes  $Q_n$ . Research in this direction began with the work of Nedela and Škoviera<sup>[61]</sup>, who, for every  $n \geq 3$ , constructed a covering map family of orientably regular embeddings of  $Q_n$  by lifting the regular orientable embeddings of  $n$ -dipoles. Dipoles are graphs consisting of two vertices joined by  $n$  parallel edges, whose regular embeddings were classified by Nedela and Škoviera<sup>[60]</sup>. In that classification, each solution  $e$  of the congruence  $e^2 \equiv 1 \pmod{n}$  yields a distinct regular map, and different solutions correspond to non-isomorphic embeddings. They conjectured that this lifting construction exhausts all orientably regular embeddings of  $Q_n$ . Later, Du, Kwak, and Nedela<sup>[26]</sup> proved that for odd  $n$ , this is indeed the case.

Subsequently, Kwon<sup>[52]</sup> discovered new families of orientably regular embeddings for all even  $n \geq 6$  by applying a switch operator, and Xu<sup>[76]</sup> established that for  $n \equiv 2 \pmod{4}$ , Kwon's embeddings are the only additional ones beyond those from the covering construction. Further new embeddings were later constructed by Catalano and Nedela<sup>[20]</sup>, who provided a characterization of all orientably-regular embeddings of  $Q_n$  in terms of certain quadrilateral identities, and introduced a construction for new regular embeddings of  $Q_n$  for all  $n$  divisible by 16, not covered by the family of embeddings previously found by Kwon. The full classification was ultimately completed by the comprehensive work of Catalano Conder, Du, Kwon, Nedela, and Wilson<sup>[19]</sup>, which settled all remaining cases.

Additionally, it was shown by Kwon and Nedela<sup>[53]</sup> that for  $n \geq 3$ , the  $n$ -cube admits no regular embeddings on non-orientable surfaces.

### Combinatorial description of regular embeddings of hypercubes

We now recall the combinatorial description of orientably regular maps in the form used in the study of regular embeddings of hypercubes. Let  $\Gamma$  be a connected graph, and let  $D = D(\Gamma)$  denote the set of arcs (or darts) of  $\Gamma$ . Thus an element of  $D$  is an ordered pair  $(u, v)$ , where  $u$  and  $v$  are adjacent vertices of  $\Gamma$ . A cellular embedding of  $\Gamma$  in an orientable closed surface may be encoded by a pair of permutations  $(R, L)$  on  $D$ . Here  $L$  is the arc-reversing involution

$$(u, v)^L = (v, u),$$

while  $R$  is a rotation whose cycles give the cyclic order of the arcs emanating from each vertex. Thus, for every vertex  $u$ , the permutation  $R$  cyclically permutes the arcs with initial vertex  $u$ . The resulting map is denoted by

$$M = (D, R, L),$$

or simply by  $M = (\Gamma; R)$ , since  $L$  is determined by the graph  $\Gamma$ .

An automorphism of  $M$  is a graph automorphism of  $\Gamma$  that preserves this rotation, i.e. a permutation of  $D$  commuting with both  $R$  and  $L$ . The map  $M$  is called *orientably regular* if its group of orientation-preserving automorphisms acts regularly on the set  $D$  of arcs. Equivalently, after fixing one arc, every other arc is reached by a unique orientation-preserving automorphism of the map.

For the  $n$ -dimensional binary hypercube  $Q_n$  we identify the vertex set with

$$V = \mathbb{F}_2^n,$$

and write  $e_0, e_1, \dots, e_{n-1}$  for the standard basis vectors of  $V$ . The arcs of  $Q_n$  are then

$$D = \{(v, v + e_j) : v \in V, j \in \mathbb{Z}/n\}.$$

In this notation the arc-reversing involution is always

$$(v, v + e_j)^L = (v + e_j, v).$$

The regular embeddings constructed by Nedela and Škoviera can be described as follows. Let  $e \in \mathbb{Z}/n$  satisfy

$$e^2 \equiv 1 \pmod{n}.$$

For  $v = (v_0, \dots, v_{n-1}) \in \mathbb{F}_2^n$ , put

$$|v| = v_0 + \dots + v_{n-1} \pmod{2}.$$

Then the corresponding map  $M(n, e) = (D, R_e, L)$  is given by

$$(v, v + e_j)^{R_e} = (v, v + e_{j+e|v|}), \quad (v, v + e_j)^L = (v + e_j, v),$$

where all subscripts are read modulo  $n$ . Thus the local rotation at an even vertex is  $j \mapsto j + 1$ , whereas the local rotation at an odd vertex is  $j \mapsto j + e$ . Du, Kwak, and Nedela proved that, when  $n$  is odd, every orientably regular embedding of  $Q_n$  is of this form.

For even  $n$ , the local rotation is no longer determined merely by the parity of the vertex. The full classification is most naturally expressed by introducing an admissible involution  $\sigma$  of the set of directions. Let

$$\rho = (0, 1, \dots, n-1)$$

be the cyclic permutation of the coordinate directions, and let  $\sigma \in S_n$  be an involution fixing 0. Following Kwon's construction, set

$$G(\sigma) = \langle \rho, \tilde{e}_0 \sigma \rangle \leq \text{Aut}(Q_n),$$

where  $\tilde{e}_0$  denotes translation by  $e_0$ . If  $G(\sigma)$  acts regularly on the arcs of  $Q_n$ , then  $\sigma$  is called *admissible*, and it determines an orientably regular embedding  $M(\sigma)$ .

To write this embedding in the  $(D, R, L)$ -language, associate to each vertex  $v \in V$  a permutation  $\theta_v \in S_n$  which records the cyclic order of the coordinate directions around  $v$ . We normalize by taking  $\theta_0 = \text{id}$ . At the vertex  $v$ , the local cyclic order of the outgoing arcs is then  $\theta_v(0), \theta_v(1), \dots, \theta_v(n-1)$ . Equivalently, the rotation  $R_\sigma$  is given by

$$(v, v + e_{\theta_v(i)})^{R_\sigma} = (v, v + e_{\theta_v(i+1)}),$$

or, for an arbitrary direction  $j$ ,

$$(v, v + e_j)^{R_\sigma} = (v, v + e_{\theta_v(\theta_v^{-1}(j)+1)}).$$

The arc-reversing involution remains

$$(v, v + e_j)^L = (v + e_j, v).$$

The permutations  $\theta_v$  are determined recursively from  $\sigma$ . Put

$$\gamma_i = \rho^{-i} \sigma \rho^i,$$

so that explicitly

$$\gamma_i(j) = \sigma(j - i) + i \pmod{n}.$$

If

$$v' = v + e_{\theta_v(i)},$$

then

$$\theta_{v'} = \rho^r \gamma_i \theta_v,$$

where  $r \in \mathbb{Z}/n$  is chosen uniquely so that

$$(\rho^r \gamma_i \theta_v)(0) = 0.$$

Equivalently,

$$r = (\gamma_i \theta_v)^{-1}(0).$$

Thus the local rotations at all vertices are computable from the single permutation  $\sigma$ .

The classification theorem of Catalano, Conder, Du, Kwon, Nedela, and Wilson can therefore be stated in the following  $(D, R, L)$ -form. Every orientably regular embedding of  $Q_n$  is isomorphic to a map

$$M(\sigma) = (D, R_\sigma, L),$$

where

$$D = \{(v, v + e_j) : v \in \mathbb{F}_2^n, j \in \mathbb{Z}/n\},$$

$$(v, v + e_j)^L = (v + e_j, v),$$

and

$$(v, v + e_j)^{R_\sigma} = (v, v + e_{\theta_v(\theta_v^{-1}(j)+1)}),$$

for some admissible involution  $\sigma$ . Moreover, distinct admissible involutions give non-isomorphic orientably regular embeddings.

More explicitly:

- When  $n$  is odd, the admissible involutions are precisely the multiplications

$$\sigma(j) = ej \pmod{n}, \quad e^2 \equiv 1 \pmod{n}.$$

In this case the above general formula reduces to the Nedela–Škovič formula

$$(v, v + e_j)^R = (v, v + e_{j+e|v|}).$$

- When  $n = 2m$  is even, the admissible involutions are precisely those involutions

$\sigma \in S_n$  fixing 0 such that

$$\sigma(i + m) = \sigma(i) + m$$

and such that the induced action on the pairs

$$B_i = \{i, i + m\}, \quad i \in \mathbb{Z}/m,$$

is multiplication by some unit  $e \in \mathbb{Z}/m$  satisfying

$$e^2 \equiv 1 \pmod{m}.$$

Thus, on the quotient set of opposite direction-pairs, the local structure is still governed by the same congruence  $e^2 \equiv 1$ , while the genuinely even-dimensional phenomena arise from possible switches inside the pairs  $B_i$ . These switches are precisely what prevent the general even-dimensional rotation from being expressible only in terms of the parity  $|v|$ , and necessitate the vertex-dependent permutations  $\theta_v$  in the formula for  $R_\sigma$ .

Equivalently, in the even-dimensional case the above description may be made more explicit by recording the possible switches inside the pairs  $B_i = \{i, i + m\}$ . Since  $\sigma$  sends each pair  $B_i$  to  $B_{ei}$ , for each  $i \in \mathbb{Z}/2m$  the value  $\sigma(i)$  is either  $ei$  or  $ei + m$  modulo  $2m$ . Hence there is a function

$$\chi_A : \mathbb{Z}/2m \rightarrow \{0, 1\}$$

such that

$$\sigma(i) = ei + m\chi_A(i) \pmod{2m}.$$

Here  $A = \{i : \chi_A(i) = 1\}$ , and  $\chi_A(i)$  records whether the image of  $i$  is shifted by  $m$  inside the target pair  $B_{ei} = \{ei, ei + m\}$ . Equivalently,  $\chi_A(i) = 0$  means  $\sigma(i) = ei$ , whereas  $\chi_A(i) = 1$  means  $\sigma(i) = ei + m$ .

With this notation, the admissible involutions are precisely of the two types appearing in the constructions of Kwon and of Catalano–Nedela.

- **Kwon type:**

$$e^2 \equiv 1 \pmod{2m},$$

and the switch function is invariant under both  $i \mapsto i + m$  and  $i \mapsto ei$ ; equivalently,

$$\chi_A(i + m) = \chi_A(i), \quad \chi_A(ei) = \chi_A(i).$$

- **Catalano–Nedela type:** this can occur only when  $16 \mid n$ . In this case

$$e^2 \equiv m + 1 \pmod{2m},$$

and the switch function satisfies the twisted condition

$$\chi_A(i + m) = \chi_A(i), \quad \chi_A(ei) \equiv \chi_A(i) + i \pmod{2}.$$

Thus the concise pair-preserving formulation above is equivalent to saying that, in even dimensions, every admissible involution belongs to one of these two families.

### Nedela–Škoviera construction visualization

We have achieved a comprehensive grasp of the Nedela–Škoviera construction, which establishes a framework for classifying regular embeddings of the hypercube graph  $Q_n$  by first analyzing the dipole graph  $D_n$ . The methodology begins with a canonical projection from  $Q_n$  to  $D_n$ , followed by a complete classification of all regular embeddings of the simpler graph  $D_n$ . These embeddings are then systematically lifted to classify the regular embeddings of  $Q_n$  itself.

The next step involves a two-stage explicit demonstration: first, presenting the embedding  $D(3, 1)$  of the dipole graph  $D_3$  in the torus  $T^2$ , and then showing how this embedding lifts to yield the corresponding embedding  $M(3, 1)$  of the hypercube graph  $Q_3$  on the torus  $T^2$ .

The regular embedding  $D(3, 1)$  of the cube graph  $D_3$  on the torus  $T^2$  is illustrated in Figure 3-1, the blue rhombus marks a fundamental domain.

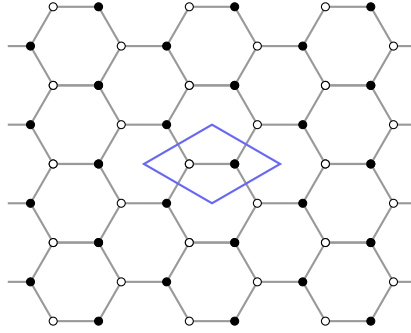


Figure 3-1 The regular embedding  $D(3, 1)$  of the cube graph  $D_3$  on the torus  $T^2$ .

The regular embedding  $M(3, 1)$  of the cube graph  $Q_3$  on the torus  $T^2$  is illustrated in Figure 3-2. The left subfigure shows the cube graph with vertices numbered 1–8; the coloring follows the bipartition given by the parity of the coordinate sum. The right subfigure is the universal cover of the torus, where the cube graph  $Q_3$  forms a hexagonal grid. The blue rhombus marks a fundamental domain.

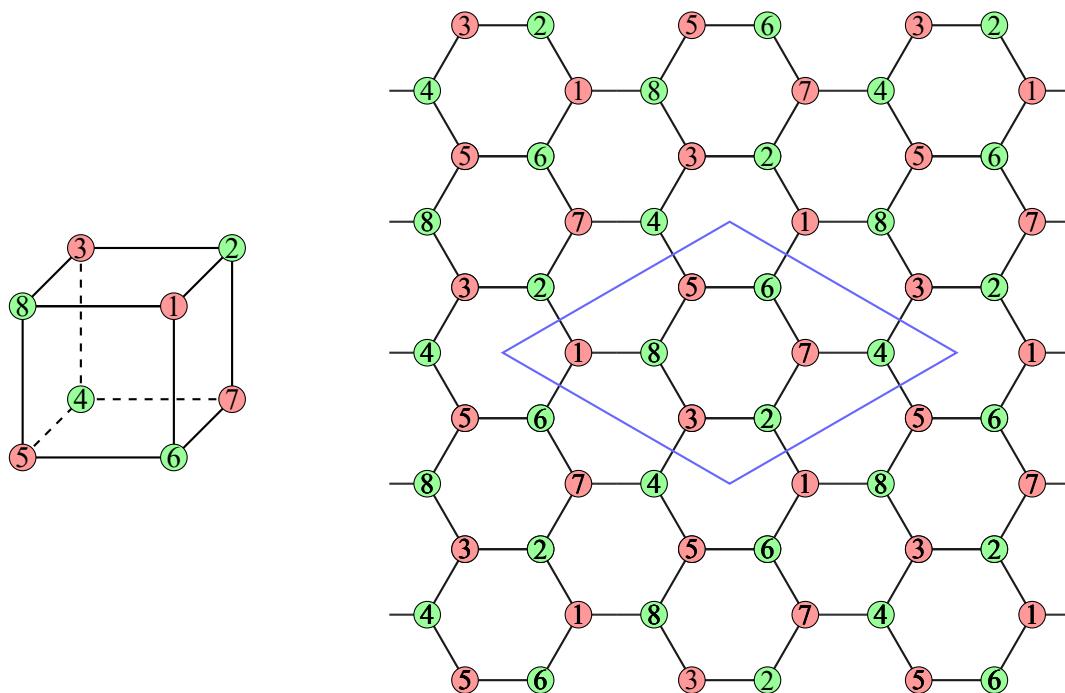


Figure 3-2 The regular embedding  $M(3, 1)$  of the cube graph  $Q_3$  on the torus  $T^2$ . Left: the cube graph with vertices different labels and colored according to the bipartition. Right: the universal cover of the torus, where the cube graph appears as a hexagonal grid; the blue quadrilateral marks a fundamental domain.

## Complete Graphs

Biggs<sup>[14]</sup> discovered all constructions of orientably regular embeddings of complete graphs in 1971, and James and Jones<sup>[41]</sup> later in 1985 proved that this list is complete, thereby completing the classification.

Biggs's construction may be described most naturally in terms of finite fields and Cayley maps. Let  $q = p^e$  be a prime power and let  $F = \mathbb{F}_q$  be a finite field. The vertices of  $K_q$  are identified with the elements of  $F$ , so that two distinct vertices  $v, w$  are joined by the difference  $w - v \in F^*$ . Choose a primitive element  $u \in F^*$ , i.e. a generator of the cyclic multiplicative group  $F^*$ . At each vertex  $v$ , impose the same cyclic ordering of its neighbours:

$$v + 1, v + u, v + u^2, \dots, v + u^{q-2}.$$

This rotation system determines an orientable embedding of  $K_q$ . Equivalently, it is the Cayley map on the additive group of  $F$  with generating set  $F^*$ , where the local rotation is induced by multiplication by  $u$ .

The resulting map is orientably regular. Its orientation-preserving automorphism

group is the affine group

$$\text{AGL}(1, F) = \{x \mapsto ax + b : a \in F^*, b \in F\},$$

which acts regularly on the directed edges of the embedded complete graph. Thus the high symmetry of the embedding is governed by the affine geometry of the finite field. Biggs proved that such embeddings exist precisely for complete graphs whose number of vertices is a prime power. James and Jones later proved the converse classification: every regular orientable embedding of a complete graph is isomorphic to one arising from this finite-field construction. More precisely, for  $q = p^e$ , the regular orientable embeddings of  $K_q$  are parametrized by primitive elements of  $\mathbb{F}_q$ , with two choices  $u$  and  $u'$  giving isomorphic oriented maps if and only if they are conjugate under the Galois group of  $\mathbb{F}_q$  over  $\mathbb{F}_p$ . Consequently, the number of isomorphism classes is  $\frac{\varphi(q-1)}{e}$ . In this sense, the classification shows that the entire family of orientably regular embeddings of complete graphs is controlled by the multiplicative structure of finite fields.

### 3.3 Homological Redundancy of Geodesics in Regular Maps

#### 3.3.1 General Constructions

Through our study in the crossroad of regular map theory in topological graph theory and hyperbolic geometric, we could found that the problem of counting the homological rank of systoles can be generalized to a broader context. The geometric objects we have studied specifically in the second section is the regular embedding maps of hypercube graphs with minimal genus represent a special case of this broader framework.

Combinatorialists have already classified all regular embeddings of hypercube graphs in an elegant list. More importantly, all these embeddings consist of even polygons as faces. This allows us to pose the same question for their dual maps and compute the homological rank of the cycles formed by their edges. If desired, we may also endow these maps with a standard hyperbolic structure so that all cycles become geodesics—though it should be noted that in this setting, they may not necessarily be the shortest geodesics.

Conversely, the regular embeddings of hypercube graphs themselves can be studied in a similar manner. We may investigate the homological relations among their edges as the dual graph. This case is more complex, as it requires distinguishing between odd and even cases. Only in the even case can all edges of the hypercube graph form closed cycles on the surface. In the odd case, we must also consider the principal diagonals of the faces.

Even in the even case, we may further study cycles formed by these principal diagonals.

Furthermore, we can enrich these topological questions with additional geometric structure. While regular maps inherently define a compact topological 2-manifold, we may endow each 2-cell with a standard hyperbolic structure modeled on a right-angled polygon. By assembling these pieces consistently, we obtain a globally hyperbolic structure on the underlying topological surface. This construction yields a distinguished point in the corresponding Teichmüller space. After analyzing the homological behavior of the edge-sets of the graph, we may then study the local geometry of hyperbolic structures around this point. Analogously, the same procedure can be carried out and analogous questions asked for the dual map of the embedding. These topics will be developed in the next chapter.

### 3.3.2 Calculation of Embeddings $M(n, -1)$ for Even $n$

**Proposition 3.1:** Table 3-1 presents the numerical computed results of the homological rank and kernel dimension of the regular embeddings of hypercube graphs  $Q_n$  for even values of  $n$ .

Table 3-1 Homological ranks of edge geodesics in the regular embedding  $M(n, -1)$  for  $n = 2k$  (even).

| $k$ | $n = 2k$ | face dim | dim $H_1$ | Geo num | Ker dim | rank in $H_1$ |
|-----|----------|----------|-----------|---------|---------|---------------|
| 2   | 4        | 15       | 2         | 8       | 6       | 2             |
| 3   | 6        | 95       | 34        | 48      | 24      | 24            |
| 4   | 8        | 511      | 258       | 256     | 96      | 160           |
| 5   | 10       | 2559     | 1538      | 1280    | 400     | 880           |
| 6   | 12       | 12287    | 8194      | 6144    | 1732    | 4412          |
| 7   | 14       | 57343    | 40962     | 28672   | 7644    | 21028         |
| 8   | 16       | 262143   | 196610    | 131072  | 33920   | 97152         |
| 9   | 18       | 1179647  | 917506    | 589824  | 150240  | 439584        |

Abbreviations: “face dim” denotes the dimension of two dimensional cellular chain.  
“Geo num” denotes the number of closed geodesics formed by edges; “Ker dim” denotes the dimension of the kernel of the Hurewicz map.

Let  $n = 2h$  be even, and set  $V = \mathbb{Z}/n$ . All indices are understood modulo  $n$ . Let  $\mathbb{F}_2$  be the field with two elements.

The set of edges is

$$E = \mathbb{Z}/n \times \mathbb{F}_2^{n-1},$$

and let  $C_1 = \mathbb{F}_2\langle E \rangle$ . The set of faces is

$$F = \mathbb{Z}/n \times \mathbb{F}_2^{n-2},$$

where the entries in the  $i$ -th and  $(i + 1)$ -st coordinates are  $*$ . Let  $C_2 = \mathbb{F}_2\langle F \rangle$ .

The differential  $d : C_2 \rightarrow C_1$  is defined by sending each two-dimensional square to the sum of its four boundary edges. Since the ground field is  $\mathbb{F}_2$ , there are no sign issues. Thus, if a face has  $*$  in the  $i$ -th and  $(i + 1)$ -st positions, then

$$d(x) = (\dots, *, 0, \dots) + (\dots, *, 1, \dots) + (\dots, 0, *, \dots) + (\dots, 1, *, \dots).$$

We also define a distinguished subset  $G \subset C_1$  consisting of all closed edge geodesics. For each  $i \in \mathbb{Z}/h$ , and for every choice of the remaining coordinates, consider the boundary of the quadrilateral obtained by varying the  $i$ -th and  $(i + h)$ -th coordinates:

$$(\dots, *, \dots, 0, \dots) + (\dots, *, \dots, 1, \dots) + (\dots, 0, \dots, *, \dots) + (\dots, 1, \dots, *, \dots).$$

Let  $\langle G \rangle$  be the  $\mathbb{F}_2$ -subspace spanned by all such elements.

**Theorem 3.1:** For each subset  $T \subseteq V$ , regard  $T$  as the set of vertices retained in the cycle graph  $C_n$ . Let  $\mathcal{R}(T)$  denote the set of maximal cyclic runs of  $T$ , equivalently its interval components in  $C_n$ .

Define a graph  $\Gamma_T$  as follows.

- (1) The vertices of  $\Gamma_T$  are the interval components  $R \in \mathcal{R}(T)$ .
- (2) For each antipodal matching edge  $\{i, i + h\}$  with  $i \in \mathbb{Z}/h$ , if  $i, i + h \in T$ , add an edge in  $\Gamma_T$  joining the two interval components containing  $i$  and  $i + h$ .
- (3) If  $i$  and  $i + h$  lie in the same interval component, this edge becomes a loop.

If  $T = \emptyset$ , set  $\beta_1(\Gamma_T) = 0$ . Then

$$\dim_{\mathbb{F}_2}(\text{Im}(d) \cap \langle G \rangle) = \sum_{T \subseteq V} \beta_1(\Gamma_T)$$

where

$$\beta_1(\Gamma_T) = |E(\Gamma_T)| - |V(\Gamma_T)| + c(\Gamma_T)$$

is the first Betti number of  $\Gamma_T$ , and  $c(\Gamma_T)$  denotes the number of connected components of  $\Gamma_T$ .

Equivalently, if

$$p(T) = \#\{\{i, i + h\} : i, i + h \in T\}, \quad r(T) = |\mathcal{R}(T)|, \quad c(T) = c(\Gamma_T),$$

then

$$\dim_{\mathbb{F}_2}(\text{Im}(d) \cap \langle G \rangle) = \sum_{T \subseteq V} (p(T) - r(T) + c(T)).$$

Moreover, if  $n = 2h \geq 4$ , then

$$\dim_{\mathbb{F}_2}(\text{Im}(d) \cap \langle G \rangle) = h 4^{h-2} + (1 + \sqrt{2})^h + (1 - \sqrt{2})^h - 2.$$

Equivalently,

$$\dim_{\mathbb{F}_2}(\text{Im}(d) \cap \langle G \rangle) = n 2^{n-5} + (1 + \sqrt{2})^{n/2} + (1 - \sqrt{2})^{n/2} - 2.$$

For the degenerate case  $n = 2$ , one has

$$\dim_{\mathbb{F}_2}(\text{Im}(d) \cap \langle G \rangle) = 1.$$

**Proof:** Let  $R = \mathbb{F}_2[x_0, \dots, x_{n-1}]/(x_i^2 = x_i)$ . A coefficient system on edges in direction  $m$  may be regarded as a Boolean function not involving the variable  $x_m$ . Hence

$$C_1 = \bigoplus_{m \in V} R_m e_m,$$

where

$$R_m = \mathbb{F}_2[x_j : j \neq m]/(x_j^2 = x_j).$$

For a subset  $S \subseteq V$ , write  $x_S = \prod_{i \in S} x_i$ . The monomial  $x_S$  can occur on an edge in direction  $m$  precisely when  $m \notin S$ . Put  $T = V \setminus S$ . Then, in the fixed monomial layer  $x_S$ , the allowable edge directions are exactly the elements of  $T$ . Thus this layer is naturally identified with

$$C_1(S) \cong \mathbb{F}_2^T = \text{span}_{\mathbb{F}_2}\{e_m : m \in T\}.$$

We now describe the restrictions of  $\text{Im}(d)$  and  $\langle G \rangle$  to this monomial layer.

Consider a face varying in the adjacent directions  $i$  and  $i+1$ . If the coefficient of this face contains the monomial  $x_S$ , then necessarily  $i, i+1 \notin S$ , or equivalently  $i, i+1 \in T$ . In the monomial layer  $x_S$ , the boundary of this face contributes the vector  $e_i + e_{i+1}$ . Therefore the contribution of  $\text{Im}(d)$  inside  $C_1(S) \cong \mathbb{F}_2^T$  is

$$A_T = \text{span}_{\mathbb{F}_2}\{e_i + e_{i+1} : i, i+1 \in T\}.$$

Similarly, the distinguished subspace  $G$  is generated by pairs of antipodal directions  $i$  and  $i+h$ . In the monomial layer  $x_S$ , such a generator is present precisely when  $i, i+h \in T$ . Hence the contribution of  $\langle G \rangle$  in this layer is

$$M_T = \text{span}_{\mathbb{F}_2}\{e_i + e_{i+h} : i \in \mathbb{Z}/h, i, i+h \in T\}.$$

Since both subspaces decompose as direct sums over the monomial layers, we have

$$\text{Im}(d) \cap \langle G \rangle = \bigoplus_{T \subseteq V} (A_T \cap M_T).$$

Consequently,

$$\dim_{\mathbb{F}_2}(\text{Im}(d) \cap \langle G \rangle) = \sum_{T \subseteq V} \dim_{\mathbb{F}_2}(A_T \cap M_T).$$

Fix a subset  $T \subseteq V$ . Consider the induced subgraph  $C_n[T]$  of the cycle graph  $C_n$ . Its connected components are precisely the maximal cyclic runs of  $T$ , namely the elements of  $\mathcal{R}(T)$ .

For each interval component  $R \in \mathcal{R}(T)$ , the adjacent-edge vectors  $e_i + e_{i+1}$  with both endpoints in  $R$  span exactly the subspace

$$\left\{ v \in \mathbb{F}_2^R : \sum_{i \in R} v_i = 0 \right\}.$$

Indeed, along an interval, the vectors  $e_i + e_{i+1}$  generate all vectors of even coordinate sum, and no vector of odd coordinate sum can be generated because every generator has coordinate sum zero. Hence

$$A_T = \left\{ v \in \mathbb{F}_2^T : \sum_{i \in R} v_i = 0 \text{ for every } R \in \mathcal{R}(T) \right\}.$$

On the other hand,  $M_T$  is generated by the antipodal edges  $\{i, i+h\}$  whose two endpoints both lie in  $T$ . Choosing a subset of these antipodal edges places a 1 at each of their endpoints. The resulting vector lies in  $A_T$  precisely when, for every interval component  $R \in \mathcal{R}(T)$ , an even number of selected antipodal-edge endpoints lies in  $R$ .

This is exactly the cycle condition in the graph  $\Gamma_T$ . The vertices of  $\Gamma_T$  are the interval components of  $T$ , and the edges of  $\Gamma_T$  are the antipodal edges fully contained in  $T$ . If the two endpoints of an antipodal edge lie in the same interval component, the corresponding edge is a loop. The condition that every interval component receives an even number of selected endpoints is precisely the condition that the selected edges form a cycle over  $\mathbb{F}_2$ , with loops contributing two endpoints at the same vertex.

Because the antipodal edges form a matching, the map from a choice of antipodal edges to the corresponding vector in  $M_T$  is injective. Therefore  $A_T \cap M_T$  is naturally identified with the  $\mathbb{F}_2$ -cycle space of  $\Gamma_T$ . It follows that

$$\dim_{\mathbb{F}_2}(A_T \cap M_T) = \beta_1(\Gamma_T).$$

Thus

$$\dim_{\mathbb{F}_2}(\text{Im}(d) \cap \langle G \rangle) = \sum_{T \subseteq V} \beta_1(\Gamma_T).$$

This proves the cycle-space formula.

We now derive the closed formula. Set

$$D_{2h} := \dim_{\mathbb{F}_2}(\text{Im}(d) \cap \langle G \rangle).$$

Let  $p(T) = |E(\Gamma_T)|$ ,  $r(T) = |V(\Gamma_T)|$ , and  $c(T) = c(\Gamma_T)$ . Since

$$\beta_1(\Gamma_T) = p(T) - r(T) + c(T),$$

we obtain

$$D_{2h} = \sum_T p(T) - \sum_T r(T) + \sum_T c(T).$$

First, compute  $\sum_T p(T)$ . There are exactly  $h$  antipodal matching edges  $\{i, i+h\}$  with  $i \in \mathbb{Z}/h$ . Fix one such edge. It is fully contained in  $T$  precisely when both of its endpoints belong to  $T$ , while the remaining  $n-2 = 2h-2$  vertices may be chosen arbitrarily. Therefore

$$\sum_T p(T) = h \cdot 2^{n-2} = h 4^{h-1}.$$

Next, compute  $\sum_T r(T)$ . The number  $r(T)$  is the number of interval components of  $T$  in the cycle graph  $C_n$ . If  $T \neq V$ , then  $r(T)$  is the number of transitions from 0 to 1 in the cyclic binary string of  $T$ :

$$r(T) = \#\{i : i \in T, i-1 \notin T\}.$$

For a fixed index  $i$ , the conditions  $i \in T$  and  $i-1 \notin T$  prescribe two vertices, and the remaining  $n-2$  vertices are arbitrary. Hence each  $i$  contributes  $2^{n-2}$ , and there are  $n = 2h$  possible indices. This gives  $n2^{n-2}$  for all subsets except that, when  $T = V$ , the transition count is 0, whereas the actual value is  $r(V) = 1$ . Thus

$$\sum_T r(T) = n2^{n-2} + 1 = 2h 4^{h-1} + 1.$$

It remains to compute  $\sum_T c(T)$ . Let  $H_h$  be the Möbius ladder graph with vertex set  $V = \mathbb{Z}/2h$ , whose edges consist of the cycle edges of  $C_{2h}$  together with the antipodal matching edges  $\{i, i+h\}$ ,  $i \in \mathbb{Z}/h$ . For a fixed  $T$ , the graph  $\Gamma_T$  is obtained from  $H_h[T]$  by contracting each interval component of  $C_n[T]$  to a point. Since contraction does not

change the number of connected components, one has

$$c(\Gamma_T) = c(H_h[T]).$$

Therefore it suffices to compute

$$C_h := \sum_{T \subseteq V} c(H_h[T]).$$

View  $H_h$  as consisting of  $h$  columns. The two vertices in the  $j$ -th column are

$$(j, 0) = j, \quad (j, 1) = j + h.$$

Each column has an internal antipodal edge. Hence, whenever a column is nonempty, its selected vertices form a connected subgraph within that column.

A column has four possible states: 0, +, −, 2, meaning, respectively, that neither vertex is chosen, only the upper vertex is chosen, only the lower vertex is chosen, or both vertices are chosen. The nonempty states are +, −, and 2.

For two adjacent nonempty columns, there are 9 ordered pairs of states. Among these pairs, the only ones that do not produce a horizontal connection are (+, −) and (−, +). Hence 7 ordered pairs produce a connection.

For a subset  $T$ , let  $a(T)$  be the number of nonempty columns, let  $b(T)$  be the number of adjacent seams at which a connection is actually present, and let  $\kappa(T) = 1$  precisely when all  $h$  columns are nonempty and all  $h$  seams produce connections; otherwise let  $\kappa(T) = 0$ . After each nonempty column is contracted to one vertex, the resulting graph is a subgraph of a cycle. Hence

$$c(H_h[T]) = a(T) - b(T) + \kappa(T).$$

Therefore

$$C_h = \sum_T a(T) - \sum_T b(T) + \sum_T \kappa(T).$$

First,

$$\sum_T a(T) = 3h 4^{h-1}.$$

Indeed, after a column is fixed to be nonempty, it has 3 possible states, while each of the remaining  $h - 1$  columns has 4 possible states.

Second,

$$\sum_T b(T) = 7h 4^{h-2}.$$

For a fixed seam, the two adjacent columns must form one of the 7 connectable ordered pairs of nonempty states, while the remaining  $h - 2$  columns are arbitrary.

It remains to compute

$$K_h := \sum_T \kappa(T).$$

This is the number of assignments of nonempty states to all columns such that every seam produces a connection.

Order the nonempty states as  $+$ ,  $-$ ,  $2$ . For ordinary adjacency, the compatibility matrix is

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

For the final seam, the Möbius twist interchanges the two layers, and the closing matrix is

$$B = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Let  $P$  be the matrix that interchanges  $+$  and  $-$ . Then  $B = AP$ . Hence

$$K_h = \text{tr}(A^{h-1}B) = \text{tr}(A^h P).$$

The matrix  $A$  has eigenvalue 1 in the antisymmetric direction  $(1, -1, 0)$ , while  $P$  has eigenvalue  $-1$  in that same direction. On the two-dimensional symmetric subspace,  $P$  acts as the identity, and the eigenvalues of  $A$  are  $1 + \sqrt{2}$  and  $1 - \sqrt{2}$ . Therefore

$$K_h = (1 + \sqrt{2})^h + (1 - \sqrt{2})^h - 1.$$

If

$$Q_h = (1 + \sqrt{2})^h + (1 - \sqrt{2})^h,$$

then  $K_h = Q_h - 1$ .

Combining the three contributions gives

$$\begin{aligned} C_h &= 3h 4^{h-1} - 7h 4^{h-2} + Q_h - 1 \\ &= 5h 4^{h-2} + Q_h - 1. \end{aligned}$$

Thus

$$\sum_T c(T) = 5h 4^{h-2} + Q_h - 1.$$

Substituting the three sums

$$\sum_T p(T) = h 4^{h-1}, \quad \sum_T r(T) = 2h 4^{h-1} + 1, \quad \sum_T c(T) = 5h 4^{h-2} + Q_h - 1$$

into the expression for  $D_{2h}$ , we obtain

$$\begin{aligned} D_{2h} &= h 4^{h-1} - (2h 4^{h-1} + 1) + (5h 4^{h-2} + Q_h - 1) \\ &= h 4^{h-2} + Q_h - 2. \end{aligned}$$

Therefore

$$D_{2h} = h 4^{h-2} + (1 + \sqrt{2})^h + (1 - \sqrt{2})^h - 2.$$

Returning to  $n = 2h$ , this becomes

$$\dim_{\mathbb{F}_2}(\text{Im}(d) \cap \langle G \rangle) = n 2^{n-5} + (1 + \sqrt{2})^{n/2} + (1 - \sqrt{2})^{n/2} - 2.$$

This proves the closed dimension formula. ■

**Remark 3.1:** The sequence

$$Q_h = (1 + \sqrt{2})^h + (1 - \sqrt{2})^h$$

is the Pell-Lucas sequence, characterized by

$$Q_0 = 2, \quad Q_1 = 2, \quad Q_h = 2Q_{h-1} + Q_{h-2}.$$

Thus, for  $n = 2h \geq 4$ , the formula may also be written as

$$\dim_{\mathbb{F}_2}(\text{Im}(d) \cap \langle G \rangle) = h 4^{h-2} + Q_h - 2.$$

The first values are

| $n$ | 2 | 4 | 6  | 8  | 10  | 12   | 14   | 16    | 18     | 20     | 22      |
|-----|---|---|----|----|-----|------|------|-------|--------|--------|---------|
| dim | 1 | 6 | 24 | 96 | 400 | 1732 | 7644 | 33920 | 150240 | 662084 | 2899820 |

### 3.3.3 General Calculation for Edge Numbers of Closed Geodesics

An embedding is algebraically characterized by a triple  $(D, R, L)$ , where  $D$  is the set of darts,  $R$  is a rotation action of order  $n$ , and  $L$  is an involution (of order 2) that reverses dart direction. The action of  $L$  is straightforward to understand. Therefore, once the rotation  $R$  is fully described by its explicit permutation action, we can derive an algorithm to construct the cellular chain complex. Within this framework, closed geodesics can be

identified as specific elements, and their behavior in homology can be computed. The entire construction is elucidated by analyzing the interplay of these two actions.

To operationalize this approach, we proceed in two stages:

(1) **Face Boundaries in the Dual Map:** First we need to find the edges of every face in the dual map. Each face  $f^*$  in the dual map corresponds to a vertex  $v$  in the original hypercube graph  $Q_n$ . The boundary of  $f^*$  exactly consists of dual edges  $e^*$  corresponding to the edges  $e$  incident to  $v$  in  $Q_n$ . These dual edges can be collected by the orbit of the rotation action  $R$ . By assigning appropriate signs to these edges, we obtain a generator that spans the boundary subgroup within the chain complex.

(2) **Geodesic Assembly from Edges:** Then we need know which edges compose a geodesic in the dual map. In the dual map, two edges  $e_1^*$  and  $e_2^*$  are consecutive in a geodesic if and only if their corresponding original edges  $e_1$  and  $e_2$  are opposite sides of a common face  $f$  in the original map. The number of sides of a face, denoted  $2m$ , is given by the formula  $\frac{4n}{\gcd(e+1, n)}$  for  $1 < e < n-1$ , with special cases  $4n$  for  $e = 1$  and  $4$  for  $e = -1$ . Let  $d_1$  and  $d_2$  be darts on  $e_1$  and  $e_2$ , respectively, oriented consistently of the second coordination value. The dart  $d_2$  is obtained from  $d_1$  by applying the composition  $L \circ R$  exactly  $m$  times (to traverse half the face perimeter), followed by a final  $L$  to correct the orientation. This defines a new operator  $A = L \circ (L \circ R)^m$ . The orbit of any dart under  $A$  traces a sequence of edges that form a geodesic in the dual map.

**Definition 3.1 (Dart set):** Fix an integer  $n \geq 2$ . Define the *Dart set* by

$$D := \mathbb{F}_2^n \times \mathbb{Z}/n\mathbb{Z}.$$

An element of  $D$  is written  $(v, i)$  with  $v \in \mathbb{F}_2^n$  and  $i \in \mathbb{Z}/n\mathbb{Z}$ .

**Definition 3.2 (Standard basis and Hamming weight parity):** Let  $\mathbf{e}_1, \dots, \mathbf{e}_n$  be the standard basis of  $\mathbb{F}_2^n$ . Indices are always understood modulo  $n$  and then reduced to  $\{1, 2, \dots, n\}$ . Write

$$\|v\| := \sum_{j=1}^n v_j \in \{0, 1, \dots, n\} \quad \text{and} \quad p(v) := \|v\| \bmod 2 \in \{0, 1\}.$$

**Definition 3.3 (The maps  $L$  and  $R$ ):** Fix  $e \in (\mathbb{Z}/n\mathbb{Z})^\times$  such that  $e^2 \equiv 1 \pmod{n}$ . Define maps  $L, R : D \rightarrow D$  by

$$L(v, i) := (v + \mathbf{e}_i, i), \quad R(v, i) := (v, i + e^{\|v\|}).$$

We use the convention  $LR := L \circ R$  (apply  $R$  first, then  $L$ ).

**Definition 3.4 (The exponent  $s(n, e)$  and the operator  $T$ ):** Define

$$s(n, e) = \begin{cases} 2n, & e \equiv 1 \pmod{n}, \\ 4, & e \equiv -1 \pmod{n}, \\ \frac{4n}{\gcd(e+1, n)}, & 1 < e < n-1. \end{cases}$$

Whenever  $s(n, e)$  is even (it is, in all cases above), define

$$T := L(LR)^{s(n, e)/2}.$$

The *period* of  $T$  means the order of  $T$  in the permutation group of  $D$ , i.e.  $\text{ord}(T) = \min\{k \geq 1 : T^k = \text{id}_D\}$ .

There are two basic lemmas.

**Lemma 3.1 (Parity flips under  $L$ ):** For all  $v \in \mathbb{F}_2^n$  and all indices  $i$ , one has

$$p(v + \mathbf{e}_i) = p(v) + 1 \pmod{2}.$$

Consequently, along iterates of  $LR$ , the parity  $p(v)$  alternates at every step.

**Proof:** Adding  $\mathbf{e}_i$  flips exactly one coordinate of  $v$ , hence changes  $\|v\|$  by  $\pm 1$ , so  $p(v)$  flips. Since each application of  $LR = L \circ R$  applies  $L$  exactly once, parity alternates. ■

**Lemma 3.2 (Two-step displacement in the second coordinate):** Assume  $e^2 \equiv 1 \pmod{n}$  and  $e \not\equiv 1 \pmod{n}$ . Let  $(v, i) \in D$ . Then in two successive applications of  $LR$ , the total increment in the  $\mathbb{Z}/n\mathbb{Z}$ -coordinate equals  $1 + e$ , regardless of the initial parity  $p(v)$ .

**Proof:** Because  $e^2 \equiv 1$ , we have  $e^{\|v\|} \in \{1, e\}$  depending only on  $p(v)$ . By the previous lemma, parity flips each step, hence the  $R$ -increments alternate between 1 and  $e$ , in one of the two orders  $(1, e)$  or  $(e, 1)$ . In either case, the two-step sum is  $1 + e$  in  $\mathbb{Z}/n\mathbb{Z}$ . ■

**The period of  $T$**

**Theorem 3.2 (Period for  $e \equiv 1$ ):** If  $e \equiv 1 \pmod{n}$ , then

$$T = L(LR)^n \text{ is an involution, i.e. } T^2 = \text{id}_D.$$

In particular,  $\text{ord}(T) = 2$ .

**Proof:** When  $e \equiv 1$ , we have  $R(v, i) = (v, i + 1)$  for all  $v$ . Then

$$(LR)(v, i) = L(v, i + 1) = (v + \mathbf{e}_{i+1}, i + 1),$$

hence

$$(LR)^n(v, i) = \left(v + \sum_{k=1}^n \mathbf{e}_{i+k}, i + n\right) = \left(v + \sum_{j=1}^n \mathbf{e}_j, i\right),$$

since  $i + n \equiv i \pmod{n}$  and  $\{\mathbf{e}_{i+1}, \dots, \mathbf{e}_{i+n}\}$  is the full standard basis. Writing  $\mathbf{s} := \sum_{j=1}^n \mathbf{e}_j$ , we get

$$T(v, i) = L((LR)^n(v, i)) = L(v + \mathbf{s}, i) = (v + \mathbf{s} + \mathbf{e}_i, i).$$

Applying  $T$  twice cancels the added vector over  $\mathbb{F}_2$ :

$$T^2(v, i) = (v + (\mathbf{s} + \mathbf{e}_i) + (\mathbf{s} + \mathbf{e}_i), i) = (v, i).$$

So  $T^2 = \text{id}_D$ . ■

**Theorem 3.3 (Period for  $e \equiv -1$ ):** If  $e \equiv -1 \pmod{n}$  (i.e.  $e = n - 1$ ), then  $s(n, e) = 4$  and

$$T = L(LR)^2$$

satisfies:

$$\text{ord}(T) = \begin{cases} 2, & n = 2, \\ 4, & n \geq 3. \end{cases}$$

**Proof:** Here  $R(v, i) = (v, i + (-1)^{\|v\|})$ , so the  $R$ -increment is  $+1$  if  $p(v) = 0$  and  $-1$  if  $p(v) = 1$ . Set  $\delta := (-1)^{\|v\|} \in \{+1, -1\}$ .

Compute:

$$(LR)(v, i) = L(v, i + \delta) = (v + \mathbf{e}_{i+\delta}, i + \delta).$$

Since applying  $L$  flips parity, the second step uses increment  $-\delta$ , and thus

$$(LR)^2(v, i) = (v + \mathbf{e}_{i+\delta} + \mathbf{e}_i, i).$$

Therefore

$$T(v, i) = L((LR)^2(v, i)) = (v + \mathbf{e}_{i+\delta}, i) = \begin{cases} (v + \mathbf{e}_{i+1}, i), & p(v) = 0, \\ (v + \mathbf{e}_{i-1}, i), & p(v) = 1. \end{cases}$$

Applying  $T$  twice flips parity between the two cases, so

$$T^2(v, i) = (v + \mathbf{e}_{i+1} + \mathbf{e}_{i-1}, i).$$

If  $n \geq 3$ , then  $i + 1 \not\equiv i - 1 \pmod{n}$ , hence  $\mathbf{e}_{i+1} \neq \mathbf{e}_{i-1}$  and  $T^2 \neq \text{id}$ . But  $(T^2)^2 = \text{id}$  because adding the same  $\mathbb{F}_2$ -vector twice cancels, so  $T^4 = \text{id}$  and  $\text{ord}(T) = 4$ . If  $n = 2$ ,

then  $i + 1 \equiv i - 1 \pmod{2}$ , hence  $\mathbf{e}_{i+1} = \mathbf{e}_{i-1}$  and  $T^2 = \text{id}$ , so  $\text{ord}(T) = 2$ .  $\blacksquare$

**Theorem 3.4 (Dichotomy for  $1 < e < n-1$ : period 2 or 4):** Assume  $1 < e < n-1$  and  $e^2 \equiv 1 \pmod{n}$ . Let

$$d := \gcd(e + 1, n), \quad g := e + 1, \quad m := \frac{n}{d}, \quad A := (LR)^{2m}.$$

Then  $A^2 = \text{id}_D$  and the operator  $T = L \circ A$  satisfies  $T^4 = \text{id}_D$ . Moreover,

$$\text{ord}(T) = 2 \iff \gcd(e + 1, n) = 2,$$

and otherwise  $\text{ord}(T) = 4$ .

**Proof: Step 1:  $A$  fixes the second coordinate.** By the two-step displacement lemma, every two steps of  $LR$  add  $1 + e = g$  to the  $\mathbb{Z}/n\mathbb{Z}$ -coordinate. Thus  $2m$  steps add  $mg$ . Since  $d = \gcd(g, n)$ , we have  $mg = (n/d)g \equiv 0 \pmod{n}$ , hence  $A$  fixes the second coordinate.

**Step 2:  $A$  is an involution.** By definition of  $s(n, e)$  for  $1 < e < n - 1$ ,

$$\frac{s(n, e)}{2} = 2m \implies A = (LR)^{s(n, e)/2}.$$

The defining property of  $s(n, e)$  is  $(LR)^{s(n, e)} = \text{id}_D$ , hence  $A^2 = (LR)^{s(n, e)} = \text{id}_D$ .

**Step 3: Explicit form of  $A$  and of  $T^2$ .** Fix  $(v, i)$ . Because each step of  $LR$  applies one  $L$ , parity flips at each step. If  $p(v) = 0$ , then the  $R$ -increments follow  $1, e, 1, e, \dots$ ; if  $p(v) = 1$ , they follow  $e, 1, e, 1, \dots$ . A direct index-tracking computation over  $2m$  steps yields:

$$A(v, i) = (v + W_{p(v)}(i), i),$$

where

$$W_0(i) = \sum_{j=0}^{m-1} \mathbf{e}_{i+1+jg} + \sum_{j=1}^m \mathbf{e}_{i+jg}, \quad W_1(i) = \sum_{j=0}^{m-1} \mathbf{e}_{i+e+jg} + \sum_{j=1}^m \mathbf{e}_{i+jg}.$$

Consequently,

$$T(v, i) = L(A(v, i)) = (v + W_{p(v)}(i) + \mathbf{e}_i, i).$$

Since  $L$  flips parity and  $A$  contains an even number  $2m$  of  $L$ -applications,  $A$  preserves parity. Hence  $T$  flips parity, so the second application uses the opposite branch  $W_{1-p(v)}(i)$ . Thus

$$T^2(v, i) = (v + V(i), i), \quad V(i) := W_0(i) + W_1(i) = \sum_{j=0}^{m-1} (\mathbf{e}_{i+1+jg} + \mathbf{e}_{i+e+jg}).$$

In particular,  $T^4 = \text{id}_D$  because  $T^2$  is translation by a fixed  $\mathbb{F}_2$ -vector and  $(v + V(i)) +$

$V(i) = v$ .

**Step 4: Criterion for period 2.** We have  $\text{ord}(T) = 2$  if and only if  $T^2 = \text{id}_D$ , i.e.  $V(i) = 0$  for all  $i$ . Since we work over  $\mathbb{F}_2$ ,  $V(i) = 0$  is equivalent to the two index sets coinciding:

$$\{i + 1 + jg : 0 \leq j \leq m - 1\} = \{i + e + jg : 0 \leq j \leq m - 1\} \subset \mathbb{Z}/n\mathbb{Z}.$$

Because  $\gcd(g, n) = d$ , the map  $j \mapsto jg$  runs through exactly the subgroup  $d\mathbb{Z}/n\mathbb{Z}$ , so each such set is precisely one congruence class modulo  $d$ :

$$\{a + jg\} = \{x \in \mathbb{Z}/n\mathbb{Z} : x \equiv a \pmod{d}\}.$$

Therefore the two sets coincide if and only if

$$i + 1 \equiv i + e \pmod{d} \iff e \equiv 1 \pmod{d} \iff d \mid (e - 1).$$

But by definition  $d \mid (e + 1)$  as well, hence  $d \mid 2$ . Since  $1 < e < n - 1$ , we have  $e \not\equiv \pm 1 \pmod{n}$ , so  $d \neq 1$ ; thus the only possibility is  $d = 2$ . This proves

$$\text{ord}(T) = 2 \iff \gcd(e + 1, n) = 2.$$

If  $\gcd(e + 1, n) \neq 2$ , then  $V(i) \neq 0$  and so  $T^2 \neq \text{id}$  while  $T^4 = \text{id}$ , hence  $\text{ord}(T) = 4$ . ■

**Corollary 3.1 (Complete classification of the 2-periodic cases):** Assume  $e^2 \equiv 1 \pmod{n}$ . Then  $T = L(LR)^{s(n,e)/2}$  is 2-periodic if and only if one of the following holds:

- (1)  $e \equiv 1 \pmod{n}$  (all  $n \geq 2$ );
- (2)  $e \equiv -1 \pmod{n}$  and  $n = 2$ ;
- (3)  $1 < e < n - 1$  and  $\gcd(e + 1, n) = 2$ .

In all remaining cases,  $T$  has period 4.

**Proof:** Combine the three theorems above. ■

## CHAPTER 4 GEOMETRIC STRUCTURES OF HYPERBOLIC SURFACES

### 4.1 Hyperbolic Structures on Surfaces

The Poincaré model of the hyperbolic plane is the upper half-plane of complex numbers, denoted by  $\mathbb{H} = \{z \in \mathbb{C} \mid \text{Im}(z) > 0\}$ , equipped with the hyperbolic metric:

$$ds^2 = \frac{1}{y^2}(dx^2 + dy^2)$$

The distance formula in this model is given by:

$$\cosh(\text{dist}(z, w)) = 1 + \frac{2|z - w|^2}{\text{Im}(z) \cdot \text{Im}(w)}$$

where  $\cosh$  denotes the hyperbolic cosine function, and  $\text{Im}(-)$  stands for the imaginary part of a complex number.

The projective special linear group is defined as:

$$\text{PSL}(2, \mathbb{R}) = \text{SL}(2, \mathbb{R}) / \{\pm I\}$$

where  $\text{SL}(2, \mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{R}, ad - bc = 1 \right\}$  is the real special linear group, and  $I$  denotes the  $2 \times 2$  identity matrix.

This group acts biholomorphically on  $\mathbb{H}$  via fractional linear transformations (Möbius transformations) of the form:

$$z \mapsto \frac{az + b}{cz + d}$$

where  $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, \mathbb{R})$ . Moreover, this action preserves the above hyperbolic metric. Furthermore,  $\text{PSL}(2, \mathbb{R})$  is exactly the full group of orientation-preserving isometries of  $\mathbb{H}$ .

Another fundamental model of the hyperbolic plane is the unit disk model. The mapping:

$$z \mapsto \frac{z - i}{z + i}, \quad z \in \mathbb{H}$$

biholomorphically maps  $\mathbb{H}$  onto the open unit disk  $D = \{w = u + iv \in \mathbb{C} \mid u^2 + v^2 < 1\}$ .

The hyperbolic metric induced on  $D$  by this mapping is:

$$ds^2 = \frac{4(du^2 + dv^2)}{(1 - (u^2 + v^2))^2}$$

In both the Poincaré upper half-plane model and the unit disk model, the geodesics are generalized circles that intersect the boundary of the model orthogonally. For convenience, we will also use  $\mathbb{H}$  to denote the abstract hyperbolic plane when there is no need to specify which concrete model is being referenced.

For background on hyperbolic geometry, one may consult the standard references: Buser<sup>[18]</sup>, Fenchel<sup>[27]</sup>, Iwayoshi et al.<sup>[38]</sup>, Iversen<sup>[39]</sup>, Ratcliffe<sup>[66]</sup>, Thurston<sup>[74]</sup>. For a focused discussion on geodesics in hyperbolic space, see Klingenberg<sup>[50]</sup>, Mirzakhani<sup>[55]</sup>.

## Teichmüller Space

The moduli problem for complex structures originates from a fundamental and profound fact: the same oriented two-dimensional differentiable manifold can often support many complex structures that are not isomorphic to each other. In other words, two closed Riemann surfaces, even if they are diffeomorphic as oriented smooth surfaces, are not necessarily biholomorphically equivalent. Therefore, a natural question arises: on a closed oriented surface of a given topological type, how many essentially different complex structures exist? This question is commonly referred to as the *Riemann moduli problem*, and it marks the historical starting point of moduli space theory and Teichmüller theory.

Let  $\Sigma_g$  be a closed oriented surface of genus  $g$ , and denote by  $\mathcal{M}_g$  the moduli space of all complex structures on  $\Sigma_g$  considered up to biholomorphic equivalence. For low genus, the structure of this space is relatively clear: when  $g = 0$ , the uniqueness of the Riemann sphere implies that  $\mathcal{M}_0$  consists of a single point; when  $g = 1$ , classical elliptic function theory and the theory of elliptic curves show that the moduli space of complex tori can be described by one complex parameter. The case  $g \geq 2$  is where the true richness of geometric and analytic content lies. In 1857, Riemann pointed out that for a closed Riemann surface of genus  $g$ , the complex structures should depend on  $3g - 3$  complex parameters. This count was obtained by viewing the surface as a branched cover of the Riemann sphere and analyzing the degrees of freedom associated with the branch points. Although this argument lacked modern rigor at the time, it accurately revealed the dimension count for higher-genus moduli spaces and laid the foundation for subsequent theoretical developments.

From a modern perspective, an effective way to study  $\mathcal{M}_g$  is to first introduce complex structures with a marking, leading to the **Teichmüller space**. Fix a surface  $\Sigma_g$  and consider all pairs  $(X, f)$ , where  $X$  is a closed Riemann surface and  $f : \Sigma_g \rightarrow X$  is an orientation-preserving homeomorphism, called a marking. Two pairs  $(X, f)$  and  $(Y, g)$  are considered equivalent if there exists a biholomorphic map  $h : X \rightarrow Y$  such that  $h \circ f$  is homotopic to  $g$ . The space of all such equivalence classes is denoted

$$\mathcal{T}_g = \mathcal{T}(\Sigma_g),$$

and is called the Teichmüller space of genus  $g$ . The group consisting of homotopy classes of markings, namely

$$\text{Mod}(\Sigma_g) = \pi_0(\text{Homeo}^+(\Sigma_g)),$$

is called the mapping class group. It acts naturally on  $\mathcal{T}_g$ , and the moduli space can be represented as its quotient:

$$\mathcal{M}_g \cong \mathcal{T}(\Sigma_g)/\text{Mod}(\Sigma_g).$$

This formulation shows that the moduli space is essentially a quotient space obtained by "forgetting the marking"; consequently, Teichmüller space plays the role of a fundamental object in modern moduli space theory.

Although the explicit notion of Teichmüller space appeared later, its rudiments were already implicit in the work of Klein and Poincaré on Fuchsian groups and automorphic functions. In particular, the Uniformization Theorem states that any closed Riemann surface of genus  $g \geq 2$  can be represented as the quotient of the upper half-plane  $\mathbb{H}$  by some torsion-free discrete group  $\Gamma \subset \text{PSL}(2, \mathbb{R})$ ,

$$X \cong \mathbb{H}/\Gamma.$$

Thus, the complex structure on a surface translates into a problem of discrete representations of Fuchsian groups, and the conformal geometry of the surface becomes closely linked to the hyperbolic geometry of  $\mathbb{H}$ . Along these lines, Fricke, Fenchel, and Nielsen systematically constructed Teichmüller space and established the so-called Fricke coordinates, using the parameters of Fuchsian group generators as coordinates. Within this framework,  $\mathcal{T}_g$  is described as a space of real dimension  $6g - 6$ . Early theory already recognized clearly that Teichmüller space should serve as an "unwrapped" version of moduli space, possessing a geometry more regular than that of  $\mathcal{M}_g$  itself.

Teichmüller's fundamental contribution lay in realizing that if one considers only

conformal maps, the moduli problem often becomes too rigid; however, by introducing *quasiconformal mappings*, the problem acquires a deformable analytic framework. Quasiconformal mappings can be characterized by the Beltrami equation

$$\partial_{\bar{z}} w = \mu \partial_z w,$$

where  $\mu$  is a Beltrami differential satisfying  $|\mu|_\infty < 1$ , describing how the complex structure deforms relative to the original complex coordinates. One of Teichmüller's key discoveries was the profound correspondence between extremal quasiconformal mappings and holomorphic quadratic differentials. This insight not only provided an intrinsic description of coordinates, geodesics, and distances on Teichmüller space but also endowed the deformation theory of complex structures with a precise analytic formulation for the first time. Based on this idea, Teichmüller introduced the distance that now bears his name and asserted that  $\mathcal{T}_g$  is homeomorphic to  $\mathbb{R}^{6g-6}$ . Although these conclusions lacked complete rigor in their original formulation, their core ideas had a decisive impact on subsequent developments.

By the late 1950s, Ahlfors and Bers laid a rigorous analytic foundation for Teichmüller theory. They systematically proved Teichmüller's main results on extremal quasiconformal mappings and moduli parameters, and further demonstrated that for  $g \geq 2$ ,  $\mathcal{T}_g$  is not only a manifold of real dimension  $6g-6$  but also carries a natural complex manifold structure, with complex dimension exactly  $3g-3$ . More concretely, fix a closed Riemann surface  $X$  of genus  $g$ , and let the vector space of holomorphic quadratic differentials on  $X$  be denoted by

$$Q(X) = H^0(X, K_X^{\otimes 2}).$$

By the Riemann–Roch theorem, one obtains

$$\dim_{\mathbb{C}} Q(X) = 3g - 3.$$

Ahlfors and Bers proved that  $\mathcal{T}_g$  can be realized via the Bers embedding as a bounded domain in some complex vector space  $Q(X)$ , thereby endowing Teichmüller space with a well-defined complex analytic structure. This result not only rigorously confirmed Riemann's dimension count but also unified complex analysis, hyperbolic geometry, and topological group actions within a single framework.

Concurrently, the mapping class group  $\text{Mod}(\Sigma_g)$  acts on  $\mathcal{T}_g$  by biholomorphic auto-

morphisms, and the action is properly discontinuous. Consequently, the quotient space

$$\mathcal{M}_g = \mathcal{T}_g / \text{Mod}(\Sigma_g)$$

naturally inherits the structure of a complex analytic space. In other words, Teichmüller space provides a local analytic model for moduli space, while the mapping class group records the discrete symmetries arising from different markings. In this sense, Teichmüller theory forms the foundation of modern moduli space theory: on one hand, it transforms the classification problem for closed Riemann surfaces into a problem of deforming complex structures and analyzing group actions; on the other hand, it provides essential tools for studying the geometric, topological, and dynamical properties of moduli space.

In summary, the development of Teichmüller space roughly followed this historical trajectory: Riemann first proposed that higher-genus moduli spaces should admit  $3g - 3$  complex parameters; the uniformization theory of Klein, Poincaré, and Koebe connected surface complex structures with Fuchsian groups; building on this, Fricke, Fenchel, and Nielsen constructed a parameter space in the real-analytic sense; subsequently, Teichmüller revealed the intrinsic geometry of this space through quasiconformal mappings and extremal problems; finally, Ahlfors and Bers established a rigorous complex analytic theory, elevating  $\mathcal{T}_g$  to a fundamental object in modern complex geometry, low-dimensional topology, and geometric group theory. It was through this historical process that Teichmüller space evolved from an auxiliary parameter space for the moduli problem into an independent and profound field of study.

Throughout these developments, deformations of Riemann surfaces have been studied from at least three interrelated perspectives:

- (1) deformations of Fuchsian groups,
- (2) deformations of conformal structures induced by hyperbolic metrics, and
- (3) deformations of complex structures induced by quasiconformal mappings.

The essence of the theory lies in exploring Teichmüller spaces from these viewpoints and clarifying the relationships among them.

## 4.2 Systole Function and Thurston Spine

The definition of topological Morse function is less restrictive than that of classical or smooth Morse function, because it only requires a local homeomorphism to a standard quadratic form, rather than a smooth diffeomorphism. This allows functions with cer-

tain degenerate or nonsmooth critical points to be included, if they are provided locally equivalent to a nondegenerate quadratic form in a topological sense.

**Definition 4.1:** Let  $M$  be an  $n$ -dimensional topological manifold. A continuous function  $f : M \rightarrow \mathbb{R}$  is called a **topological Morse function** if every point  $p \in M$  is either a regular point or a critical point with respect to  $f$ , characterized by the following local coordinate descriptions:

A point  $p$  is **regular** if there exists an open neighborhood  $U$  containing  $p$  that admits a homeomorphic parametrization  $\phi : U \rightarrow \mathbb{R}^n$ . The  $n$  real numbers  $(x_1, \dots, x_n) \in \mathbb{R}^n$  are the parameters that uniquely label points in  $U$ . After possibly reordering the coordinates, we have:

$$f \circ \phi^{-1}(x) - f(p) = x_n^2.$$

A point  $p$  is **critical** if there exists an **index**  $j \in \mathbb{Z}$ , with  $0 \leq j \leq n$ , and a homeomorphic parametrization  $\phi : U \rightarrow \mathbb{R}^n$  of an open neighborhood  $U$  of  $p$ , such that under the image coordinates  $(x_1, \dots, x_n) \in \mathbb{R}^n$ , the function  $f$  can be locally expressed as a non-degenerate quadratic form:

$$f \circ \phi^{-1}(x) - f(p) = \sum_{i=1}^{n-j} x_i^2 - \sum_{i=n-j+1}^n x_i^2$$

Here, the index  $j$  is defined as the “number of negative square terms” in this quadratic form, also called the “index of the critical point”.

**Example 4.1:** Consider the real line  $M = \mathbb{R}$  with various functions:

The function  $f(x) = x^3$  has vanishing derivative hence is a degenerate critical point at  $x = 0$  in the classical definition. However, the homeomorphism  $\phi^{-1}(x) = x^{1/3}$  satisfies

$$f(\phi^{-1}(x)) = (x^{1/3})^3 = x,$$

showing that  $f$  is locally just the linear coordinate  $x$ . Hence  $x = 0$  is a topologically regular point, illustrating the topological notion is strictly weaker than the smooth one.

The function  $f(x) = x^4$  is not a classical Morse function because its second derivative vanishes at  $x = 0$ . Nevertheless, an appropriate homeomorphism like

$$\phi^{-1}(z) = \text{sgn}(z) \cdot \sqrt{|z|}$$

transforms  $f$  locally into  $z^2$ , i.e.  $f(\phi^{-1}(z)) = z^2$ . Thus  $f$  is a topological Morse function of index 0 at the origin.

The absolute value  $f(x) = |x|$  is not smooth at 0, so it cannot be a classical Morse function. Yet the homeomorphism

$$\phi^{-1}(z) = \operatorname{sgn}(z) \cdot z^2$$

yields

$$f(\phi^{-1}(z)) = |\operatorname{sgn}(z) \cdot z^2| = z^2,$$

proving that  $|x|$  is also a topological Morse function of index 0 at the origin.

A broader class of examples: The pointwise minimum of finitely many smooth functions,

$$f(x) := \min_{i=1}^N f_i(x),$$

though typically non-differentiable, always furnishes topological Morse functions. This shows that the topological approach easily includes more kinds of functions that commonly appear in variational problems and optimization.

**Definition 4.2:** The *systole function*  $\operatorname{sys}(X)$  is defined as the length of the shortest essential simple closed geodesic on a hyperbolic surface  $X$ . It can be expressed as the minimum of a family of smooth geodesic-length functions  $\ell_\gamma(X)$ :

$$\operatorname{sys}(X) = \min_{\gamma} \ell_\gamma(X).$$

Because it is defined as a minimum, the function fails to be smooth at points where several distinct geodesics simultaneously realize the shortest length. At such points it develops “corners” (nonsmooth behavior), although it remains continuous.

Akrout<sup>[1]</sup> proved that the systole function, despite being nonsmooth as a minimum of geodesic-length functions, is a *topological Morse function* on both Teichmüller space and moduli space. Its critical points are characterized by a *eutacticity condition*: the origin lies in the (relative) interior of the convex hull of the Weil–Petersson gradients of the geodesic-length functions corresponding to systolic curves. Moreover, the Morse index at such a point equals the rank of these gradients.

Let  $S(X)$  denote the set of all curves realizing the systole at  $X$ . Consider their Weil–Petersson gradients  $\nabla \ell_\gamma(X)$ . The *eutactic condition* states that the origin 0 lies in the interior of the convex hull of these gradients (within the subspace they span). Equivalently, there exist positive weights such that the weighted sum of these gradients vanishes. Geometrically, this means that no infinitesimal deformation can simultaneously increase the lengths of all systolic curves; the systole is therefore “locked” at that point. This condition

is necessary and sufficient for  $X$  to be a critical point of the systole function.

In topological Morse theory, the *index* of a critical point is the dimension of the space of descending directions. For the systole function, one has the important formula

$$\text{index}(X) = \dim \text{span}\{\nabla \ell_\gamma(X) : \gamma \in S(X)\}.$$

Thus, the index equals the dimension of the linear span of the gradients of the systolic curves, reflecting the number of independent constraints imposed at that point.

A collection of simple closed curves is said to be *filling* if their complement on the surface consists only of disks (or topological polygons). It is known that if  $X$  is a critical point of the systole function, then  $S(X)$  must be filling. However, the converse does not hold: the filling condition alone does not imply the eutactic condition. Therefore, the set  $X_g$  of surfaces whose systoles are filling strictly contains the set of critical points. This distinction plays an important role in understanding the geometric structure of moduli space.

The *Thurston spine* is the subset of Teichmüller space consisting of points whose systoles decompose the surface into polygons. It forms a geometric “skeleton” of Teichmüller space. This set is larger than the set of critical points, as it includes all surfaces satisfying the combinatorial cutting condition. The Thurston spine provides a powerful framework for understanding the topology, geometry, and dynamics of Teichmüller space, and further illustrates the rich structure of the systole function as a geometric Morse function.

Theorem 37 in Schaller<sup>[72]</sup> provides a symmetry-based criterion for critical points of the systole function on Teichmüller space. It states that if a closed hyperbolic surface  $M$  has the property that its full isometry group acts transitively on the set of systolic curves  $S(M)$ , then  $M$  is a critical point of the systole function. The underlying reason is that transitivity forces all systolic geodesics to be geometrically equivalent, so their Weil-Petersson gradients form a single orbit under the isometry group. By symmetry, the sum (and hence a positive weighted combination) of these gradients must vanish, which implies that the origin lies in the interior of their convex hull; in other words, the eutactic condition is automatically satisfied. Thus, high symmetry guarantees the balance condition required for criticality, showing that many highly symmetric hyperbolic surfaces, such as those arising from triangle groups, are natural examples of critical points of the systole function.

### 4.3 Special Hyperbolic Structures from Regular Maps

#### Structure equations of polygons

The hyperbolic upper half-plane model has been defined as  $\mathbb{H}^2 = \{z \in \mathbb{C} \mid \text{Im}(z) > 0\}$ , equipped with the hyperbolic metric  $ds^2 = \frac{dx^2 + dy^2}{y^2}$  for  $z = x + iy$ . The group of orientation-preserving hyperbolic isometries of  $\mathbb{H}^2$  is isomorphic to the projective special linear group  $\text{PSL}(2, \mathbb{R}) = \text{SL}(2, \mathbb{R})/\{\pm I\}$ , where  $\text{SL}(2, \mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{R}, ad - bc = 1 \right\}$ . Elements of  $\text{PSL}(2, \mathbb{R})$  act on  $\mathbb{H}^2$  in the form of fractional linear transformations:

For  $[A] \in \text{PSL}(2, \mathbb{R})$  with  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, \mathbb{R})$ ,

$$z \cdot [A] = \frac{az + b}{cz + d}, \quad \forall z \in \mathbb{H}^2.$$

The symbol  $[A]$  denotes the equivalence class of  $A$  modulo  $\pm I$ , since  $A$  and  $-A$  induce the same fractional linear transformation. Where no confusion is likely to arise, we may omit the brackets indicating the equivalence class.

**Theorem 4.1 (Rotation about  $i$  by Angle  $\theta$ ):** The orientation-preserving hyperbolic rotation of  $\mathbb{H}^2$  about the point imaginary unit  $i$  by angle  $\theta \in \mathbb{R}$  is induced by the element

$$\left[ \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \\ -\sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \right] \in \text{PSL}(2, \mathbb{R}).$$

Here  $\theta > 0$  corresponds to counterclockwise rotation, and  $\theta < 0$  corresponds to clockwise rotation.

**Proof:** We just need to verify three properties of this transformation: 1. the matrix is in  $\text{SL}(2, \mathbb{R})$ ; 2. the transformation fixes point  $i$  (i.e. making it to be the center of rotation); 3. the transformation acts as a rotation by  $\theta$  at the neighborhood of point  $i$  in the sense of conformal map.

Compute the determinant of  $R_\theta = \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \\ -\sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$ , we can easily get:

$$\det(R_\theta) = \cos \frac{\theta}{2} \cdot \cos \frac{\theta}{2} - \sin \frac{\theta}{2} \cdot (-\sin \frac{\theta}{2}) = \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} = 1.$$

Thus  $R_\theta \in \text{SL}(2, \mathbb{R})$ , so  $[R_\theta] \in \text{PSL}(2, \mathbb{R})$ .

Calculate the fractional linear transformation induced by  $R_\theta$  at  $z = i$ , and multiply the numerator and denominator by the complex conjugate of the denominator ( $\cos \theta +$

$i \sin \theta$ ) to simplify:

$$\begin{aligned}
 R_\theta \cdot i &= \frac{\cos \frac{\theta}{2} \cdot i + \sin \frac{\theta}{2}}{-\sin \frac{\theta}{2} \cdot i + \cos \frac{\theta}{2}} \\
 &= \frac{(\cos \frac{\theta}{2} \cdot i + \sin \frac{\theta}{2})(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})}{(\cos \frac{\theta}{2})^2 + (\sin \frac{\theta}{2})^2} \\
 &= i \left( \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} \right) + \left( -\cos \frac{\theta}{2} \sin \frac{\theta}{2} + \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right) \\
 &= i.
 \end{aligned}$$

Hence  $i$  is a fixed point of the transformation, as required for a rotation about  $i$ .

For a holomorphic map  $f : \mathbb{C} \rightarrow \mathbb{C}$ , the angle of rotation at a fixed point  $z_0$  is the argument of the derivative  $f'(z_0)$ . Therefore, we compute the derivative of the fractional linear transformation  $f(z) = R_\theta \cdot z$ :

$$f'(z) = \frac{\det(R_\theta)}{(cz + d)^2} = \frac{1}{(-\sin \frac{\theta}{2} \cdot z + \cos \frac{\theta}{2})^2}$$

Evaluate at  $z = i$ :

$$f'(i) = \frac{1}{(-i \sin \frac{\theta}{2} + \cos \frac{\theta}{2})^2} = \frac{1}{(\cos \frac{\theta}{2} - i \sin \frac{\theta}{2})^2} = \left( \cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)^2 = e^{i\theta}.$$

The hyperbolic metric is conformally equivalent to the Euclidean metric on  $\mathbb{C}$ , so the hyperbolic rotation angle at  $i$  is just the argument of  $f'(i)$ . Thus the hyperbolic rotation angle is  $\theta$ , as desired.

Since  $R_\theta$  and  $-R_\theta$  induce the same transformation (i.e.,  $[R_\theta] = [-R_\theta]$ ), the rotation is uniquely represented by  $[R_\theta]$  in  $\text{PSL}(2, \mathbb{R})$ . ■

**Theorem 4.2 (Hyperbolic Translation Upward by Length  $l$ ):** The orientation-preserving hyperbolic translation of  $\mathbb{H}^2$  along the imaginary axis with length  $l \in \mathbb{R}$  is induced by the element

$$\left[ \begin{pmatrix} e^{l/2} & 0 \\ 0 & e^{-l/2} \end{pmatrix} \right] \in \text{PSL}(2, \mathbb{R}).$$

Here  $l > 0$  corresponds to upward translation (increasing imaginary part), and  $l < 0$  corresponds to downward translation (decreasing imaginary part). The magnitude  $|l|$  is the hyperbolic displacement along the imaginary axis.

**Proof:** First, recall the hyperbolic arc length formula of a curve  $\gamma(t) = x(t) + iy(t) \subset$

$\mathbb{H}^2$  ( $t \in [a, b] \subset \mathbb{R}$ ) is integration under the hyperbolic metric:

$$L(\gamma) = \int_a^b \sqrt{\frac{dx(t)^2 + dy(t)^2}{y(t)^2}} = \int_a^b \frac{\|\gamma'(t)\|_{\mathbb{E}}}{\text{Im}(\gamma(t))} dt,$$

where  $\|\cdot\|_{\mathbb{E}}$  is the Euclidean norm on  $\mathbb{R}^2$ . So we only need to verify three properties: 1. the matrix is in  $\text{SL}(2, \mathbb{R})$ ; 2. the transformation acts as a translation along the imaginary axis; 3. the translation moves hyperbolic length  $l$ .

Let  $T_l = \begin{pmatrix} e^{l/2} & 0 \\ 0 & e^{-l/2} \end{pmatrix}$ . Compute its determinant:

$$\det(T_l) = e^{l/2} \cdot e^{-l/2} - 0 \cdot 0 = e^0 = 1.$$

Thus  $T_l \in \text{SL}(2, \mathbb{R})$ , so  $[T_l] \in \text{PSL}(2, \mathbb{R})$ .

Evaluate the fractional linear transformation induced by  $T_l$  on an arbitrary  $z = x + iy \in \mathbb{H}^2$ :

$$T_l \cdot z = \frac{e^{l/2} \cdot z + 0}{0 \cdot z + e^{-l/2}} = e^{l/2} \cdot e^{l/2} z = e^l z.$$

For a point on the imaginary axis  $z = iy$  ( $y > 0$ ), this simplifies to  $T_l \cdot iy = e^l \cdot iy$ , which is a pure imaginary axis map (upward along the positive imaginary axis if  $l > 0$ ). For a general point  $z = x + iy$ ,  $T_l \cdot z = e^l x + ie^l y$ , which preserves Euclidean lines parallel to the imaginary axis and maps the imaginary axis to itself, this is the defining property of a hyperbolic translation along the imaginary axis in  $\mathbb{H}^2$ .

To confirm the translation length is  $l$ , consider the canonical curve along the imaginary axis:  $\gamma(t) = ie^t$  for  $t \in \mathbb{R}$ . Compute its hyperbolic length from  $t = 0$  (i.e.,  $\gamma(0) = i$ ) to  $t = l$  (i.e.,  $\gamma(l) = ie^l$ ):

$$\begin{aligned} L(\gamma|_{[0,l]}) &= \int_0^l \frac{\|\gamma'(t)\|_{\mathbb{E}}}{\text{Im}(\gamma(t))} dt \\ &= \int_0^l \frac{e^t}{e^t} dt = \int_0^l 1 dt = l. \end{aligned}$$

The transformation  $T_l$  maps  $\gamma(t) = ie^t$  to  $T_l \cdot \gamma(t) = e^l \cdot ie^t = ie^{t+l} = \gamma(t+l)$ . Thus  $T_l$  shifts the curve  $\gamma$  by hyperbolic length  $l$  along the imaginary axis, which is exactly the hyperbolic translation in the problem statement.

Finally,  $T_l$  and  $-T_l$  induce the same fractional linear transformation (i.e.,  $[T_l] = [-T_l]$ ), so the upward hyperbolic translation is uniquely represented by  $[T_l]$  in  $\text{PSL}(2, \mathbb{R})$ . ■

In the upper half-plane model  $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$  of the hyperbolic plane,

the isometry group is  $\text{PSL}(2, \mathbb{R}) = \text{SL}(2, \mathbb{R})/\{\pm I\}$ . Consider a geodesic polygon with  $n$  sides, with vertices  $P_1, P_2, \dots, P_n$  listed counterclockwise, edges  $l_i$  connecting  $P_{i-1}$  and  $P_i$  (indices modulo  $n$ ), interior angles  $\alpha_i$  at vertices  $P_i$ , and exterior angles defined as  $\theta_i = \pi - \alpha_i$ .

To derive the closure condition, we place the polygon in  $\mathbb{H}$  and successively apply isometries. Initially, we arrange edge  $l_1$  to lie along the imaginary axis, with vertex  $P_n$  aligned at the imaginary unit  $i$ . The transformation after this initial state is denoted by  $\gamma \in \text{PSL}(2, \mathbb{R})$ . The initial transformation is  $\gamma_0 = I$ .

The process of translating and rotating around the polygon until it returns to its original position proceeds as follows:

(1) Translate along the positive imaginary axis by distance  $l_1$ , corresponding to right multiplying by the translation matrix  $T(l_1)$ , yielding  $\gamma_1 = T(l_1)$ . At this stage, vertex  $P_1$  is aligned at the point  $i$ .

(2) Rotate counterclockwise about point  $i$  by the exterior angle  $\theta_1$ , corresponding to right multiplying by the rotation matrix  $R(\theta_1)$ , yielding  $\gamma_2 = T(l_1)R(\theta_1)$ . Edge  $l_2$  now lies along the imaginary axis.

(3) Translate along the positive imaginary axis by distance  $l_2$ , corresponding to right multiplying by  $T(l_2)$ , yielding  $\gamma_3 = T(l_1)R(\theta_1)T(l_2)$ .

(4) Continue this process, alternating translations and rotations for a total of  $n$  translations and  $n$  rotations, until all edges and vertices have been traversed. This procedure is illustrated in Figure 4-1.

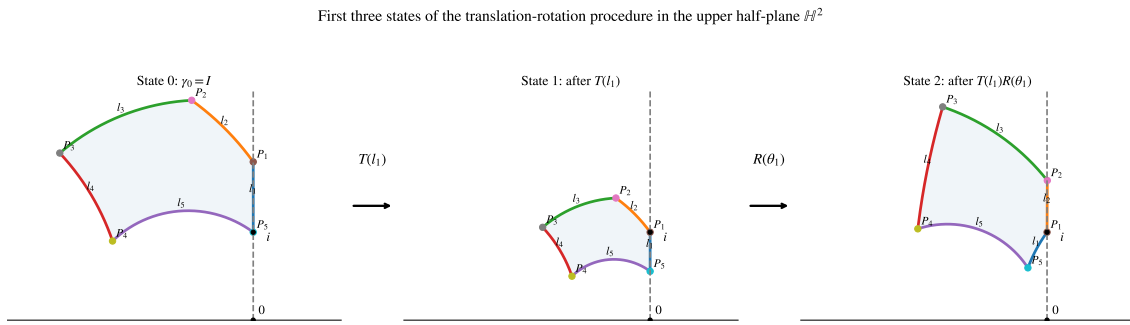


Figure 4-1 The first two steps of translation and rotation for a polygon

The resulting transformation matrix, which incorporates every edge length and vertex angle, is

$$T(l_1)R(\theta_1)T(l_2)R(\theta_2) \cdots T(l_n)R(\theta_n).$$

The condition that the polygon closes and returns to its original position requires that  $[\gamma_{2n}] = [I] \in \text{PSL}(2, \mathbb{R})$ ; that is,

$$\gamma_{2n} = T(l_1)R(\theta_1)T(l_2)R(\theta_2) \cdots T(l_n)R(\theta_n) = \pm I.$$

In fact, the resulting matrix is always  $-I$ . Here,

$$T(l) = \begin{pmatrix} e^{l/2} & 0 \\ 0 & e^{-l/2} \end{pmatrix}, \quad R(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}.$$

This matrix equation, understood in  $\text{PSL}(2, \mathbb{R})$  (allowing for an overall sign), imposes three real constraints. These three constraints determine the geometry of the polygon.

**Theorem 4.3:** Let  $n \geq 5$  be a positive integer. In the hyperbolic plane with curvature  $K = -1$ , the side length  $s$  of a regular  $n$ -gon with all interior angles equal to right angles is given by

$$s = 2 \operatorname{arcosh}\left(\sqrt{2} \cos \frac{\pi}{n}\right).$$

**Proof:** Consider the hyperbolic plane with curvature  $K = -1$ , and take a regular  $n$ -gon with each interior angle equal to  $\alpha = \frac{\pi}{2}$  and side length  $s$ . The geometric configuration is illustrated in Figure 4-2.

By symmetry, connect the center  $O$  of the polygon to two adjacent vertices  $A$  and  $B$ , forming an isosceles hyperbolic triangle  $\triangle OAB$  where

$$AB = s, \quad \angle A = \angle B = \frac{\alpha}{2} = \frac{\pi}{4}, \quad \angle O = \frac{2\pi}{n}.$$

Apply the angular form of the law of cosines to the hyperbolic triangle  $\triangle OAB$ :

$$\cos \angle O = -\cos \angle A \cos \angle B + \sin \angle A \sin \angle B \cosh(AB).$$

Substituting the known angles gives:

$$\cos \frac{2\pi}{n} = -\left(\cos \frac{\pi}{4}\right)^2 + \left(\sin \frac{\pi}{4}\right)^2 \cosh s = -\frac{1}{2} + \frac{1}{2} \cosh s.$$

Rearranging yields:

$$\cosh s = 1 + 2 \cos \frac{2\pi}{n}.$$

Thus the side length formula is:

$$s = \operatorname{arcosh}\left(1 + 2 \cos \frac{2\pi}{n}\right).$$

Using the identity  $\cosh s = 2 \cosh^2 \frac{s}{2} - 1$ , we have

$$2 \cosh^2 \frac{s}{2} - 1 = 1 + 2 \cos \frac{2\pi}{n} \Rightarrow \cosh \frac{s}{2} = \sqrt{1 + \cos \frac{2\pi}{n}}.$$

Since  $\sqrt{1 + \cos \theta} = \sqrt{2} \cos \frac{\theta}{2}$  (for  $\theta \in [0, \pi]$ ), taking  $\theta = \frac{2\pi}{n}$  gives

$$\cosh \frac{s}{2} = \sqrt{2} \cos \frac{\pi}{n}.$$

Therefore the equivalent form is:

$$s = 2 \operatorname{arcosh} \left( \sqrt{2} \cos \frac{\pi}{n} \right).$$

Note that in the hyperbolic plane, the sum of the interior angles of a regular  $n$ -gon must be less than  $(n - 2)\pi$ . Here the sum of interior angles is  $n \cdot \frac{\pi}{2}$ , so we require  $\frac{n\pi}{2} < (n - 2)\pi$ , i.e.,  $n > 4$ . When  $n = 4$ ,  $\cos(2\pi/4) = \cos(\pi/2) = 0$ , yielding  $\cosh s = 1$  and hence  $s = 0$ , which degenerates to a point. Therefore, hyperbolic right-angled regular polygons actually exist only for  $n \geq 5$ . ■

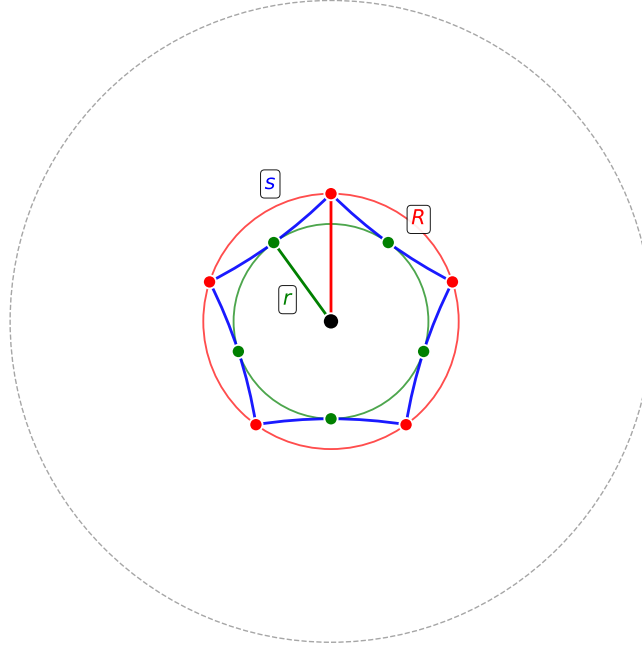


Figure 4-2 Regular right-angled pentagon ( $n = 5$ ) in the Poincaré disk model, drawn to scale. Blue arcs are geodesic edges; red and green circles are the circumcircle (radius  $R$ ) and incircle (radius  $r$ ), respectively.

**Theorem 4.4:** In the hyperbolic plane with curvature  $K = -1$ , let  $P$  be a regular  $n$ -gon with interior angle  $\alpha = \pi/2$  and side length  $s$  ( $n \geq 5$ ). Let  $O$  be its center. Then

1. Circumradius (center to vertex)  $R$  satisfies:

$$\sinh R = \frac{\sqrt{\cos(2\pi/n)}}{\sin(\pi/n)}, \quad \cosh R = \cot \frac{\pi}{n}, \quad R = \operatorname{arcosh} \left( \cot \frac{\pi}{n} \right)$$

2. Inradius (center to midpoint of side)  $r$  satisfies:

$$\sinh r = \frac{\sqrt{\cos(2\pi/n)}}{\sqrt{2} \sin(\pi/n)}, \quad \cosh r = \frac{1}{\sqrt{2} \sin(\pi/n)}, \quad r = \operatorname{arcosh} \left( \frac{1}{\sqrt{2} \sin(\pi/n)} \right)$$

**Proof:** It is known in Theorem 4.3 that  $\cosh s = 1 + 2 \cos \frac{2\pi}{n}$ , hence

$$\sinh s = 2 \sqrt{\cos \frac{2\pi}{n} (1 + \cos \frac{2\pi}{n})} = 2\sqrt{2} \cos \frac{\pi}{n} \sqrt{\cos \frac{2\pi}{n}}.$$

As illustrated in Figure 4-2,  $\triangle OAB$  satisfies  $OA = OB = R$ ,  $AB = s$ ,  $\angle AOB = 2\pi/n$ . Since  $OA$  bisects the interior angle  $\pi/2$ ,  $\angle OAB = \angle OBA = \pi/4$ . Applying the hyperbolic law of sines:

$$\frac{\sin(\pi/4)}{\sinh R} = \frac{\sin(2\pi/n)}{\sinh s}$$

Substituting  $\sinh s$  and simplifying yields  $\sinh R = \sqrt{\cos(2\pi/n)}/\sin(\pi/n)$ . Then  $\cosh R = \sqrt{1 + \sinh^2 R} = \cot(\pi/n)$  using  $\cos(2\pi/n) = 1 - 2 \sin^2(\pi/n)$ . Hence

$$R = \operatorname{arcosh}(\cot(\pi/n)).$$

Let  $M$  be the midpoint of  $AB$ , in the right hyperbolic triangle  $\triangle OAM$ ,  $\angle M = \pi/2$ ,  $\angle A = \pi/4$ ,  $OA = R$ ,  $OM = r$ . By the law of sines:

$$\frac{\sin(\pi/4)}{\sinh r} = \frac{\sin(\pi/2)}{\sinh R}$$

Thus  $\sinh r = \sin(\pi/4) \sinh R = \frac{1}{\sqrt{2}} \sinh R = \frac{\sqrt{\cos(2\pi/n)}}{\sqrt{2} \sin(\pi/n)}$ . Using  $\cosh r = \sqrt{1 + \sinh^2 r}$  and  $\cos(2\pi/n) = 1 - 2 \sin^2(\pi/n)$  gives  $\cosh r = \frac{1}{\sqrt{2} \sin(\pi/n)}$ , so

$$r = \operatorname{arcosh} \left( \frac{1}{\sqrt{2} \sin(\pi/n)} \right).$$

■

**Corollary 4.1 (Asymptotic Behavior):** In the hyperbolic plane  $K = -1$ , consider a regular  $n$ -gon ( $n \geq 5$ ) with interior angle  $\alpha = \pi/2$ . As  $n \rightarrow \infty$ , The side length  $s_n$  converges to a positive constant:

$$s_n \rightarrow s_\infty = 2 \ln(1 + \sqrt{2}).$$

Both the circumradius  $R_n$  and the inradius  $r_n$  diverge to infinity, growing logarithmically

in  $n$ . However, their difference tends to a finite limit:

$$R_n - r_n \rightarrow \ln \sqrt{2}.$$

**Proof:** From  $\cosh s = 1 + 2 \cos \frac{2\pi}{n}$ . As  $n \rightarrow \infty$ ,  $\cos \frac{2\pi}{n} \rightarrow 1$ , one has  $\cosh s \rightarrow 3$ . Therefore, side length  $s_n \rightarrow \operatorname{arcosh} 3 = 2 \ln(1 + \sqrt{2})$ , a constant.

For circumradius  $R_n$ ,  $\cosh R = \cot \frac{\pi}{n} \sim \frac{n}{\pi}$  for large  $n$ . Since  $\cosh R_n \sim \frac{1}{2}e^{R_n}$ , we get  $\frac{1}{2}e^{R_n} \sim \frac{n}{\pi}$ ,  $R \sim \ln \frac{2n}{\pi}$ . For inradius  $r_n$ ,  $\cosh r_n = \frac{1}{\sqrt{2} \sin(\pi/n)} \sim \frac{n}{\sqrt{2}\pi}$ . Hence  $\frac{1}{2}e^{r_n} \sim \frac{n}{\sqrt{2}\pi}$ ,  $r \sim \ln \frac{\sqrt{2}n}{\pi}$ .

At the end we check the difference of radii.  $R_n - r_n \sim \ln \frac{2n}{\pi} - \ln \frac{\sqrt{2}n}{\pi} = \ln \sqrt{2}$ . The limit  $\ln \sqrt{2}$  follows directly from the leading terms. ■

$$\text{Let } M(l, \theta) = T(l)R(\theta) = \begin{pmatrix} e^{l/2} & 0 \\ 0 & e^{-l/2} \end{pmatrix} \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix} = \begin{pmatrix} e^{\frac{l}{2}} \cos \frac{\theta}{2} & -e^{\frac{l}{2}} \sin \frac{\theta}{2} \\ e^{-\frac{l}{2}} \sin \frac{\theta}{2} & e^{-\frac{l}{2}} \cos \frac{\theta}{2} \end{pmatrix}.$$

According to Theorem 4.3, the shape parameters of a right-angled regular polygon are  $\theta = \pi/2$ ,  $l = 2 \operatorname{arcosh}(\sqrt{2} \cos \frac{\pi}{n})$ . At this given shape, we compute the values of the transformation  $T(l)R(\theta)$  and their partial derivatives as binary functions.

**Corollary 4.2:** At the right-angled regular  $n$ -gon, the parameters take the values  $\theta = \pi/2$ ,  $l = s = 2 \operatorname{arcosh}(\sqrt{2} \cos \frac{\pi}{n})$ . We can compute the values of the transformation  $T(l)R(\theta)$  and their partial derivatives as binary functions as follows:

$$\begin{aligned} M\left(s, \frac{\pi}{2}\right) &= \frac{\sqrt{2}}{2} \begin{pmatrix} a & -a \\ a^{-1} & a^{-1} \end{pmatrix}, \\ \frac{\partial M}{\partial \theta} \Big|_{(s, \frac{\pi}{2})} &= \frac{\sqrt{2}}{4} \begin{pmatrix} -a & -a \\ a^{-1} & -a^{-1} \end{pmatrix}, \\ \frac{\partial M}{\partial l} \Big|_{(s, \frac{\pi}{2})} &= \frac{\sqrt{2}}{4} \begin{pmatrix} a & -a \\ -a^{-1} & -a^{-1} \end{pmatrix}, \end{aligned}$$

where:

$$a = \sqrt{2} \cos \frac{\pi}{n} + \sqrt{\cos \frac{2\pi}{n}} \quad (> 1), \quad a^{-1} = \sqrt{2} \cos \frac{\pi}{n} - \sqrt{\cos \frac{2\pi}{n}}.$$

**Proof:** Given the matrix  $M(l, \theta) = T(l) R(\theta)$ , where

$$T(l) = \begin{pmatrix} e^{l/2} & 0 \\ 0 & e^{-l/2} \end{pmatrix}, \quad R(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}.$$

First, clarify:

$$M(l, \theta) = \begin{pmatrix} e^{l/2} \cos(\theta/2) & -e^{l/2} \sin(\theta/2) \\ e^{-l/2} \sin(\theta/2) & e^{-l/2} \cos(\theta/2) \end{pmatrix}.$$

Substitute the given values of the parameters:

$$\theta = \frac{\pi}{2}, \quad l = 2 \operatorname{arcosh}(\sqrt{2} \cos(\pi/n)).$$

Let:

$$a = e^{l/2} = \sqrt{2} \cos \frac{\pi}{n} + \sqrt{\cos \frac{2\pi}{n}}, \quad a^{-1} = e^{-l/2} = \sqrt{2} \cos \frac{\pi}{n} - \sqrt{\cos \frac{2\pi}{n}}.$$

Substituting the values  $l = s$  and  $\theta = \pi/2$ , the transformation matrix can be written as:

$$M\left(s, \frac{\pi}{2}\right) = \frac{\sqrt{2}}{2} \begin{pmatrix} a & -a \\ a^{-1} & a^{-1} \end{pmatrix}.$$

Taking the partial derivative with respect to  $\theta$ , we obtain:

$$\frac{\partial M}{\partial \theta} = \frac{1}{2} \begin{pmatrix} -e^{l/2} \sin(\theta/2) & -e^{l/2} \cos(\theta/2) \\ e^{-l/2} \cos(\theta/2) & -e^{-l/2} \sin(\theta/2) \end{pmatrix}.$$

Taking the partial derivative with respect to  $l$ , we obtain:

$$\frac{\partial M}{\partial l} = \frac{1}{2} \begin{pmatrix} e^{l/2} \cos(\theta/2) & -e^{l/2} \sin(\theta/2) \\ -e^{-l/2} \sin(\theta/2) & -e^{-l/2} \cos(\theta/2) \end{pmatrix}.$$

Evaluating the partial derivative with respect to  $\theta$  at the given values:

$$\left. \frac{\partial M}{\partial \theta} \right|_{(s, \pi/2)} = \frac{\sqrt{2}}{4} \begin{pmatrix} -a & -a \\ a^{-1} & -a^{-1} \end{pmatrix}.$$

Evaluating the partial derivative with respect to  $l$  at the given values:

$$\left. \frac{\partial M}{\partial l} \right|_{(s, \pi/2)} = \frac{\sqrt{2}}{4} \begin{pmatrix} a & -a \\ -a^{-1} & -a^{-1} \end{pmatrix}.$$

■

**Lemma 4.1 (Derivative of Matrix Product):** Let  $A : U \rightarrow \mathbb{R}^{m \times n}$  and  $B : U \rightarrow \mathbb{R}^{n \times p}$  be matrix-valued functions defined on an open set  $U \subseteq \mathbb{R}^q$ , whose entries  $A_{ik}(x)$  and

$B_{kj}(x)$  are continuously differentiable functions of the variable  $x = (x_1, \dots, x_q)$ . Define  $C(x) = A(x)B(x)$  with entries

$$C_{ij}(x) = \sum_{k=1}^n A_{ik}(x)B_{kj}(x), \quad i = 1, \dots, m, j = 1, \dots, p.$$

Then for each component index  $(i, j)$  and each partial derivative variable  $x_\ell$  ( $\ell = 1, \dots, q$ ),

$$\frac{\partial C_{ij}}{\partial x_\ell}(x) = \sum_{k=1}^n \left[ \frac{\partial A_{ik}}{\partial x_\ell}(x)B_{kj}(x) + A_{ik}(x)\frac{\partial B_{kj}}{\partial x_\ell}(x) \right].$$

**Proof:** Fix indices  $i, j$  and a coordinate direction  $x_\ell$ . By definition,

$$C_{ij}(x) = \sum_{k=1}^n A_{ik}(x)B_{kj}(x).$$

Since differentiation is linear and the sum is finite,

$$\frac{\partial C_{ij}}{\partial x_\ell}(x) = \sum_{k=1}^n \frac{\partial}{\partial x_\ell} [A_{ik}(x)B_{kj}(x)].$$

For each fixed  $k$ , the term  $A_{ik}(x)B_{kj}(x)$  is a product of two real-valued differentiable functions of  $x$ . By the ordinary product rule from multivariable calculus,

$$\frac{\partial}{\partial x_\ell} [A_{ik}(x)B_{kj}(x)] = \frac{\partial A_{ik}}{\partial x_\ell}(x)B_{kj}(x) + A_{ik}(x)\frac{\partial B_{kj}}{\partial x_\ell}(x).$$

Substituting this into the sum gives

$$\frac{\partial C_{ij}}{\partial x_\ell}(x) = \sum_{k=1}^n \left[ \frac{\partial A_{ik}}{\partial x_\ell}(x)B_{kj}(x) + A_{ik}(x)\frac{\partial B_{kj}}{\partial x_\ell}(x) \right].$$

Thus the formula holds for each  $i, j, \ell$ .

If we denote by  $\frac{\partial A}{\partial x_\ell}$  the matrix of entrywise partial derivatives  $\left[ \frac{\partial A_{ik}}{\partial x_\ell} \right]_{i,k}$ , then the above componentwise equality can be written in matrix form as

$$\frac{\partial}{\partial x_\ell}(AB) = \frac{\partial A}{\partial x_\ell}B + A\frac{\partial B}{\partial x_\ell}.$$

This is exactly the product rule for matrix-valued functions, applied componentwise and summed. ■

**Lemma 4.2 (Derivative for Products of Several Matrix-Valued Functions):**

Let  $A_1 : U \rightarrow \mathbb{R}^{m_0 \times m_1}$ ,  $A_2 : U \rightarrow \mathbb{R}^{m_1 \times m_2}$ , ...,  $A_N : U \rightarrow \mathbb{R}^{m_{N-1} \times m_N}$  be matrix-valued functions defined on an open set  $U \subseteq \mathbb{R}^q$ , whose entries are continuously differentiable functions of  $x = (x_1, \dots, x_q)$ . Define the product

$$P(x) = A_1(x)A_2(x) \cdots A_N(x) \in \mathbb{R}^{m_0 \times m_N}.$$

Then for each partial derivative variable  $x_\ell$  ( $\ell = 1, \dots, q$ ),

$$\frac{\partial P}{\partial x_\ell}(x) = \sum_{i=1}^N A_1(x) \cdots A_{i-1}(x) \left( \frac{\partial A_i}{\partial x_\ell}(x) \right) A_{i+1}(x) \cdots A_N(x).$$

Equivalently, for each entry index  $(r, s)$  with  $r \in \{1, \dots, m_0\}$ ,  $s \in \{1, \dots, m_N\}$ ,

$$\frac{\partial P_{rs}}{\partial x_\ell}(x) = \sum_{i=1}^N \sum_{\substack{k_1, \dots, k_{N-1} \\ \text{matching dimensions}}} (A_1)_{r, k_1} \cdots (A_{i-1})_{k_{i-2}, k_{i-1}} \left( \frac{\partial A_i}{\partial x_\ell} \right)_{k_{i-1}, k_i} (A_{i+1})_{k_i, k_{i+1}} \cdots (A_N)_{k_{N-1}, s}.$$

**Proof:** We proceed by induction on  $N$ , the number of matrices.

Base case  $N = 2$ . Let  $P = A_1 A_2$ . By the standard matrix product rule (which follows from the scalar product rule applied entrywise, as proven in the previous theorem),

$$\frac{\partial P}{\partial x_\ell} = \frac{\partial A_1}{\partial x_\ell} A_2 + A_1 \frac{\partial A_2}{\partial x_\ell}.$$

This matches the claimed formula with  $i = 1, 2$ .

Assume the formula holds for a product of  $N - 1$  matrices. Write

$$P = (A_1 A_2 \cdots A_{N-1}) A_N = Q A_N,$$

where  $Q = A_1 \cdots A_{N-1}$ . By the product rule for two matrices,

$$\frac{\partial P}{\partial x_\ell} = \frac{\partial Q}{\partial x_\ell} A_N + Q \frac{\partial A_N}{\partial x_\ell}.$$

Apply the induction hypothesis to  $\frac{\partial Q}{\partial x_\ell}$ :

$$\frac{\partial Q}{\partial x_\ell} = \sum_{i=1}^{N-1} A_1 \cdots A_{i-1} \left( \frac{\partial A_i}{\partial x_\ell} \right) A_{i+1} \cdots A_{N-1}.$$

Multiplying on the right by  $A_N$  gives

$$\frac{\partial Q}{\partial x_\ell} A_N = \sum_{i=1}^{N-1} A_1 \cdots A_{i-1} \left( \frac{\partial A_i}{\partial x_\ell} \right) A_{i+1} \cdots A_{N-1} A_N.$$

The second term  $Q \cdot \frac{\partial A_N}{\partial x_\ell}$  is simply  $A_1 \cdots A_{N-1} \cdot \frac{\partial A_N}{\partial x_\ell}$ , which corresponds to the  $i = N$  term in the sum. Adding them together, we obtain

$$\frac{\partial P}{\partial x_\ell} = \sum_{i=1}^N A_1 \cdots A_{i-1} \left( \frac{\partial A_i}{\partial x_\ell} \right) A_{i+1} \cdots A_N.$$

This completes the induction. ■

**Hyperbolic right-angled regular polygon: side-length–exterior-angle coordinates and geometric structure equations**

For a hyperbolic polygon, the side lengths and interior angles must satisfy the structure equation

$$M_1 M_2 \cdots M_n = -I.$$

Define  $F := M_1 M_2 \cdots M_n$ . Then the matrix equation  $F = -I$  is equivalent to four real equations:  $F_{11} = -1$ ,  $F_{12} = 0$ ,  $F_{21} = 0$ ,  $F_{22} = -1$ . Since  $F \in \text{SL}(2, \mathbb{R})$ , we have  $\det(F) = 1$ , and among these four real equations there is one constraint:

$$F_{11}F_{22} - F_{12}F_{21} = 1. \quad (4-1)$$

Thus, only three real equations serve as constraints.

Let the variable  $x$  represent any length  $l_i$  or angle  $\theta_j$ . Taking the total differential of equation (4-1) yields:

$$\frac{\partial F_{11}}{\partial x} dx + \frac{\partial F_{22}}{\partial x} dx = 0,$$

which indicates that in the tangent space, the gradients  $\nabla F_{11}$  and  $\nabla F_{22}$  are linearly dependent. In the following, we will only consider  $F_\delta$ ,  $\delta \in \{(1, 1), (1, 2), (2, 1)\}$ .

We may consider an infinitesimal deformation near a given hyperbolic structure on the surface, which corresponds to taking the total differential of  $F = -I$ . From this, we obtain linear relations among the infinitesimal elements of each length  $l_i$  and angle  $\theta_j$ . The space of all parameters is denoted by:

$$\mathcal{P}_{l,\theta} := \mathbb{R}^n \times \mathbb{R}^n \ni (l_i, \theta_j).$$

Near a point in  $\mathcal{P}_{l,\theta}$ , the total differential yields the linear relations:

$$dF_\delta = \sum_{i=1}^n \frac{\partial F_\delta}{\partial l_i} dl_i + \sum_{j=1}^n \frac{\partial F_\delta}{\partial \theta_j} d\theta_j = 0, \quad \delta \in \{(1, 1), (1, 2), (2, 1)\}. \quad (4-2)$$

The subspace spanned by these relations corresponds to the tangent space of the hyperbolic structure equation  $F = -I$  at that point.

Now, suppose our polygon is a right-angled regular  $n$ -gon, so all real parameters take the values  $l_i = s$ ,  $\theta_j = \pi/2$ . The values of the transformation matrices and their partial derivatives are independent of the index  $i$ .

$$M_i\left(s, \frac{\pi}{2}\right) = M = M\left(s, \frac{\pi}{2}\right) = \frac{\sqrt{2}}{2} \begin{pmatrix} a & -a \\ a^{-1} & a^{-1} \end{pmatrix},$$

$$\left(\frac{\partial M_i}{\partial l_i}\right)\Big|_{(s, \frac{\pi}{2})} = M_l = \frac{\partial M}{\partial \theta}\Big|_{(s, \frac{\pi}{2})} = \frac{\sqrt{2}}{4} \begin{pmatrix} -a & -a \\ a^{-1} & -a^{-1} \end{pmatrix},$$

$$\left(\frac{\partial M_i}{\partial \theta_i}\right)\Big|_{(s, \frac{\pi}{2})} = M_\theta = \frac{\partial M}{\partial l}\Big|_{(s, \frac{\pi}{2})} = \frac{\sqrt{2}}{4} \begin{pmatrix} a & -a \\ -a^{-1} & -a^{-1} \end{pmatrix},$$

where:

$$a = \sqrt{2} \cos \frac{\pi}{n} + \sqrt{\cos \frac{2\pi}{n}} \quad (> 1), \quad a^{-1} = \sqrt{2} \cos \frac{\pi}{n} - \sqrt{\cos \frac{2\pi}{n}}.$$

According to Lemma 4.2, the differential of each element can be computed using the Leibniz rule for products of several matrix-valued functions. Let the variable  $x_i$  be either  $l_i$  or  $\theta_j$ :

$$\begin{aligned} \left.\frac{\partial F}{\partial x_i}\right|_{(s, \frac{\pi}{2})} &= \left.\frac{\partial}{\partial x_i} (M_1 M_2 \cdots M_n)\right|_{(s, \pi/2)} \\ &= \sum_{i=1}^n M_1 \cdots M_{i-1} \left(\frac{\partial M_i}{\partial x_i}\right) M_{i+1} \cdots M_n \Big|_{(s, \frac{\pi}{2})} \\ &= M^{i-1} \cdot \left(\frac{\partial M_i}{\partial x_i}\right)\Big|_{(s, \frac{\pi}{2})} \cdot M^{n-i} \\ &= M^{i-1} \cdot M_x \cdot M^{n-i}. \end{aligned}$$

For each matrix entry, we obtain three differential constraints:

$$\left.\frac{\partial F_\delta}{\partial x_i}\right|_{(s, \frac{\pi}{2})} = \left(\left.\frac{\partial F}{\partial x_i}\right|_{(s, \frac{\pi}{2})}\right)_\delta = (M^{i-1} \cdot M_x \cdot M^{n-i})_\delta, \quad \delta \in \{(1, 1), (1, 2), (2, 1)\}. \quad (4-3)$$

**Theorem 4.5:** Let  $(P, \langle \cdot, \cdot \rangle)$  be a finite-dimensional real inner product space, let  $V \subset P$  be a subspace, and let  $s : P \rightarrow \mathbb{R}$  be a linear functional. Then

1. There exists a unique vector  $\nabla s \in P$  such that

$$s(x) = \langle \nabla s, x \rangle, \quad \forall x \in P.$$

$\nabla s$  is called the **gradient** of  $s$  in  $P$  (Riesz representation vector).

2. There exists a unique vector  $\nabla(s|_V) \in V$  such that

$$s(v) = \langle \nabla(s|_V), v \rangle, \quad \forall v \in V.$$

$\nabla(s|_V)$  is called the **gradient** of  $s|_V$  in  $V$ .

Let  $\pi_V : P \rightarrow V$  be the orthogonal projection induced by  $\langle \cdot, \cdot \rangle$ . Then

$$\nabla(s|_V) = \pi_V(\nabla s).$$

In other words: The gradient of the restricted functional equals the orthogonal projection of the gradient of the original functional onto the subspace.

**Proof:** The orthogonal projection  $\pi_V : P \rightarrow V$  is uniquely determined by the condition:

$$\langle \pi_V(x) - x, v \rangle = 0, \quad \forall v \in V.$$

Equivalently:  $x - \pi_V(x) \in V^\perp$ .

Next, we verify that  $\pi_V(\nabla s)$  satisfies the representation property for  $s|_V$ .

Take any  $v \in V$ . By the property of orthogonal projection,

$$\nabla s = \pi_V(\nabla s) + \eta, \quad \eta \in V^\perp.$$

Then

$$s(v) = \langle \nabla s, v \rangle = \langle \pi_V(\nabla s) + \eta, v \rangle = \langle \pi_V(\nabla s), v \rangle + \langle \eta, v \rangle.$$

Since  $\eta \in V^\perp$  and  $v \in V$ , we have  $\langle \eta, v \rangle = 0$ . Therefore

$$s(v) = \langle \pi_V(\nabla s), v \rangle, \quad \forall v \in V.$$

Finally, regarding uniqueness, the Riesz representation vector of  $s|_V$  in  $V$  is unique, and  $\pi_V(\nabla s) \in V$  satisfies this representation property, so  $\nabla(s|_V) = \pi_V(\nabla s)$ . This completes the proof.  $\blacksquare$

**Remark 4.1:** In the above proposition, the definition of “gradient” depends on the given inner product  $\langle \cdot, \cdot \rangle$  on  $P$ . If the inner product changes, both  $\nabla s$  and  $\nabla(s|_V)$  will change, but the relationship  $\nabla(s|_V) = \pi_V(\nabla s)$  still holds for the orthogonal projection induced by the same inner product.

The  $\pi_V$  in the proposition must be the orthogonal projection (relative to the inner product defining the gradient). If an oblique projection is used, the conclusion generally does not hold, because an oblique projection cannot guarantee  $\langle \eta, v \rangle = 0$  for all  $v \in V$ .

Coordinate representation (taking  $P = \mathbb{R}^m$  with the standard inner product as an example): Suppose  $s(x) = a^\top x$ , then  $\nabla s = a$ . Let  $Q \in \mathbb{R}^{m \times n}$  be a matrix whose

columns form an orthonormal basis of  $V$  (so  $Q^T Q = I_n$ ). Then

$$\pi_V(x) = Q Q^T x.$$

Hence

$$\nabla(s|_V) = \pi_V(a) = Q Q^T a.$$

Direct verification: for any  $v = Qy \in V$ ,

$$s(v) = a^T Qy = (Q^T a)^T y = (Q Q^T a)^T Qy = \langle \pi_V(a), v \rangle.$$

**Theorem 4.6:** Let  $(P, \langle \cdot, \cdot \rangle_P)$  be an  $m$ -dimensional real inner product space, and let  $(W, \langle \cdot, \cdot \rangle_W)$  be a  $k$ -dimensional real inner product space. Let  $J : P \rightarrow W$  be a linear map, and denote  $V = \ker J \subset P$  as the null space of  $J$ . Denote  $V^\perp = \{x \in P \mid \langle x, v \rangle_P = 0, \forall v \in V\}$  as the orthogonal complement of  $V$  in  $P$  with respect to the inner product  $\langle \cdot, \cdot \rangle_P$ .

Then

$$V^\perp = \{J^\dagger(w) \mid w \in W\}$$

where  $J^\dagger : W \rightarrow P$  is the *adjoint map* of  $J$  with respect to the given inner products, defined by:

$$\langle J^\dagger(w), p \rangle_P = \langle w, J(p) \rangle_W, \quad \forall p \in P, \forall w \in W.$$

Equivalently, if we take orthonormal bases of  $P$  and  $W$ , represent  $J$  by a matrix  $A \in \mathbb{R}^{k \times m}$ , represent the inner product  $\langle \cdot, \cdot \rangle_P$  by a symmetric positive definite matrix  $M \in \mathbb{R}^{m \times m}$ , and represent the inner product  $\langle \cdot, \cdot \rangle_W$  by a symmetric positive definite matrix  $N \in \mathbb{R}^{k \times k}$ , then

$$V^\perp = \{M^{-1} A^T N y \mid y \in \mathbb{R}^k\}.$$

In particular, when  $P = \mathbb{R}^m$  and  $W = \mathbb{R}^k$  both have the standard inner product (i.e.,  $M = I_m, N = I_k$ ),

$$V^\perp = \{A^T y \mid y \in \mathbb{R}^k\} = \text{row}(A) = \text{col}(A^T).$$

**Proof:** Suppose the inner product on  $P$  is  $\langle p, q \rangle_P = p^T M q$ , with  $M$  symmetric positive definite. Suppose the inner product on  $W$  is  $\langle w, z \rangle_W = w^T N z$ , with  $N$  symmetric positive definite. Let the matrix representation of  $J$  under the standard bases be  $A \in \mathbb{R}^{k \times m}$ , i.e.,  $J(p) = Ap$ .

The adjoint map  $J^\dagger : W \rightarrow P$  is determined by the definition  $\langle J^\dagger(w), p \rangle_P =$

$\langle w, J(p) \rangle_w$ .

Let  $J^\dagger(w) = Xw$ , where  $X \in \mathbb{R}^{m \times k}$  is to be determined. Then

$$\langle J^\dagger(w), p \rangle_p = (Xw)^\top M p = w^\top X^\top M p.$$

$$\langle w, J(p) \rangle_w = w^\top N(Ap) = w^\top N A p.$$

Since this holds for all  $w \in \mathbb{R}^k$  and  $p \in \mathbb{R}^m$ , we have

$$X^\top M = N A \quad \Rightarrow \quad X^\top = N A M^{-1} \quad \Rightarrow \quad X = M^{-1} A^\top N.$$

Therefore

$$J^\dagger(w) = M^{-1} A^\top N w, \quad \forall w \in \mathbb{R}^k.$$

Take any  $w \in W$ , and let  $x = J^\dagger(w)$ . For any  $v \in V = \ker J$  (i.e.,  $Av = 0$ ), we have

$$\langle x, v \rangle_p = (M^{-1} A^\top N w)^\top M v = w^\top N A M^{-1} M v = w^\top N(Av) = w^\top N \cdot 0 = 0.$$

Thus  $x \in V^\perp$ . Therefore  $J^\dagger(W) \subseteq V^\perp$ .

By the dimension formula:

$$\dim V = \dim \ker J = m - \text{rank } J.$$

$$\dim V^\perp = m - \dim V = \text{rank } J.$$

And  $J^\dagger$  is the adjoint of  $J$ , so they have the same rank:  $\text{rank } J^\dagger = \text{rank } J$ . Hence  $\dim J^\dagger(W) = \text{rank } J^\dagger = \text{rank } J = \dim V^\perp$ .

Combining  $J^\dagger(W) \subseteq V^\perp$  with equal dimensions yields

$$J^\dagger(W) = V^\perp.$$

Thus we obtain:

$$V^\perp = \{J^\dagger(w) \mid w \in W\} = \{M^{-1} A^\top N w \mid w \in \mathbb{R}^k\}.$$

When  $M = I_m$  and  $N = I_k$ ,

$$J^\dagger(w) = A^\top w.$$

In this case, the column space of  $A^\top \in \mathbb{R}^{m \times k}$  is precisely  $\text{col}(A^\top) = \text{row}(A)$ . Therefore

$$V^\perp = \text{col}(A^\top) = \{A^\top y \mid y \in \mathbb{R}^k\}.$$

The orthogonal complement of the null space equals the row space. This completes the proof. ■

**Theorem 4.7:** Let  $P$  be a finite-dimensional real linear space, and let  $V \subset P$  be a sub-

space. Let  $\pi_V : P \rightarrow V$  be a (fixed, arbitrary) projection onto  $V$ , and let  $\ker \pi_V$  be the kernel of the projection (i.e., the space of the projection direction). Let  $\{s_1, \dots, s_k\} \subset P$  be a set of linearly independent vectors, and denote  $S = \text{span}\{s_1, \dots, s_k\}$ . Consider the projected set of vectors  $\{\pi_V(s_1), \dots, \pi_V(s_k)\} \subset V$ . Then we have the following conclusions:

1. The projected set of vectors is linearly dependent if and only if  $S \cap \ker \pi_V \neq \{0\}$ , i.e., the original space  $S$  has a nontrivial intersection with the kernel of the projection.

2. The dimension lost by the projection (i.e., the dimension compressed during the projection) is exactly equal to  $\dim(S \cap \ker \pi_V)$ . The dimension of the space spanned after projection is:

$$\dim \pi_V(S) = \dim S - \dim(S \cap \ker \pi_V).$$

**Proof:** Let  $T : S \rightarrow V$  be the linear map obtained by restricting the projection map  $\pi_V$  to  $S$ , i.e.,  $T(s) = \pi_V(s)$  for any  $s \in S$ .

By the fundamental theorem of linear algebra (rank-nullity theorem):

$$\dim S = \dim \text{Im } T + \dim \ker T.$$

Here,  $\text{Im } T = \pi_V(S)$  is the image space after projection, and  $\ker T = S \cap \ker \pi_V$  is the kernel of the projection restricted to  $S$ .

Therefore:

$$\dim \pi_V(S) = \dim S - \dim(S \cap \ker \pi_V).$$

If  $S \cap \ker \pi_V \neq \{0\}$ , then  $\dim(S \cap \ker \pi_V) \geq 1$ , and hence

$$\dim \pi_V(S) \leq k - 1 < k = \dim S.$$

This means the  $k$  projected vectors cannot span a  $k$ -dimensional space, so they must be linearly dependent (since  $k$  linearly independent vectors would span a  $k$ -dimensional space).

Conversely, if the projected set of vectors is linearly dependent, then  $\dim \pi_V(S) < k$ . From the dimension formula we get

$$\dim(S \cap \ker \pi_V) = \dim S - \dim \pi_V(S) \geq 1,$$

hence  $S \cap \ker \pi_V \neq \{0\}$ .

This proves the equivalence:

$$\{\pi_V(s_1), \dots, \pi_V(s_k)\} \text{ is linearly dependent} \iff S \cap \ker \pi_V \neq \{0\}.$$

The dimension formula follows directly from the rank-nullity theorem:

$$\dim \pi_V(S) = \dim S - \dim(S \cap \ker \pi_V).$$

This is the formula for the dimension of the space spanned after projection. The lost dimension is defined as the difference between the dimension of the original space and the dimension of the image space after projection:

$$\text{Lost dimension} = \dim S - \dim \pi_V(S).$$

Substituting the dimension formula:

$$\text{Lost dimension} = \dim S - [\dim S - \dim(S \cap \ker \pi_V)] = \dim(S \cap \ker \pi_V).$$

Thus, the lost dimension is exactly equal to the dimension of the intersection of the original space  $S$  with the projection kernel  $\ker \pi_V$ . The compression originates from the directions lying in the projection kernel being completely “flattened.” ■

**Remark 4.2:** The above proposition holds for any linear projection  $\pi_V$  (not necessarily orthogonal). As long as  $\pi_V$  is a linear map satisfying  $\pi_V|_V = \text{id}_V$ , the kernel  $\ker \pi_V$  is the space of the projection direction.

If  $\pi_V$  is an orthogonal projection, then  $\ker \pi_V = V^\perp$ . In this case, the proposition becomes:

$$\dim \pi_V(S) = \dim S - \dim(S \cap V^\perp).$$

The lost dimension equals the dimension of the intersection of  $S$  with the orthogonal complement of  $V$ .

## 4.4 Deformation of Hyperbolic Structure

If one considers the set of shortest geodesics that fill a hyperbolic structure, then the entire Teichmüller space deformation retracts onto this subspace. Moreover, the critical points of the systole function, viewed as a topological Morse function, lie in the Thurston spine, so the index of these critical points can be used to study the dimension and cell structure of the spine. Schmutz Schaller<sup>[72]</sup> showed that surfaces with high symmetries are precisely the critical points. In their research for minimal filling sets that decompose the surface, An, Ihringer and Irmer<sup>[2]</sup> computed cases for  $n$  from 5 to 8 and conjectured that the index is  $n \cdot 2^{n-3} - (n + 3)$ . We use a different method to obtain the hyperbolic structures; this construction is consistent and holds for all  $n$ . Using this method, we can solve the problem and the results agree with the conjecture.

### Calculation method

If we require that, within the closed surface obtained from the dual decomposition, the four edges still constitute geodesics, each of which may vary in length, and that at each vertex the four interior angles sum to  $\pi$ , then we have  $n \cdot 2^{n-1}$  real parameters for the edge lengths and  $n \cdot 2^{n-2}$  real parameters for the angles. There are  $2^n$  faces, each subject to three constraints, so the total dimension is

$$n \cdot 2^{n-1} + n \cdot 2^{n-2} - 3 \cdot 2^n = 3(n-4)2^{n-2} = 6g - 6 = \dim \mathcal{M}_g,$$

which provides an alternative parametrization of moduli space. This perspective can serve as a starting point for computing the index of the systole function. Denote the  $n \cdot 2^{n-1}$  edge lengths by  $l_i$  and the  $n \cdot 2^{n-2}$  angles by  $\theta_j$ .

**The systole function index of surface  $M(n, -1)^* = Z(C_n, (D^1, S^0))^*$**

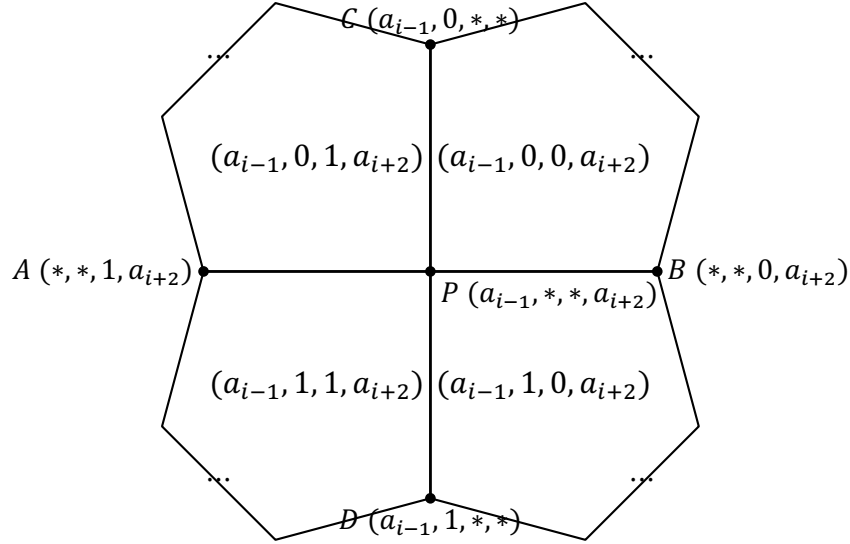


Figure 4-3 Local geometric configuration around a point  $P$ , showing its four adjacent points  $A, B, C, D$ , four incident edges, and four incident faces. For clarity, coordinates are displayed only for indices from  $i-1$  to  $i+2$ . Due to the perigon  $360^\circ$  at  $P$  and the geodesic nature of its edges, adjacent faces meeting at  $P$  have supplementary dihedral angles, while opposite faces share the same angle.

Next, for the surface  $M(n, -1)^* = Z(C_n, (D^1, S^0))^*$  with  $n \geq 5$ , we study the linear dependence relations among the gradients of the length functions corresponding to each systole on the surface, within the big parametrization space that includes the tangent space of the Teichmüller space.

First, we introduce the encoding rules for the set of objects. Given an integer  $n \geq 5$ , we use the symbol set  $\{0, 1, *\}$  on a string of length  $n$ , and encode three types of objects as follows:

1. Face objects:

$$\mathfrak{F} = \mathbb{F}_2^n.$$

A face  $f \in \mathfrak{F}$  is denoted as a binary vector  $f = (a_1, \dots, a_n)$ , where  $a_i \in \{0, 1\}$ . The total number of faces is  $|\mathfrak{F}| = 2^n$ .

2. Edge objects:

$$\mathfrak{E} = \mathbb{Z}/n \times \mathbb{F}_2^{n-1}.$$

An edge can be understood as a string with “exactly one position equal to  $*$ ”: for each  $i \in \mathbb{Z}/n$ , the edge  $(l, i)$  has the  $i$ -th position equal to  $*$ , and the remaining  $n - 1$  positions are 0/1. The total number of edges is

$$|\mathfrak{E}| = n \cdot 2^{n-1}.$$

3. Point objects:

$$\mathfrak{P} = \mathbb{Z}/n \times \mathbb{F}_2^{n-2}.$$

A point can be understood as a string with “two adjacent positions equal to  $*$ ”: for each  $i \in \mathbb{Z}/n$ , the point  $(\theta, i)$  has the  $i$ -th and  $(i + 1)$ -th positions equal to  $*$  (indices are cyclic), and the remaining  $n - 2$  positions are 0/1. The total number of points is

$$|\mathfrak{P}| = n \cdot 2^{n-2}.$$

### Linear algebra data: $2 \times 2$ fundamental matrices and local weights

Define the parameters

$$a = \sqrt{2} \cos \frac{\pi}{n} + \sqrt{\cos \frac{2\pi}{n}}, \quad a^{-1} = \sqrt{2} \cos \frac{\pi}{n} - \sqrt{\cos \frac{2\pi}{n}},$$

and construct three  $2 \times 2$  real matrices  $M, M_l, M_\theta$  as following,

$$M = \frac{\sqrt{2}}{2} \begin{pmatrix} a & -a \\ a^{-1} & a^{-1} \end{pmatrix}, \quad M_l = \frac{\sqrt{2}}{4} \begin{pmatrix} -a & -a \\ a^{-1} & -a^{-1} \end{pmatrix}, \quad M_\theta = \frac{\sqrt{2}}{4} \begin{pmatrix} a & -a \\ -a^{-1} & -a^{-1} \end{pmatrix}.$$

For each position  $i = 1, \dots, n$  and type  $x \in \{l, \theta\}$ , define

$$F_{(x,i)} := M^{i-1} M_x M^{n-i}.$$

Fix a set of entry extractions

$$\Delta = \{11, 12, 21\},$$

indicating that we extract the three entries  $(1, 1), (1, 2), (2, 1)$  from the  $2 \times 2$  matrices.

### Constructing the matrix $V$

Construct the matrix

$$V \in \mathbb{R}^{(3 \cdot 2^n) \times (|\mathfrak{E}| + |\mathfrak{P}|)},$$

whose rows are indexed by  $(f, \delta) \in \mathfrak{F} \times \Delta$  and columns by  $\mathfrak{E} \cup \mathfrak{P}$ .

Define the incidence function  $\chi$  as follows: for an edge or a point element,  $\chi(x, f) = 1$  if  $x$  is incident to face  $f$ , and 0 otherwise. The geometric notion of incidence is captured systematically using symbolic coordinates, with the precise conditions given below:

- An edge  $e \in \mathfrak{E}$  is incident to a face  $f$  if and only if, after ignoring the position marked with  $*$  in  $e$ , the remaining coordinates exactly match the corresponding 0/1 entries of  $f$ .
- A point  $p \in \mathfrak{P}$  is incident to a face  $f$  if and only if, after ignoring the two positions marked with  $*$  in  $p$ , the remaining coordinates exactly match the corresponding 0/1 entries of  $f$ .

Then define the matrix entries:

- If the column corresponds to an edge  $e = (l, i, \cdot) \in \mathfrak{E}$ , then

$$V((f, \delta), e) = \chi(e, f) \cdot (F_{(l, i)})_\delta.$$

- If the column corresponds to a point  $p = (\theta, i, \cdot) \in \mathfrak{P}$ , then

$$V((f, \delta), p) = \chi(p, f) \cdot (-1)^{a_i + a_{i+1}} \cdot (F_{(\theta, i)})_\delta.$$

### Constructing the circle-constraint matrix $S$

Construct the matrix

$$S \in \mathbb{R}^{(|\mathfrak{E}|/4) \times (|\mathfrak{E}| + |\mathfrak{P}|)}.$$

Each row corresponds to a “circle” (specially it is exactly all the systoles): fix a star position  $i$ , and fix the 0/1 values of all  $n - 3$  positions except  $i - 1, i, i + 1$ ; then let the binary values at positions  $i - 1$  and  $i + 1$  vary freely, yielding four edges

$$(0/1, *, 0/1)$$

(i.e., the three positions  $i - 1, i, i + 1$  are respectively 0/1, \*, 0/1), which form one circle. For the four edge columns corresponding to this circle, the entries are set to 1; all other edge columns are set to 0, and all point columns (corresponding to  $\mathfrak{P}$ ) are identically 0. Thus each row contains exactly four 1’s, and the number of rows is

$$|\mathfrak{E}|/4 = n \cdot 2^{n-3}.$$

### Final computation goal

The next step is to compute the dimension of the intersection of the row spaces of  $V$  and  $S$ . By vertically concatenating  $V$  and  $S$ , we obtain

$$A = \begin{pmatrix} V \\ S \end{pmatrix},$$

and compute its numerical rank. The dimension of the intersection of the row spaces can then be obtained using the following formula:

$$\text{rank}(\text{Row}(V) \cap \text{Row}(S)) = \text{rank}(S) + \text{rank}(V) - \text{rank}(A).$$

The row space of  $V$  is the normal space of the tangent space of the Teichmüller space, and the row space of  $S$  is the space spanned by the gradients of the length functions of the systoles. The intersection of these two spaces corresponds to the directions in which the length functions of the systoles remain unchanged under infinitesimal deformations. The index of the critical point corresponding to the surface  $M(n, -1)^*$  can be calculated as

$$\text{index} = \#S - \text{rank}(V \cap S).$$

### Numerical calculation result

The rank of real number value matrix  $A$  can be determined using a threshold criterion (tol) for the SVD: singular values  $\sigma \leq \text{tol}$  are considered zero.

Singular Value Decomposition (SVD) provides a theoretically complete and numerically stable method for determining the rank of a matrix. For any matrix  $A \in \mathbb{R}^{m \times n}$ , its singular value decomposition can be written as  $A = U\Sigma V^T$ , where  $U$  and  $V$  are orthogonal matrices, and  $\Sigma$  is a diagonal matrix whose diagonal entries  $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{\min(m,n)} \geq 0$  are called singular values. Fundamental results in linear algebra establish that the rank of a matrix equals the number of its non-zero singular values. This conclusion stems from the fact that the invertibility of  $U$  and  $V$  guarantees  $\text{rank}(A) = \text{rank}(\Sigma)$ , and the rank of the diagonal matrix  $\Sigma$  is precisely the number of its non-zero diagonal entries. Geometrically, singular values characterize the scaling factors of the linear transformation in each principal direction; the number of non-zero singular values is exactly the dimension of the transformed space, which is the rank of the matrix.

In numerical computing environments, due to rounding errors in floating-point arithmetic, theoretically zero singular values manifest as extremely small non-zero values (typically on the order of machine precision  $\epsilon_{\text{mach}} \approx 10^{-16}$ ). Therefore, we need to introduce threshold-based criteria to distinguish numerically non-zero singular values from those that are effectively zero. A commonly used threshold selection strategy is based on ma-

chine precision and matrix dimensions: an absolute threshold  $\tau = \epsilon_{\text{mach}} \cdot \max(m, n) \cdot \sigma_1$ , where  $\sigma_1$  is the largest singular value. This choice is justified because the numerical error in SVD is of order  $O(\epsilon_{\text{mach}} \|A\|)$ . A more robust approach involves analyzing the distribution of the singular value spectrum to identify significant jumps in the ratio of consecutive singular values  $\sigma_i/\sigma_{i+1}$ —if there exists an  $r$  such that  $\sigma_{r+1}/\sigma_1 \ll 1$  and  $\sigma_r/\sigma_{r+1} \gg 1$ , then the rank of the matrix can be determined as  $r$ . This gap-based singular value determination method effectively overcomes the uncertainties introduced by floating-point precision and provides numerically stable rank estimation.

Endowing each right-angled regular polygon of the surface  $X = Z(C_n, (D^1, S^0))^*$  with the standard hyperbolic metric, we computed  $\text{index}(X)$  for  $n$  in the range from 5 to 13. The results are shown in Table 4-1.

Table 4-1 Calculation result of rank and index

| $n$ | row(V) | row(S) | shape(A)       | rank(A) | rank( $V \cap S$ ) | index |
|-----|--------|--------|----------------|---------|--------------------|-------|
| 5   | 96     | 20     | [116, 120]     | 108     | 8                  | 12    |
| 6   | 192    | 48     | [240, 288]     | 231     | 9                  | 39    |
| 7   | 384    | 112    | [496, 672]     | 486     | 10                 | 102   |
| 8   | 768    | 256    | [1024, 1536]   | 1013    | 11                 | 245   |
| 9   | 1536   | 576    | [2112, 3456]   | 2100    | 12                 | 564   |
| 10  | 3072   | 1280   | [4352, 7680]   | 4339    | 13                 | 1267  |
| 11  | 6144   | 2816   | [8960, 16896]  | 8946    | 14                 | 2802  |
| 12  | 12288  | 6144   | [18432, 36864] | 18417   | 15                 | 6129  |
| 13  | 24576  | 13312  | [37888, 79872] | 37872   | 16                 | 13296 |

## Preliminaries on the Walsh–Hadamard Transform

The Walsh–Hadamard transform (WHT) is a discrete orthogonal transform whose basis functions take only the values  $+1$  and  $-1$ . It can be implemented using only additions and subtractions, and a fast algorithm (FWHT) requires  $N \log_2 N$  operations, making it useful in signal processing, image compression, and quantum computing.

**Definition 4.3 (Hadamard matrix):** Let  $N = 2^n$ . The Hadamard matrix  $\mathbf{H}_n$  is defined recursively by

$$\mathbf{H}_1 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \mathbf{H}_{n+1} = \mathbf{H}_n \otimes \mathbf{H}_1 = \begin{bmatrix} \mathbf{H}_n & \mathbf{H}_n \\ \mathbf{H}_n & -\mathbf{H}_n \end{bmatrix},$$

where  $\otimes$  denotes the Kronecker product. All entries of  $\mathbf{H}_n$  are  $\pm 1$ .

**Definition 4.4 (Forward Walsh–Hadamard transform):** For an input vector  $\mathbf{x} = (x_0, \dots, x_{N-1}) \in \mathbb{R}^N$ , the WHT coefficients  $y_k$  are given by

$$y_k = \sum_{i=0}^{N-1} x_i (-1)^{\langle i, k \rangle_2}, \quad k = 0, 1, \dots, N-1,$$

where  $\langle i, k \rangle_2 = \sum_{j=0}^{n-1} b_j(i) b_j(k)$  is the dot product of the binary representations of  $i$  and  $k$  taken modulo 2. Equivalently, it is the parity (even or odd) of the number of 1-bits in the bitwise AND of  $i$  and  $k$ .

**Definition 4.5 (Inverse transform and decomposition into basis vectors):**

The inverse transform recovers the original vector:

$$x_i = \frac{1}{N} \sum_{k=0}^{N-1} y_k (-1)^{\langle i, k \rangle_2}, \quad i = 0, 1, \dots, N-1.$$

Define the basis vector  $\mathbf{w}_k$  by  $(\mathbf{w}_k)_i = (-1)^{\langle i, k \rangle_2}$ . Then

$$\mathbf{x} = \sum_{k=0}^{N-1} \frac{y_k}{N} \mathbf{w}_k.$$

Thus  $y_k/N$  is the weight of  $\mathbf{w}_k$  in the representation of  $\mathbf{x}$ .

**Normalised form.** In the normalised version the transform matrix is  $\frac{1}{\sqrt{N}} \mathbf{H}_n$ , which is orthogonal. The forward transform becomes

$$y_k = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} x_i (-1)^{\langle i, k \rangle_2},$$

and the inverse is

$$x_i = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} y_k (-1)^{\langle i, k \rangle_2}.$$

In this case  $\mathbf{x} = \sum_{k=0}^{N-1} y_k \mathbf{b}_k$  with  $\mathbf{b}_k$  being the orthonormal basis vectors  $(\mathbf{b}_k)_i = \frac{1}{\sqrt{N}} (-1)^{\langle i, k \rangle_2}$ .

**Example 4.2:** For  $n = 2$ ,  $N = 4$ , unnormalised case, the Hadamard matrix is

$$\mathbf{H}_2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix},$$

with rows

$$\mathbf{w}_0 = (1, 1, 1, 1), \mathbf{w}_1 = (1, -1, 1, -1), \mathbf{w}_2 = (1, 1, -1, -1), \mathbf{w}_3 = (1, -1, -1, 1).$$

The forward transform computes  $y_k = \langle \mathbf{x}, \mathbf{w}_k \rangle$ , and the inverse is  $\mathbf{x} = \frac{1}{4} \sum_{k=0}^3 y_k \mathbf{w}_k$ .

## 4.5 Proof of the Index Formula Theorem

We first review the problem statement of index theorem. Let  $A = \begin{pmatrix} V \\ S \end{pmatrix}$ . We wish to compute

$$\dim(\text{Row}(V) \cap \text{Row}(S)).$$

By the dimension formula from linear algebra,

$$\dim(\text{Row}(V) \cap \text{Row}(S)) = \text{rank}(V) + \text{rank}(S) - \text{rank}(A).$$

Hence the problem reduces to evaluating

$$\text{rank}(V) + n2^{n-3} - \text{rank} \begin{pmatrix} V \\ S \end{pmatrix}.$$

**Theorem 4.8 (Index formula):** For the matrices  $V$  and  $S$  defined in the context of Teichmüller space and systole length functions, the dimension of the intersection of their row spaces satisfies

$$\dim(\text{Row}(V) \cap \text{Row}(S)) = n + 3.$$

Hence, using the index formula  $\text{index} = \#S - \dim(\text{Row}(V) \cap \text{Row}(S))$  and noting that  $\#S = \text{Row}(S) = n2^{n-3}$ , we obtain

$$\text{index} = n2^{n-3} - (n + 3).$$

The proof is given in the following context.

### Writing Row Combinations as Functions

All matrix entries lie in the field

$$\mathbb{Q} \left( \sqrt{2}, \cos \frac{\pi}{n}, \sqrt{\cos \frac{2\pi}{n}} \right).$$

Denote this field by

$$K_n = \mathbb{Q} \left( \sqrt{2}, \cos \frac{\pi}{n}, \sqrt{\cos \frac{2\pi}{n}} \right).$$

Since  $K_n \subset \mathbb{R}$  and the rank of a finite matrix is unchanged under field extensions, we can discuss row spaces and dimensions over  $K_n$ .

Take an arbitrary linear combination of the rows of  $V$ . Assign to each row  $(f, \delta)$  a coefficient  $\lambda_{f,\delta} \in K_n$ . Equivalently, for each  $\delta \in \Delta$  we have a function

$$\lambda_\delta : \mathbb{F}_2^n \rightarrow K_n.$$

If a face  $f$  has coordinates  $f = (a_1, \dots, a_n)$ , define

$$\lambda_\delta(a_1, \dots, a_n) = \lambda_{f,\delta}.$$

Thus a row combination of  $V$  is determined by three functions

$$\lambda_{11}, \lambda_{12}, \lambda_{21} : \mathbb{F}_2^n \rightarrow K_n.$$

Define

$$v(a) = (\lambda_{11}(a), \lambda_{12}(a), \lambda_{21}(a)) \in K_n^3.$$

Also define

$$L_i = ((F_{(l,i)})_{11}, (F_{(l,i)})_{12}, (F_{(l,i)})_{21}) \in K_n^3,$$

and

$$\Theta_i = ((F_{(\theta,i)})_{11}, (F_{(\theta,i)})_{12}, (F_{(\theta,i)})_{21}) \in K_n^3.$$

### Edge and Point Coordinates of a Row Combination

An edge with star position  $i$  is incident to two faces, corresponding to  $a_i = 0$  and  $a_i = 1$ . Therefore the value of the row combination of  $V$  on that edge is

$$\sum_{a_i \in \mathbb{F}_2} L_i \cdot v(a_1, \dots, a_n).$$

Thus the edge coordinate is essentially a sum along the  $i$ -th coordinate followed by an inner product with the weight  $L_i$ .

A point with star positions  $i$  and  $i + 1$  is incident to four faces. Because the point column also contains a factor  $(-1)^{a_i + a_{i+1}}$ , the value of the row combination of  $V$  on that

point is

$$\sum_{a_i, a_{i+1} \in \mathbb{F}_2} (-1)^{a_i + a_{i+1}} \cdot \Theta_i \cdot v(a_1, \dots, a_n).$$

### Description of the Row Space of $S$

$S$  is non-zero only on edge coordinates; it is identically zero on all point coordinates.

Fix a star position  $i$ . A row of  $S$  places the four edges of the form

$$\prod_{a_{i-1}, a_{i+1} \in \mathbb{F}_2} (a_{i-1}, *, a_{i+1})$$

into the same circle. Hence a vector belongs to  $\text{Row}(S)$  if and only if:

- (1) all point coordinates are 0;
- (2) for each star position  $i$ , the edge coordinate depends only on the variables other than  $i-1, i, i+1$ , i.e. it does not depend on  $a_{i-1}$  and  $a_{i+1}$ , as a function  $\mathbb{F}_2^{[n] \setminus \{i\}} \rightarrow K_n$ , its value is invariant when  $a_{i-1}$  and  $a_{i+1}$  vary.

### Walsh–Hadamard Decomposition

On  $\mathbb{F}_2^n$  we use the Walsh–Hadamard characters as a basis. Define the character

$$\chi_J(a) = (-1)^{\sum_{j \in J} a_j}, \quad J \subseteq \{1, \dots, n\}.$$

Every function  $\lambda_\delta$  can be uniquely written as

$$\lambda_\delta(a) = \sum_{J \subseteq \{1, \dots, n\}} \hat{\lambda}_{\delta, J} \chi_J(a).$$

For each  $J$  set define a vector  $v_J$  by

$$v_J = (\hat{\lambda}_{11, J}, \hat{\lambda}_{12, J}, \hat{\lambda}_{21, J}) \in K_n^3.$$

Hence we have

$$v(a) = \sum_{J \subseteq [n]} v_J \chi_J(a).$$

The Walsh-Hadamard decomposition reduces the problem to studying a separate three-dimensional linear algebra problem for each  $J$ .

### Separating Fourier Modes in Edge and Point Summations

After expanding

$$v(a) = \sum_{J \subseteq [n]} v_J \chi_J(a), \quad \chi_J(a) = (-1)^{\sum_{j \in J} a_j},$$

the edge and point coordinates treat each Fourier mode  $\chi_J$  independently. Fix one mode  $v_J \chi_J(a)$  and examine its contribution to the edge and point coordinates, the conditions defining  $\text{Row}(S)$  can be expressed in terms of the Fourier support  $J$ .

### Fourier Modes Contribution in Edge Coordinates

**Edge contribution.** Fix a star position  $i$ . An edge is described by

$$(a_1, \dots, a_{i-1}, *, a_{i+1}, \dots, a_n).$$

Its edge coordinate is

$$y_i^l(u) = \sum_{a_i \in \mathbb{F}_2} L_i \cdot v(a),$$

where  $u$  denotes all coordinates except  $a_i$ . Restrict on the Fourier mode  $v_J \chi_J(a)$ :

$$y_{i,J}^l(u) = \sum_{a_i \in \mathbb{F}_2} L_i \cdot v_J \chi_J(a) = (L_i \cdot v_J) \sum_{a_i \in \mathbb{F}_2} \chi_J(a).$$

• **If  $i \in J$ :** then  $\chi_J(a) = (-1)^{a_i} (-1)^{\sum_{j \in J \setminus \{i\}} a_j}$ . The sum over  $a_i$  gives  $\sum_{a_i=0}^1 (-1)^{a_i} = 0$ . Hence no output from mode  $J$  on edge  $i$ .

• **If  $i \notin J$ :**  $\chi_J(a)$  does not depend on  $a_i$ , so

$$\sum_{a_i=0}^1 \chi_J(a) = 2\chi_J(u).$$

Thus  $y_{i,J}^l(u) = 2(L_i \cdot v_J) \chi_J(u)$  the output coefficient is proportional to  $L_i \cdot v_J$  up to the non-zero factor 2. Here we need to use the circle condition of  $\text{Row}(S)$ .

**Circle condition of  $\text{Row}(S)$  on edges.**  $\text{Row}(S)$  requires that for a fixed star  $i$ , the edge value does not depend on the two neighbouring variables  $a_{i-1}$  and  $a_{i+1}$ . Now  $\chi_J(u)$  depends exactly on the variables in  $J$ . Since  $i \notin J$  in the second case above, we distinguish:

• **If  $J \cap \{i-1, i+1\} = \emptyset$ :** then  $\chi_J(u)$  does not involve  $a_{i-1}$  or  $a_{i+1}$ ; the circle condition is satisfied. Thus the output is allowed, and we obtain a contribution proportional to  $L_i \cdot v_J$ .

• **If  $J \cap \{i-1, i+1\} \neq \emptyset$ :** then  $\chi_J(u)$  would depend on  $a_{i-1}$  or  $a_{i+1}$ , violating the circle condition. To avoid this, the coefficient must vanish:

$$\boxed{i \notin J, J \cap \{i-1, i+1\} \neq \emptyset \implies L_i \cdot v_J = 0.}$$

## Fourier Modes Contribution in Point Coordinates

**Point contribution.** Fix a point with adjacent stars  $(i, i + 1)$ . Its coordinate is

$$y_i^\theta(w) = \sum_{a_i, a_{i+1} \in \mathbb{F}_2} (-1)^{a_i + a_{i+1}} \Theta_i \cdot v(a),$$

where  $w$  denotes the remaining variables. Restrict on the Fourier mode  $v_J \chi_J(a)$ :

$$y_{i,J}^\theta(w) = (\Theta_i \cdot v_J) \sum_{a_i, a_{i+1} \in \mathbb{F}_2} (-1)^{a_i + a_{i+1}} \chi_J(a).$$

• **If  $i, i + 1 \in J$ :** then  $\chi_J(a)$  contains the factor  $(-1)^{a_i + a_{i+1}}$ . The product  $(-1)^{a_i + a_{i+1}} \chi_J(a) = (-1)^{a_i + a_{i+1}} (-1)^{a_i + a_{i+1}} = 1$  cancels the oscillation, so the double sum is non-zero. Hence

$$y_{i,J}^\theta(w) = 4(\Theta_i \cdot v_J) \chi_{J \setminus \{i, i+1\}}(w).$$

Because  $\text{Row}(S)$  is zero on every point coordinate, we must have

$$i, i + 1 \in J \Rightarrow \Theta_i \cdot v_J = 0.$$

• **If  $i$  and  $i + 1$  are not both in  $J$ :** then at least one of the two variables still appears with a single oscillating factor, and the sum over that variable gives zero. For example, if  $i \in J$  but  $i + 1 \notin J$ , then

$$\sum_{a_i, a_{i+1} \in \mathbb{F}_2} (-1)^{a_i + a_{i+1}} \chi_J(a) = \sum_{a_i, a_{i+1} \in \mathbb{F}_2} (-1)^{a_i + a_{i+1}} (-1)^{a_i} = 0.$$

Hence the whole double sum vanishes automatically, and no extra condition is imposed.

**Concise rule of thumb.** For a fixed Fourier mode with support  $J$ , these are the edge and point conditions.

|                                                      |                                                             |
|------------------------------------------------------|-------------------------------------------------------------|
| $i \in J$                                            | $\Rightarrow$ edge automatically zero;                      |
| $i \notin J, J \cap \{i - 1, i + 1\} = \emptyset$    | $\Rightarrow$ edge output allowed $\propto L_i \cdot v_J$ ; |
| $i \notin J, J \cap \{i - 1, i + 1\} \neq \emptyset$ | $\Rightarrow L_i \cdot v_J = 0$ ;                           |
| $i, i + 1 \in J$                                     | $\Rightarrow \Theta_i \cdot v_J = 0$ ;                      |
| otherwise                                            | $\Rightarrow$ point automatically zero.                     |

### Three-Dimensional Linear Algebra for a Fixed Fourier Mode

For a fixed  $J \subseteq \{1, \dots, n\}$ , define the constraint space

$$C_J = \text{span}\left(\{\Theta_i : i, i+1 \in J\} \cup \{L_i : i \notin J, J \cap \{i-1, i+1\} \neq \emptyset\}\right) \subseteq K_n^3.$$

Then valid  $v_J$  must satisfy

$$c \cdot v_J = 0, \quad \forall c \in C_J.$$

On the other hand, the positions that actually allow outputs belonging to  $\text{Row}(S)$  are those with

$$i-1 \notin J, \quad i \notin J, \quad i+1 \notin J.$$

Define the output space

$$O_J = \text{span}\{L_i : i-1, i, i+1 \notin J\} \subseteq K_n^3.$$

The output coefficient is proportional to  $L_i \cdot v_J$ , if non-zero valid  $v_J$  is vertical to all vectors in  $O_J$ , it cannot contribute to the intersection dimension.

Thus the contribution of Fourier mode  $J$  to the dimension of the intersection is

$$\begin{aligned} d_J &= \text{rank}(C_J^\perp) - \text{rank}(C_J + O_J)^\perp \\ &= (3 - \text{rank}(C_J)) - (3 - \text{rank}(C_J + O_J)) \\ &= \text{rank}(C_J + O_J) - \text{rank}(C_J). \end{aligned}$$

Consequently,

$$\dim(\text{Row}(V) \cap \text{Row}(S)) = \sum_{J \subseteq \{1, \dots, n\}} d_J.$$

### Basic Facts about $L_i$ and $\Theta_i$

We will use three basic facts about  $L_i$  and  $\Theta_i$ .

**Fact 4.6 (Fact A:  $L_i$  forms a rotation orbit):** The matrix  $M$  satisfies

$$\det M = 1, \quad \text{tr } M = 2 \cos \frac{\pi}{n}.$$

Hence the eigenvalues of  $M$  are  $e^{\pm i\pi/n}$ . It follows that  $L_i$  can be written as

$$L_i = P \begin{pmatrix} 1 \\ \cos \frac{2\pi i}{n} \\ \sin \frac{2\pi i}{n} \end{pmatrix},$$

where  $P$  is an invertible  $3 \times 3$  matrix.

Therefore any three distinct  $L_i$  are linearly independent. That is, if  $i, j, k$  are pairwise distinct then

$$\det(L_i, L_j, L_k) \neq 0.$$

**Fact 4.7 (Fact B:  $\Theta_i$  is a non-degenerate linear combination of two adjacent  $L$ 's):** A direct computation gives a non-degenerate linear relation:

$$\Theta_i = \gamma L_i + \eta L_{i+1},$$

with

$$\gamma = -\frac{a^4 + 1}{a^4 - 1}, \quad \eta = \frac{2a^2}{a^4 - 1}.$$

For  $n \geq 5$  we have  $a^4 \neq 1$ , so  $\gamma \neq 0$ ,  $\eta \neq 0$ .

**Proof:** The core of the computation is the local matrix identity

$$M_\theta = \gamma M_l + \eta M M_l M^{-1}.$$

Indeed, since  $F_{(\theta,i)} = M^{i-1} M_\theta M^{n-i}$  and  $F_{(l,i)} = M^{i-1} M_l M^{n-i}$ ,  $F_{(l,i+1)} = M^{i-1} (M M_l M^{-1}) M^{n-i}$ , the identity immediately gives

$$F_{(\theta,i)} = \gamma F_{(l,i)} + \eta F_{(l,i+1)}.$$

Extracting the three entries (11), (12), (21) yields  $\Theta_i = \gamma L_i + \eta L_{i+1}$ .

Now verify the local identity. A direct multiplication gives

$$M M_l M^{-1} = \frac{\sqrt{2}}{4} \begin{pmatrix} -a^{-1} & -a^3 \\ a^{-3} & -a \end{pmatrix}.$$

Removing the common factor  $\frac{\sqrt{2}}{4}$ , the relation  $M_\theta = \gamma M_l + \eta M M_l M^{-1}$  becomes

$$\begin{pmatrix} a & -a \\ -a^{-1} & -a^{-1} \end{pmatrix} = \gamma \begin{pmatrix} -a & -a \\ a^{-1} & -a^{-1} \end{pmatrix} + \eta \begin{pmatrix} -a^{-1} & -a^3 \\ a^{-3} & -a \end{pmatrix}.$$

Comparing the (1, 1) and (1, 2) entries:

$$a = -\gamma a - \eta a^{-1}, \quad -a = -\gamma a - \eta a^3.$$

Subtracting the two eliminates  $\gamma$ :

$$-\eta(a^3 - a^{-1}) = -2a \implies \eta = \frac{2a}{a^3 - a^{-1}} = \frac{2a^2}{a^4 - 1}.$$

Substituting back gives

$$\gamma = 1 - \eta a^2 = 1 - \frac{2a^4}{a^4 - 1} = -\frac{a^4 + 1}{a^4 - 1}.$$

Thus

$$\gamma = -\frac{a^4 + 1}{a^4 - 1}, \quad \eta = \frac{2a^2}{a^4 - 1}.$$

The other matrix entries are automatically consistent. ■

**Fact 4.8 (Fact C: Three vectors in an interval are linearly independent):** If  $2 \leq k \leq n - 3$ , then

$$\Theta_i, \quad L_{i-1}, \quad L_{i+k}$$

are linearly independent. In fact, using Fact A and Fact B one computes

$$\det(\Theta_i, L_{i-1}, L_{i+k}) = C_n \sin \frac{\pi}{n} \sin \frac{(k+1)\pi}{n} \sin \frac{(k+2)\pi}{n},$$

where  $C_n \neq 0$ . For  $2 \leq k \leq n - 3$  each of the three sine factors is non-zero, hence the determinant is non-zero.

### Contribution of Each Fourier Mode

**Case 1:  $J = \emptyset$ .** There are no point constraints and no circle constraints, so  $C_\emptyset = 0$ . All edge positions are allowed, thus

$$O_\emptyset = \text{span}\{L_1, \dots, L_n\}.$$

By Fact A,  $\text{rank}(O_\emptyset) = 3$ . Hence  $d_\emptyset = 3$ .

**Case 2:  $|J| = 1$ .** Let  $J = \{j\}$ . Since  $J$  contains no adjacent pair, there are no point constraints.

The circle conditions give two constraints. For star position  $j - 1$ , the right neighbour  $j$  belongs to  $J$ , so we must have  $L_{j-1} \cdot v_j = 0$ . For star position  $j + 1$ , the left neighbour  $j$  belongs to  $J$ , so we must have  $L_{j+1} \cdot v_j = 0$ . Hence

$$C_J = \text{span}\{L_{j-1}, L_{j+1}\}.$$

By Fact A,  $\text{rank}(C_J) = 2$ .

Allowed output positions are those  $i$  with  $i \notin \{j - 1, j, j + 1\}$ . Since  $n \geq 5$ , such  $i$  exist. Taking any such  $i$ , the vectors  $L_{j-1}, L_{j+1}, L_i$  are three distinct  $L$  vectors; by Fact A they are linearly independent. Therefore  $\text{rank}(C_J + O_J) = 3$ , and  $d_J = 3 - 2 = 1$ . There are  $n$  such  $J$ 's, so the total contribution from all first-order Fourier modes is  $n$ .

**Case 3:  $|J| \geq 2$ .** We show that in this case  $d_J = 0$ .

If  $O_J = 0$  then clearly  $d_J = 0$ . So we may assume  $O_J \neq 0$ ; then the complement of

$J$  contains three consecutive positions.

If the circle constraints already involve at least three distinct  $L_i$ , then by Fact A  $\text{rank}(C_J) = 3$ . Since  $K_n^3$  is only three-dimensional, adding  $O_J$  cannot increase the rank, so  $d_J = 0$ .

The only remaining possibility is that  $J$  is a cyclic interval. Write

$$J = \{j, j+1, \dots, j+k-1\}, \quad k \geq 2.$$

Because  $O_J \neq 0$ , the complement contains three consecutive positions, hence  $k \leq n-3$ .

In this case  $C_J$  contains at least the three vectors

$$L_{j-1}, \quad L_{j+k}, \quad \Theta_j.$$

Here  $L_{j-1}$  and  $L_{j+k}$  come from the circle constraints at the ends of the interval, and  $\Theta_j$  comes from the point constraint because  $j, j+1 \in J$ .

By Fact C,  $\Theta_j, L_{j-1}, L_{j+k}$  are linearly independent. Hence  $\text{rank}(C_J) = 3$  and consequently  $d_J = 0$ .

Thus for  $|J| \geq 2$  we always have  $d_J = 0$ .

### Dimension of the Intersection

From the three cases,

$$d_\emptyset = 3, \quad d_J = 1 \text{ if } |J| = 1, \quad d_J = 0 \text{ if } |J| \geq 2.$$

Therefore

$$\dim(\text{Row}(V) \cap \text{Row}(S)) = \sum_{J \subseteq \{1, \dots, n\}} d_J = 3 + n.$$

Hence

$$\dim(\text{Row}(V) \cap \text{Row}(S)) = n + 3.$$

### A Natural Basis for the Intersection

We now write down a natural basis for the intersection.

**Circle row vectors.** Fix a star position  $i$  and fix the values of all coordinates except  $i-1, i, i+1$ :

$$t \in \mathbb{F}_2^{\{1, \dots, n\} \setminus \{i-1, i, i+1\}}.$$

Denote the corresponding circle row vector of  $S$  by  $s_{i,t}$ . It takes the value 1 on the four edges of that circle and 0 elsewhere. Thus

$$\text{Row}(S) = \text{span}\{s_{i,t}\}.$$

**Three basis vectors from the constant Fourier mode.** Take the standard basis of  $K_n^3$ :

$$e_{11} = (1, 0, 0), \quad e_{12} = (0, 1, 0), \quad e_{21} = (0, 0, 1).$$

For each  $\delta \in \{11, 12, 21\}$ , define

$$B_\delta^{(0)} = \sum_{i,t} (L_i)_\delta s_{i,t}.$$

In coordinate language,  $B_\delta^{(0)}(l, i, u) = (L_i)_\delta$  on every edge, and  $B_\delta^{(0)}(p) = 0$  for all points  $p \in \mathfrak{P}$ . These three vectors correspond to the constant functions  $\lambda(a) = \frac{1}{2}e_\delta$  on the faces, hence they belong to  $\text{Row}(V)$ . Moreover, they are clearly linear combinations of the circle row vectors of  $S$ , so they also lie in  $\text{Row}(S)$ .

**$n$  basis vectors from the first-order Fourier modes.** For each  $j \in \mathbb{Z}/n\mathbb{Z}$ , set

$$q_j = L_{j-1} \times L_{j+1}.$$

Because  $L_{j-1}$  and  $L_{j+1}$  are linearly independent,  $q_j \neq 0$  and  $L_{j-1} \cdot q_j = 0$ ,  $L_{j+1} \cdot q_j = 0$ .

Define

$$B^{(j)} = \sum_{\substack{i,t \\ j \notin \{i-1, i, i+1\}}} (L_i \cdot q_j) (-1)^{t_j} s_{i,t}.$$

In edge coordinates,

$$B^{(j)}(l, i, u) = \begin{cases} (L_i \cdot q_j) (-1)^{u_j}, & j \notin \{i-1, i, i+1\}, \\ 0, & j \in \{i-1, i, i+1\}. \end{cases}$$

and  $B^{(j)}(p) = 0$  for all points  $p$ .

This vector corresponds to the first-order Walsh mode  $\lambda(a) = \frac{1}{2}q_j(-1)^{a_j}$  on the faces. The choice  $q_j = L_{j-1} \times L_{j+1}$  is forced by the fact that the first-order mode  $(-1)^{a_j}$  affects the circle-constant conditions at stars  $j-1$  and  $j+1$ , so we need  $L_{j-1} \cdot q_j = 0$ ,  $L_{j+1} \cdot q_j = 0$ . The cross product exactly gives a non-zero direction orthogonal to both of these vectors.

Thus a natural basis for the intersection is

$$\mathcal{B} = \{B_{11}^{(0)}, B_{12}^{(0)}, B_{21}^{(0)}, B^{(1)}, \dots, B^{(n)}\}.$$

where

$$B_{\delta}^{(0)} = \sum_{i,t} (L_i)_{\delta} s_{i,t}, \quad \delta \in \{11, 12, 21\},$$

and

$$B^{(j)} = \sum_{\substack{i,t \\ j \notin \{i-1, i, i+1\}}} (L_i \cdot (L_{j-1} \times L_{j+1})) (-1)^{t_j} s_{i,t}.$$

This basis contains  $3 + n$  elements.

**Spanning and linear independence.** By the Walsh–Fourier decomposition, any row combination can be written as

$$\lambda(a) = \sum_{J \subseteq \{1, \dots, n\}} v_J (-1)^{\sum_{r \in J} a_r}.$$

The intersection conditions decompose mode by mode. The previous calculations show:

- $J = \emptyset$  contributes 3 dimensions;
- $J = \{j\}$  each contributes 1 dimension;
- $|J| \geq 2$  contribute 0 dimensions.

Hence the intersection contains only constant and first-order Fourier modes. The vectors  $B_{11}^{(0)}, B_{12}^{(0)}, B_{21}^{(0)}$  constructed above span the constant mode part, and the vectors  $B^{(j)}$  ( $j = 1, \dots, n$ ) span the first-order part.

Moreover, different Walsh–Fourier modes are independent; they cannot cancel each other. Within the constant mode the three vectors are linearly independent because  $\text{span}\{L_i : i \in \mathbb{Z}/n\mathbb{Z}\} = K_n^3$ . Therefore these  $n + 3$  vectors are linearly independent.

We conclude

$$\text{Row}(V) \cap \text{Row}(S) = \text{span}\{B_{11}^{(0)}, B_{12}^{(0)}, B_{21}^{(0)}, B^{(1)}, \dots, B^{(n)}\}.$$

In particular,

$$\dim(\text{Row}(V) \cap \text{Row}(S)) = n + 3.$$

## 4.6 Relation between Homological Independence and Gradient Independence

**Theorem 4.9:** Let  $W := \text{Row}_{\mathbb{R}}(V) \cap \text{Row}_{\mathbb{R}}(S)$ . From the Walsh-Hadamard decomposition we have  $W = W_0 \oplus W_1$ , with  $\dim W_0 = 3$ ,  $\dim W_1 = n$ .

More precisely,  $W_1 = \bigoplus_{j \in \mathbb{Z}/n} W_j$ , each  $W_j$  being one-dimensional and corresponding to the Fourier character  $\chi_j(a) = (-1)^{aj}$ .

Define the real coefficient version of the homological independence space of systoles, the kernel of Hurwitz map  $\tilde{h} \otimes \mathbb{R}$ :

$$U_{\mathbb{R}}^{\text{ind}} := \text{span}_{\mathbb{R}}\{\tilde{d}x_I : I \subset \mathbb{Z}/n \text{ is a non-empty independent set}\}.$$

Then we have

$$W_0 \subset U_{\mathbb{R}}^{\text{ind}}.$$

However, in general  $W \cap U_{\mathbb{R}}^{\text{ind}} \neq W_0$ . In fact,

$$W \cap U_{\mathbb{R}}^{\text{ind}} = W_0 \oplus W_1^{\text{sym}},$$

where  $W_1^{\text{sym}} \subset W_1$  is one-dimensional. Consequently,

$$\dim(W \cap U_{\mathbb{R}}^{\text{ind}}) = 4.$$

Thus a generic element of  $W_1$  does not belong to  $U_{\mathbb{R}}^{\text{ind}}$ , but there is a special one-dimensional subspace of  $W_1$  that does.

**Proof:** We first give an explicit description of  $W_0$ . Recall

$$L_i = ((F_{(l,i)})_{11}, (F_{(l,i)})_{12}, (F_{(l,i)})_{21}) \in \mathbb{R}^3.$$

A zero-order Fourier mode corresponds to a constant vector  $v(a) \equiv q \in \mathbb{R}^3$ . Its value on an edge of type  $i$  is  $2L_i \cdot q$ , while all point coordinates vanish because  $\sum_{a_i, a_{i+1}} (-1)^{a_i + a_{i+1}} = 0$ .

Let  $D_i := \tilde{d}x_i$ . Since  $dx_i$  equals 1 on every edge of direction  $i$  and 0 otherwise,  $D_i$  is precisely the sum of all  $S$ -circles of direction  $i$ . Because  $\{i\}$  is an independent set in the cycle graph  $C_n$ , we have  $D_i \in U_{\mathbb{R}}^{\text{ind}}$ . Hence

$$W_0 = \left\{ \sum_{i \in \mathbb{Z}/n} 2(L_i \cdot q) D_i : q \in \mathbb{R}^3 \right\},$$

and therefore  $W_0 \subset U_{\mathbb{R}}^{\text{ind}}$ .

**Differential forms description of  $U_{\mathbb{R}}^{\text{ind}}$ .** Any  $u \in U_{\mathbb{R}}^{\text{ind}}$  can be written as  $u = dP$  with

$$P = \sum_{I \text{ independent}} c_I x_I, \quad c_I \in \mathbb{R}.$$

Thus the component on an edge of direction  $m$  is  $u_m = \partial_m P$ . Because  $P$  consists of monomials whose supports are independent sets,  $u_m$  does not depend on  $x_{m-1}, x_{m+1}$ ; this is exactly the  $S$ -circle condition.

Moreover, any  $u = dP$  satisfies the mixed partial compatibility condition

$$\partial_r u_m = \partial_m u_r,$$

since  $\partial_r u_m = \partial_r \partial_m P = \partial_m \partial_r P = \partial_m u_r$ . This condition will restrict which elements of  $W_1$  can lie in  $U_{\mathbb{R}}^{\text{ind}}$ .

**Form of the first-order Fourier part  $W_1$ .** Fix  $j \in \mathbb{Z}/n$ . An element of  $W_j$  is given by a vector  $q_j \in \mathbb{R}^3$  satisfying

$$L_{j-1} \cdot q_j = 0, \quad L_{j+1} \cdot q_j = 0,$$

hence  $W_j$  is one-dimensional. Its component on an edge of direction  $m$  is

$$y_m^{(j)} = \begin{cases} 2(L_m \cdot q_j) \chi_j, & m \notin \{j-1, j, j+1\}, \\ 0, & m \in \{j-1, j, j+1\}. \end{cases}$$

A general element of  $W_1$  is  $y = \sum_{j \in \mathbb{Z}/n} y^{(j)}$ , i.e.

$$y_m = \sum_{\substack{j \in \mathbb{Z}/n \\ j \notin \{m-1, m, m+1\}}} 2(L_m \cdot q_j) \chi_j,$$

where  $\chi_j = 1 - 2x_j$ .

**Restriction from the mixed partial condition.** If  $y \in W_1 \cap U_{\mathbb{R}}^{\text{ind}}$ , then  $\partial_r y_m = \partial_m y_r$ .

For non-adjacent  $m, r$  one computes

$$\partial_r y_m = -4(L_m \cdot q_r), \quad \partial_m y_r = -4(L_r \cdot q_m),$$

so we must have

$$\boxed{L_m \cdot q_r = L_r \cdot q_m \quad \text{for all non-adjacent } m, r.}$$

Now use Fact A: under an invertible linear transformation the vectors  $L_i$  become

$$(1, \cos(2\pi i/n), \sin(2\pi i/n)).$$

For fixed  $j$ , the one-dimensional subspace determined by  $L_{j-1} \cdot q_j = L_{j+1} \cdot q_j = 0$  can be expressed as

$$L_m \cdot q_j = \lambda_j \left( \cos \frac{2\pi(m-j)}{n} - \cos \frac{2\pi}{n} \right), \quad \lambda_j \in \mathbb{R}.$$

Substituting into  $L_m \cdot q_r = L_r \cdot q_m$  yields

$$\lambda_r \left( \cos \frac{2\pi(m-r)}{n} - \cos \frac{2\pi}{n} \right) = \lambda_m \left( \cos \frac{2\pi(r-m)}{n} - \cos \frac{2\pi}{n} \right).$$

Because  $\cos \frac{2\pi(m-r)}{n} = \cos \frac{2\pi(r-m)}{n}$  and for non-adjacent  $m, r$  the factor  $\cos \frac{2\pi(m-r)}{n} - \cos \frac{2\pi}{n} \neq 0$ , we obtain  $\lambda_m = \lambda_r$ . The graph of non-adjacent pairs is connected for  $n \geq 5$ , hence all  $\lambda_j$  are equal. Consequently,

$$\dim(W_1 \cap U_{\mathbb{R}}^{\text{ind}}) \leq 1.$$

**Existence of this one-dimensional subspace.** Take all  $\lambda_j = \lambda$  equal. Define

$$h_{mj} = \cos \frac{2\pi(m-j)}{n} - \cos \frac{2\pi}{n}.$$

Set

$$y_m = \sum_{\substack{j \in \mathbb{Z}/n \\ j \notin \{m-1, m, m+1\}}} \lambda h_{mj} \chi_j.$$

Because  $h_{mj} = h_{jm}$ , this  $y$  can be written as the differential of an independent-set polynomial. Explicitly, let

$$P = \sum_{m \in \mathbb{Z}/n} A x_m + 2\lambda \sum_{\substack{m < j \\ m, j \text{ non-adjacent}}} h_{mj} x_m x_j,$$

where

$$A = \lambda \sum_{\substack{j \in \mathbb{Z}/n \\ j \notin \{m-1, m, m+1\}}} h_{mj}$$

is independent of  $m$  by cyclic symmetry. Then

$$\partial_m P = A + 2\lambda \sum_{\substack{j \in \mathbb{Z}/n \\ j \notin \{m-1, m, m+1\}}} h_{mj} x_j.$$

Since  $\chi_j = 1 - 2x_j$ , we have  $\lambda h_{mj} \chi_j = \lambda h_{mj} - 2\lambda h_{mj} x_j$ . Hence

$$\partial_m P = \sum_{\substack{j \in \mathbb{Z}/n \\ j \notin \{m-1, m, m+1\}}} \lambda h_{mj} \chi_j = y_m.$$

Thus  $y = dP$ . The polynomial  $P$  contains only linear terms  $x_m$  and non-adjacent quadratic terms  $x_m x_j$ , whose supports are independent sets. Therefore  $y \in U_{\mathbb{R}}^{\text{ind}}$ . Hence

$$\dim(W_1 \cap U_{\mathbb{R}}^{\text{ind}}) = 1.$$

Denote this one-dimensional space by  $W_1^{\text{sym}}$ .

We have  $W = W_0 \oplus W_1$  with  $W_0 \subset U_{\mathbb{R}}^{\text{ind}}$  and

$$W_1 \cap U_{\mathbb{R}}^{\text{ind}} = W_1^{\text{sym}}, \quad \dim W_1^{\text{sym}} = 1.$$

Therefore

$$W \cap U_{\mathbb{R}}^{\text{ind}} = W_0 \oplus W_1^{\text{sym}}.$$

In particular,

$$\dim(W \cap U_{\mathbb{R}}^{\text{ind}}) = 4.$$

■

## CHAPTER 5 DEGENERACY SURFACES IN QUANTUM PHYSICS

### 5.1 Quantum Theory

Condensed matter physics studies the collective behavior of systems composed of a very large number of particles, such as electrons and ions in crystalline solids. A central objective is to understand the energy spectrum, quantum states, and elementary excitations of such systems. The basic mathematical language for this purpose is quantum mechanics, in which the physical state space is modeled by a Hilbert space and the dynamics are determined by a Hamiltonian operator<sup>[71]</sup>.

The Hamiltonian, originating in classical mechanics and generalized in quantum theory, represents the total energy of a physical system. In quantum mechanics, the Hamiltonian  $\hat{H}$  is an operator acting on a Hilbert space, and the time evolution of the system is governed by the Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = \hat{H} |\psi\rangle,$$

where  $|\psi\rangle$  denotes the wave function, or more generally the quantum state, of the system. The Hamiltonian determines both the time evolution of the system and its allowed energy levels. In condensed matter systems, it typically contains the kinetic energies of electrons and ions together with their mutual interactions, including electron-electron, electron-ion, and ion-ion interactions<sup>[32,71]</sup>.

#### 5.1.1 Energy Bands

For crystalline solids, translational symmetry plays a fundamental role. The periodic structure of a crystal lattice implies that the single-particle Hamiltonian is invariant under lattice translations. Bloch's theorem states that, in a periodic potential, the wave function of an electron can be written in the form

$$\psi_{n,\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n,\mathbf{k}}(\mathbf{r}),$$

where  $u_{n,\mathbf{k}}(\mathbf{r})$  has the same periodicity as the crystal lattice<sup>[5,15,49]</sup>. Here  $\mathbf{k}$  is the crystal wavevector, which lives in reciprocal space and labels Bloch states.

Because of lattice periodicity, it is sufficient to study the system within the first Brill-

loun zone, which is a fundamental domain of the reciprocal lattice<sup>[5,49]</sup>. After Bloch decomposition, the Hamiltonian can be represented as a family of matrices  $H(\mathbf{k})$ , called Bloch Hamiltonians. For each  $\mathbf{k}$ , one solves the eigenvalue problem

$$H(\mathbf{k})|u_n(\mathbf{k})\rangle = E_n(\mathbf{k})|u_n(\mathbf{k})\rangle.$$

The eigenvalues  $E_n(\mathbf{k})$ , as functions of  $\mathbf{k}$ , form the *energy bands* of the system. Thus, the band structure describes how the electron energy varies with crystal momentum.

The separation between neighboring bands may open *band gaps*, which are crucial for distinguishing metals, semiconductors, and insulators. The electronic properties of a material are largely determined by the position of the Fermi level relative to the energy bands: if the Fermi level crosses a band, the system typically behaves as a metal, whereas if it lies inside a band gap, the system may behave as an insulator or a semiconductor<sup>[5,49]</sup>. Moreover, the dispersion relation  $E_n(\mathbf{k})$  also determines important dynamical quantities. For example, its gradient gives the group velocity of a wave packet,

$$\mathbf{v}_n(\mathbf{k}) = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_n(\mathbf{k}),$$

while its local curvature is related to the effective mass of quasiparticles<sup>[5,49]</sup>.

In modern condensed matter theory, one is often interested not only in the eigenvalues  $E_n(\mathbf{k})$ , but also in the eigenstates  $|u_n(\mathbf{k})\rangle$ . These may carry geometric and topological information, such as Berry phases and topological invariants, which play an essential role in the study of topological materials<sup>[5,71]</sup>. In practice, the analysis of a crystalline system usually begins with the construction of an effective Hamiltonian in momentum space, and the solution of its eigenvalue problem yields the corresponding band structure.

### 5.1.2 Hermiticity and Pseudo-Hermitian Quantum Mechanics

When only finitely many bands are relevant, the effective Hamiltonian at each  $\mathbf{k}$ -point is represented by a finite-dimensional matrix. In conventional closed quantum systems, this matrix is required to be Hermitian:

$$H(\mathbf{k}) = H^\dagger(\mathbf{k}), \quad H_{ij}(\mathbf{k}) = \overline{H_{ji}(\mathbf{k})}.$$

Hermiticity guarantees real expectation values of observables and unitary time evolution. In the finite-dimensional case, it also implies that the eigenvalues are real and that the eigenvectors can be chosen to form a complete orthonormal basis<sup>[32,71]</sup>.

However, many realistic physical systems are effectively open and may exchange energy or particles with their environment. This naturally leads to non-Hermitian de-

scriptions, which arise, for instance, in optical and photonic systems with gain and loss, in dissipative systems, and in effective descriptions of resonances<sup>[12,57]</sup>. In general, a non-Hermitian Hamiltonian may have complex eigenvalues. Physically, the real part often describes energy or oscillation frequency, while the imaginary part is related to amplification, attenuation, or finite lifetime<sup>[36,57]</sup>.

Unlike the Hermitian case, non-Hermitian Hamiltonians generally require one to distinguish between right and left eigenvectors:

$$H|\psi_n^R\rangle = E_n|\psi_n^R\rangle, \quad H^\dagger|\psi_n^L\rangle = \overline{E_n}|\psi_n^L\rangle.$$

These vectors typically form a biorthogonal system rather than an orthonormal basis, and this biorthogonal structure is fundamental in non-Hermitian spectral theory<sup>[57,59]</sup>. Another characteristic feature of non-Hermitian systems is the existence of *exceptional points*, at which both eigenvalues and eigenvectors coalesce. At such points the Hamiltonian becomes non-diagonalizable and develops Jordan block structure, leading to phenomena with no Hermitian analogue<sup>[13,36]</sup>.

A particularly important class of non-Hermitian Hamiltonians is provided by *pseudo-Hermitian* operators. A Hamiltonian  $H$  is called  $\eta$ -pseudo-Hermitian if there exists an invertible Hermitian operator  $\eta$  such that

$$\eta H \eta^{-1} = H^\dagger.$$

This means that  $H$  is Hermitian with respect to the modified inner product

$$\langle\psi, \phi\rangle_\eta := \langle\psi|\eta|\phi\rangle.$$

Pseudo-Hermiticity does not automatically imply that the whole spectrum is real; more generally, it implies a complex-conjugation symmetry of the spectrum, and under suitable additional conditions the spectrum may become entirely real<sup>[58-59]</sup>. This framework is closely related to  $\mathcal{PT}$ -symmetric quantum mechanics, where a non-Hermitian Hamiltonian invariant under the combined action of parity and time reversal may exhibit a wholly real spectrum in the unbroken phase, while complex-conjugate pairs of eigenvalues appear after symmetry breaking<sup>[12,59]</sup>.

In non-Hermitian band theory, several new phenomena arise that have no direct counterpart in ordinary Hermitian systems. One of the most notable is the *non-Hermitian skin effect*, in which a macroscopic number of eigenstates become exponentially localized near the boundary under open boundary conditions. This reveals a striking sensitivity of non-Hermitian spectra to boundary conditions and motivates the use of generalized Brillouin

zones and non-Bloch band theory<sup>[13,77]</sup>.

For the particular choice

$$\eta = \text{diag}(-1, 1, 1),$$

the modified inner product is indefinite, with signature  $(-, +, +)$ . This is exactly the signature of the Minkowski metric in  $2 + 1$ -dimensional spacetime, so the geometry induced by  $\eta$  is naturally analogous to that of  $2 + 1$ -dimensional Minkowski space. Hence, the underlying space is an indefinite inner-product space rather than a standard Hilbert space.

## 5.2 Ruled Surfaces

### 5.2.1 Classification and Properties of Ruled Surfaces

**Definition 5.1:** A surface formed by the movement of a straight line along a curve in 3-dimensional Euclidean space is called a **ruled surface**. A parametric vector representation of a ruled surface can be written in the following form:

$$\mathbf{r}(u, v) = \mathbf{c}(u) + v \cdot \mathbf{l}(u)$$

both  $\mathbf{c}(u)$  and  $\mathbf{l}(u)$  should be differentiable and  $\mathbf{l}(u) \neq \mathbf{0}$  for any  $u$ .

Straight lines belonging to a ruled surface are called rectilinear generatrices or rectilinear generators or rulings. A curve intersecting all rulings of the surface is called a directrix curve or a director curve or a directing curve.

**Definition 5.2 (tangent developable surface):** A **tangent developable surface** is a particular kind of developable surface obtained from a curve  $\mathbf{c}$  in Euclidean space  $\mathbb{R}^3$  and swept out by its tangent lines  $\mathbf{c}'(u)$  as  $\mathbf{l}(u)$ . That means any tangent developable surface is described by a parametric representation:

$$\mathbf{r}(u, v) = \mathbf{c}(u) + \mathbf{c}'(u) \cdot v$$

**Example 5.1:** As an example, consider the tangent developable surface with a helix as its directing curve  $\mathbf{c}(u)$ . Figure 5-1 shows its image. The parametric equation for the helix is:

$$\begin{cases} x = a \cos u \\ y = a \sin u \\ z = \frac{h}{2\pi} u \end{cases}$$

The parameter  $a$  is the constant radius of the circular component of the helix, defining

the distance from the helix to its central axis, the  $z$ -axis in this case. The parameter  $h$  represents the pitch of the helix, which is the linear distance the helix advances along the  $z$ -axis in one full revolution.

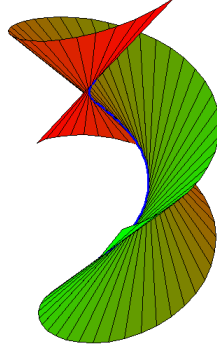


Figure 5-1 The tangent developable of a helix, cited from Wikipedia: Tangent developable surface

**Definition 5.3 (developable surface):** The **developable surface** is a surface in 3-dimensional Euclidean space with **zero** Gaussian curvature.

**Theorem 5.1 (Euler 1772):** Any tangent developable surface is a developable surface. And that every developable surface is of one of generalized cones and the cylinders and tangent developable surface.

### 5.2.2 Ruledness of Discriminant Surfaces

#### Prototype: General cubic polynomial

First of all, we will investigate the discriminant surface of a generic degree-3 polynomial. The equation is  $f(\lambda) = \lambda^3 + x\lambda^2 + y\lambda + z = 0$ , where  $x, y, z$  are parameters.  $f(\lambda) = 0$  has a zero with multiplicity at least 2, iff  $(f, f') \neq 1$ . The discriminant can be calculated from the resultant of  $f$  and  $f'$ ,

$$\Delta(x, y, z) = x^2y^2 - 4x^3z - 4y^3 + 18xyz - 27z^2.$$

Now we will show that the discriminant surface  $\Delta = 0$  is a ruled surface. Suppose  $t$  is the common root of  $f$  and  $f'$ , then

$$\begin{cases} t^3 + xt^2 + yt + z = 0 \\ 3t^2 + 2xt + y = 0 \end{cases} \quad (5-1)$$

regard  $t$  as a new parameter,  $y$  and  $z$  can be expressed by  $x$  and  $t$ :

$$\begin{cases} y = -3t^2 - 2xt \\ z = 2t^3 + xt^2 \end{cases} \quad (5-2)$$

Since the discriminant surface can be parameterized in the following form,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -3t^2 \\ 2t^3 \end{bmatrix} + x \begin{bmatrix} 1 \\ -2t \\ t^2 \end{bmatrix}, \quad (5-3)$$

this form suggests that the surface is a ruled surface.

If  $f(\lambda)$  has a zero  $t$  with multiplicity 3.  $f$  can be written as  $f(\lambda) = (\lambda - t)^3 = \lambda^3 - 3t\lambda^2 + 3t^2\lambda - t^3$ . We have correspondence of coefficients

$$\begin{cases} x = -3t \\ y = 3t^2 \\ z = -t^3 \end{cases} \quad (5-4)$$

This parameterized curve is a cuspidal edge on the discriminant surface.

**Proposition 5.1:** The discriminant surface of the generic cubic polynomial is a tangent developable surface.

**Proof:** Let  $\gamma(t)$  be a parameterized smooth curve described as above triple multiplicity curve,  $\gamma(t) = (-3t, 3t^2, -t^3)$ . Then the tangent developable surface defined by  $\gamma$  is parameterized by the following map

$$(t, s) \mapsto \gamma(t) + s\gamma'(t).$$

Taking this coefficients in the cubic polynomial we can obtain the decomposition

$$f(\lambda) = \lambda^3 - (3t + 3s)\lambda^2 + (3t^2 + 6ts)\lambda - (t^3 + 3t^3s) = (\lambda - t)^2(\lambda - (t + 3s)).$$

This suggests that the tangent developable surface given by  $(t, s) \mapsto \gamma(t) + s\gamma'(t)$  is just the discriminant surface. ■

**Corollary 5.1:** The discriminant surface of generic cubic polynomial is a developable surface.

### 5.2.3 Three Energy Band Systems and Examples

Four swallowtails (sw4):

The characteristic polynomial of matrix

$$\begin{bmatrix} 1 & f_1 & f_2 \\ -f_1 & -1 & f_3 \\ -f_2 & f_3 & -1 \end{bmatrix}$$

is  $g(\lambda) = \lambda^3 + \lambda^2 + (f_1^2 + f_2^2 - f_3^2 - 1)\lambda + (f_1^2 + f_2^2 + f_3^2 + 2f_1f_2f_3 - 1)$ . The discriminant of  $g(\lambda) = 0$  is  $\Delta = 4f_1^4 - 4f_1^6 + 8f_1^2f_2^2 - 12f_1^4f_2^2 + 4f_2^4 - 12f_1^2f_2^4 - 4f_2^6 + 64f_1f_2f_3 - 72f_1^3f_2f_3 - 72f_1f_2^3f_3 + 64f_3^2 - 80f_1^2f_3^2 + 12f_1^4f_3^2 - 80f_2^2f_3^2 - 84f_1^2f_2^2f_3^2 + 12f_2^4f_3^2 - 144f_1f_2f_3^3 - 32f_3^4 - 12f_1^2f_3^4 - 12f_2^2f_3^4 + 4f_3^6$ .

If  $g(\lambda)$  have multiple roots at  $t$ , then  $(\lambda - t)|(g, g')$ . Using the result from Example 1, We have correspondence of coefficients

$$\begin{cases} x = 1 \\ y = -3t^2 - 2xt = f_1^2 + f_2^2 - f_3^2 - 1 \\ z = 2t^3 + xt^2 = f_1^2 + f_2^2 + f_3^2 + 2f_1f_2f_3 - 1 \end{cases} \quad (5-5)$$

Eliminating this system of equations, with  $t$  and  $f_3$  as parameters, we can simplify and obtain the following results:

$$\begin{cases} f_1^2 + f_2^2 = f_3^2 - 3t^2 - 2t + 1 \\ f_1f_2 = \frac{t(t+1)^2}{f_3} - f_3 \end{cases} \quad (5-6)$$

Calculate the squares of sum and difference of  $f_1$  and  $f_2$ , finally obtain the following factorizations:

$$\begin{cases} (f_1 + f_2)^2 = \frac{f_3^3 - 2f_3^2 - (3t^2 + 2t - 1)f_3 + 2t(t+1)^2}{f_3} \\ \quad = \frac{(f_3 + 2t)(f_3 - t - 1)^2}{f_3} \\ (f_1 - f_2)^2 = \frac{f_3^3 + 2f_3^2 - (3t^2 + 2t - 1)f_3 - 2t(t+1)^2}{f_3} \\ \quad = \frac{(f_3 - 2t)(f_3 + t + 1)^2}{f_3} \end{cases} \quad (5-7)$$

Let  $2 \sin^2 \theta \cdot f_3 = f_3 + 2t$  (It shows that  $t = -\frac{1}{2}f_3 \cos 2\theta$ ). However, we will discuss two extra cases where the equation may not hold.

One case is  $f_3 = 0$ , (5-7) implies that  $t = 0$  or  $t = -1$ . We only consider  $t = -1$ , since the  $t = 0$  case is contained in the next main case.  $t = -1$  leads to  $f_1 = f_2 = 0$ .

The other case is the quadratic terms on both sides of one equation of (5-7) are zero,

which shows that the left can be negative. Consider

$$f_1 + f_2 = f_3 - t - 1 = 0$$

in (5-7), then

$$4f_1^2(t+1) = (-t+1) \cdot 4(t+1)^2 \quad (5-8)$$

Thus,  $(t+1)(f_1^2 + t^2 - 1) = 0$ . When  $t+1 = 0$ , it has been discussed above. A parametrization of  $f_1$ ,  $f_2$  and  $f_3$  can be

$$\begin{cases} f_1 = \cos \psi \\ f_2 = -\cos \psi \\ f_3 = 1 + \sin \psi \end{cases} \quad (5-9)$$

In the same way, consider

$$f_1 - f_2 = f_3 + t + 1 = 0$$

in (5-7), then

$$\begin{cases} f_1 = \cos \psi \\ f_2 = \cos \psi \\ f_3 = -1 - \sin \psi \end{cases} \quad (5-10)$$

For general case,

$$\begin{cases} (f_1 + f_2)^2 = 2 \sin^2 \theta (f_3 - t - 1)^2 \\ (f_1 - f_2)^2 = 2 \cos^2 \theta (f_3 + t + 1)^2 \end{cases} \quad (5-11)$$

There are four cases for  $f_1 + f_2$  and  $f_1 - f_2$  as for the signs. For case 1,

$$\begin{cases} f_1 + f_2 = \sqrt{2} \sin \theta (f_3 - t - 1) \\ f_1 - f_2 = \sqrt{2} \cos \theta (f_3 + t + 1) \end{cases} \quad (5-12)$$

we obtain

$$\begin{aligned} f_1 &= \frac{\sqrt{2}}{2} \sin \theta (f_3 - t - 1) + \frac{\sqrt{2}}{2} \cos \theta (f_3 + t + 1) \\ &= \frac{\sqrt{2}}{2} (\sin \theta + \cos \theta) f_3 + \frac{\sqrt{2}}{2} (\cos \theta - \sin \theta) (t + 1) \\ &= \sin(\theta + \frac{\pi}{4}) f_3 + \cos(\theta + \frac{\pi}{4}) (t + 1) \\ &= \sin(\theta + \frac{\pi}{4}) f_3 + \cos(\theta + \frac{\pi}{4}) (-\frac{1}{2} \cos 2\theta f_3 + 1) \\ &= (\sin(\theta + \frac{\pi}{4}) - \frac{1}{2} \cos(\theta + \frac{\pi}{4}) \cos 2\theta) f_3 + \cos(\theta + \frac{\pi}{4}) \end{aligned}$$

$$\begin{aligned}
 &= (\sin(\theta + \frac{\pi}{4}) - \cos^2(\theta + \frac{\pi}{4}) \sin(\theta + \frac{\pi}{4}))f_3 + \cos(\theta + \frac{\pi}{4}) \\
 &= \sin^3(\theta + \frac{\pi}{4})f_3 + \cos(\theta + \frac{\pi}{4})
 \end{aligned}$$

Replace  $\theta + \frac{\pi}{4}$  by  $\varphi$ , then

$$f_1 = \sin^3 \varphi f_3 + \cos \varphi \quad (5-13)$$

For  $f_2$ ,

$$\begin{aligned}
 f_2 &= \frac{\sqrt{2}}{2} \sin \theta (f_3 - t - 1) - \frac{\sqrt{2}}{2} \cos \theta (f_3 + t + 1) \\
 &= \frac{\sqrt{2}}{2} (\sin \theta - \cos \theta) f_3 - \frac{\sqrt{2}}{2} (\cos \theta + \sin \theta) (t + 1) \\
 &= -\cos(\theta + \frac{\pi}{4}) f_3 - \sin(\theta + \frac{\pi}{4}) (t + 1) \\
 &= -\cos \varphi f_3 - \sin \varphi (-\frac{1}{2} \sin 2\varphi f_3 + 1) \\
 &= (-\cos \varphi + \sin^2 \varphi \cos \varphi) f_3 - \sin \varphi \\
 &= -\cos^3 \varphi f_3 - \sin \varphi
 \end{aligned} \quad (5-14)$$

Thus, for case 1,

$$\begin{cases} f_1 = \sin^3 \varphi f_3 + \cos \varphi \\ f_2 = -\cos^3 \varphi f_3 - \sin \varphi \end{cases} \quad (5-15)$$

**Remark 5.1:** For the other cases, they can change the solutions to  $f_1, f_2$  by the following two methods: (1) change the signs of both  $f_1$  and  $f_2$ ; (2) exchange the results of  $f_1$  and  $f_2$ .

**Remark 5.2:** Since the discriminant surface can be paramaterized in the form

$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = \begin{bmatrix} \cos \varphi \\ -\sin \varphi \\ 0 \end{bmatrix} + f_3 \begin{bmatrix} \sin^3 \varphi \\ -\cos^3 \varphi \\ 1 \end{bmatrix}, \quad (5-16)$$

it is a ruled surface.

### Stratification Structure Analysis

In a three-band system with three control parameters, the degeneracy set may form geometric objects of different dimensions in parameter space, such as surfaces, isolated lines, intersection lines, caspidal edges, or isolated points. A key distinction is whether the degeneracy is non-defective or exceptional. In a non-defective degeneracy, the eigenvalues become degenerate while the corresponding eigenstates remain linearly independent,

so the Hamiltonian is still diagonalizable. In contrast, at an exceptional degeneracy, both the eigenvalues and the eigenstates coalesce, and the Hamiltonian becomes defective. Accordingly, a nodal line (NL) is an ordinary line degeneracy with independent eigenstates, while an exceptional line (EL) and an exceptional surface (ES) consist of defective degeneracies. A non-defective intersection line (NIL) refers to an intersection line of degeneracy manifolds where the eigenstates remain linearly independent. Finally, a meeting point (MP) is a special point where different degeneracy structures meet or reconnect, and its local spectral properties depend on the specific model.

Table 5-1 Types of degeneracies in a three-band system with three parameters.

| Abbreviation | Name                            | Eigenvalues                | Eigenstates          |
|--------------|---------------------------------|----------------------------|----------------------|
| ES           | exceptional surface             | coalescing                 | coalescing           |
| NIL          | non-defective intersection line | degenerate                 | linearly independent |
| NL           | nodal line                      | degenerate                 | linearly independent |
| EL           | exceptional line                | coalescing                 | coalescing           |
| MP           | meeting point                   | multiple degeneracies meet | model-dependent      |

According to prototype, the EL parts consist of 3 multiplicity. We suppose the 3 multiply root is  $t$ , hence  $x = -3t = 1$ . And we have already set  $2t = -f_3 \cos(2\theta) = -f_3 \sin(2\varphi)$ , so  $f_3 = \frac{2}{3 \sin(2\varphi)}$ .

$$f_1 = \frac{2 \sin^3 \varphi}{3 \sin(2\varphi)} + \cos \varphi = \frac{1}{3 \cos \varphi} + \frac{2}{3} \cos \varphi$$

$$f_2 = -\frac{2 \cos^3 \varphi}{3 \sin(2\varphi)} - \sin \varphi = -\frac{1}{3 \sin \varphi} - \frac{2}{3} \sin \varphi$$

Therefore, the EL parts can be described as parameter curves:

$$\begin{cases} f_1 = \frac{1}{3 \cos \varphi} + \frac{2}{3} \cos \varphi \\ f_2 = -\frac{1}{3 \sin \varphi} - \frac{2}{3} \sin \varphi \\ f_3 = \frac{2}{3 \sin 2\varphi} \end{cases} \quad (5-17)$$

Actually, this curve is naturally divided into four parts at  $\varphi = 0, \frac{\pi}{2}, \pi, \frac{3}{4}\pi$ , since the denominator is 0. And the minimum absolution of  $f_3$  is reached by  $\varphi = \frac{\pi}{4}, \frac{3}{4}\pi, \frac{5}{4}\pi, \frac{7}{4}\pi$ .

These 4 points are

$$\left(\frac{2}{3}\sqrt{2}, -\frac{2}{3}\sqrt{2}, \frac{2}{3}\right), \left(-\frac{2}{3}\sqrt{2}, \frac{2}{3}\sqrt{2}, -\frac{2}{3}\right), \left(\frac{2}{3}\sqrt{2}, \frac{2}{3}\sqrt{2}, -\frac{2}{3}\right), \left(-\frac{2}{3}\sqrt{2}, -\frac{2}{3}\sqrt{2}, -\frac{2}{3}\right),$$

which contribute to 4 MPs.

Eigenstates of ES and EL, the 2-multiply real root is We know in these parts, there is a good parametrization in  $\varphi$  and  $f_3$ . We have assumed that  $t = -\frac{1}{2}f_3 \sin 2\varphi$  is a at least 2-multiply root. The other root is  $-1 + f_3 \sin 2\varphi$ . So the eigenvalues of the Hamiltonian matrix are

$$-\frac{1}{2}f_3 \sin 2\varphi, -\frac{1}{2}f_3 \sin 2\varphi, -1 + f_3 \sin 2\varphi,$$

and their corresponding eigenstates are

$$\begin{bmatrix} 1 \\ -\cos \varphi \\ \sin \varphi \end{bmatrix}, \begin{bmatrix} -4f_3 \cos(2\varphi) \\ 4 \sin \varphi + f_3(-5 \cos \varphi + \cos(3\varphi)) \\ -4 \cos \varphi + f_3(5 \sin \varphi + \sin(3\varphi)) \end{bmatrix}$$

It is easy to observe on the region ES, the 2 multiple roots has only 1 corresponding eigenvector.

When  $-\frac{1}{2}f_3 \sin 2\varphi = -1 + f_3 \sin 2\varphi$  i.e.  $f_3 = \frac{2}{3 \sin(2\varphi)}$  all roots merge into a 3 multiple root as we have shown. In this case, the 3 multiple root is  $-\frac{1}{3}$ . Considering the eigenstates, there are two situations on ELs and MPs regions. If  $\varphi \notin \{0, \frac{\pi}{2}, \pi, \frac{3}{4}\pi, \frac{\pi}{4}, \frac{3}{4}\pi, \frac{5}{4}\pi, \frac{7}{4}\pi\} + 2k\pi, k \in \mathbb{Z}$ , i.e. in the region ELs, then the eigenvalue  $-\frac{1}{3}$  have just 1 corresponding eigenvector

$$\begin{bmatrix} 1 \\ -\cos \varphi \\ \sin \varphi \end{bmatrix}.$$

If  $\varphi \in \{\frac{\pi}{4}, \frac{3}{4}\pi, \frac{5}{4}\pi, \frac{7}{4}\pi\} + 2k\pi, k \in \mathbb{Z}$ , i.e. at the 4 MPs, then the eigenvalue  $-\frac{1}{3}$  have 2 different corresponding eigenvectors,

$$\begin{bmatrix} -1/\sqrt{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1 \end{bmatrix}.$$

**Theorem 5.2:** The discriminant surface  $sw_4$  is a tangent developable surface.

**Proof:** EL part in (5-17) is exactly the corresponding curve  $\gamma(\varphi)$  of tangent developable surface  $sw_4$ . We can verify that  $\Delta(\gamma(\varphi) + s\gamma'(\varphi)) = 0$ . ■

We have a line  $l_\varphi$  in 3D space with coordinates  $(f_1, f_2, f_3)$ , parameterized by  $t = f_3$

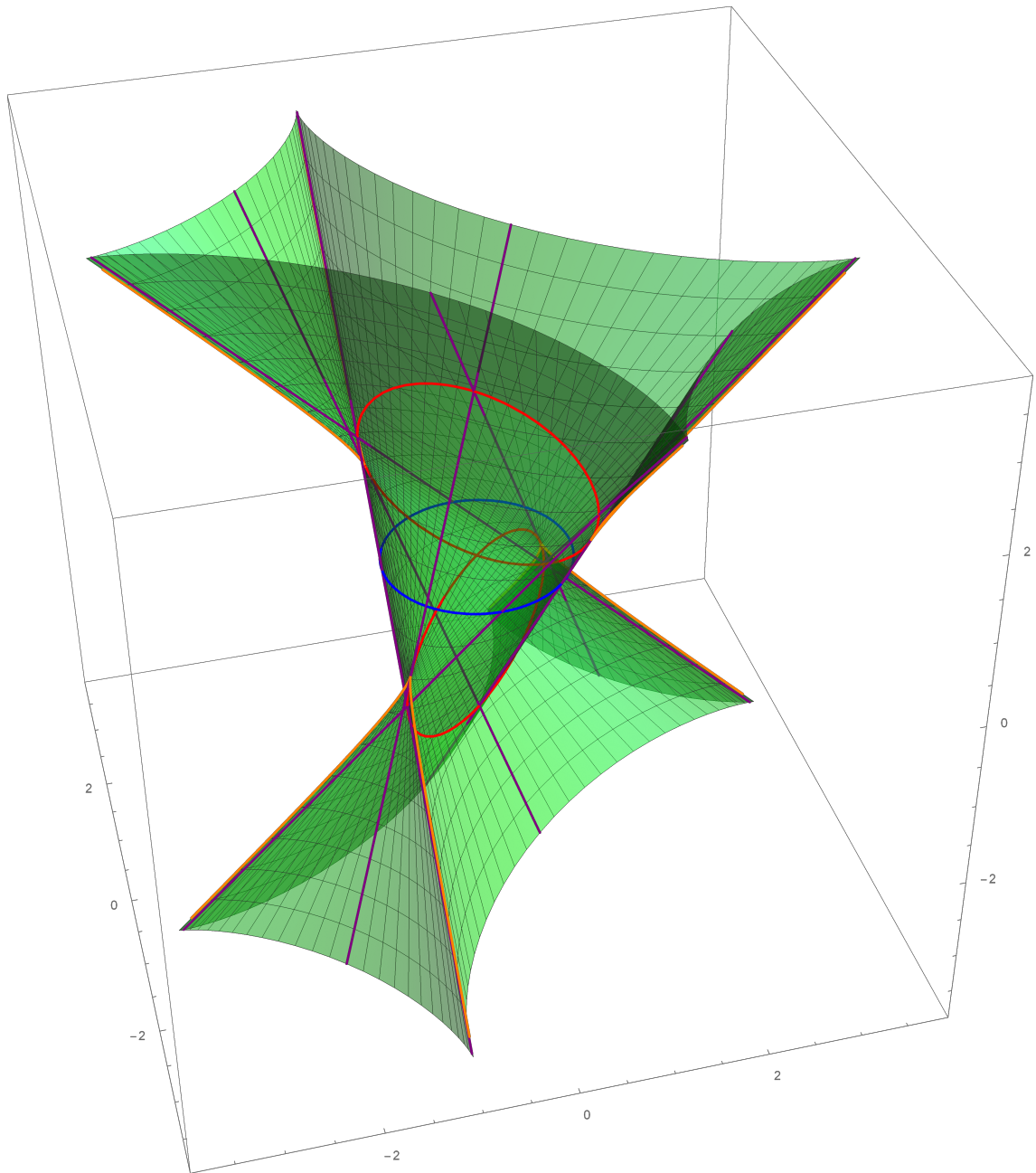


Figure 5-2 This figure illustrates a ruled algebraic surface with four swallowtail singularities and three ellipses.

and fixed  $\varphi$ :

$$f_1 = \cos \varphi + t \sin^3 \varphi, \quad f_2 = -\sin \varphi - t \cos^3 \varphi, \quad f_3 = t.$$

As  $\varphi$  varies, the family of such lines forms a ruled surface  $\Sigma$ . We want the tangent plane  $T_\varphi$  to  $\Sigma$  along the line  $l_\varphi$ .

**Proposition 5.2:** The tangent plane  $T_\varphi$  to the surface  $\Sigma$  along the line  $l_\varphi$  is given by the equation

$$-\cos \varphi f_1 + \sin \varphi f_2 + \sin \varphi \cos \varphi f_3 + 1 = 0,$$

or

$$f_3 = \frac{f_1}{\sin \varphi} - \frac{f_2}{\cos \varphi} - \frac{1}{\sin \varphi \cos \varphi}$$

**Proof:** The surface  $\Sigma$  is parametrized by  $(\varphi, t)$  as:

$$\mathbf{R}(\varphi, t) = \begin{pmatrix} \cos \varphi + t \sin^3 \varphi \\ -\sin \varphi - t \cos^3 \varphi \\ t \end{pmatrix}.$$

The two tangent vectors are:

$$\mathbf{R}_\varphi = \frac{\partial \mathbf{R}}{\partial \varphi} = \begin{pmatrix} -\sin \varphi + 3t \sin^2 \varphi \cos \varphi \\ -\cos \varphi + 3t \cos^2 \varphi \sin \varphi \\ 0 \end{pmatrix},$$

$$\mathbf{R}_t = \frac{\partial \mathbf{R}}{\partial t} = \begin{pmatrix} \sin^3 \varphi \\ -\cos^3 \varphi \\ 1 \end{pmatrix}.$$

The normal vector  $\mathbf{n} = \mathbf{R}_\varphi \times \mathbf{R}_t$  is computed as follows. Let

$$A = -\sin \varphi + 3t \sin^2 \varphi \cos \varphi, \quad B = -\cos \varphi + 3t \cos^2 \varphi \sin \varphi.$$

Then

$$\mathbf{n} = \begin{pmatrix} B \\ -A \\ -A \cos^3 \varphi - B \sin^3 \varphi \end{pmatrix} = \begin{pmatrix} -\cos \varphi + 3t \cos^2 \varphi \sin \varphi \\ \sin \varphi - 3t \sin^2 \varphi \cos \varphi \\ \sin \varphi \cos \varphi - 3t \sin^2 \varphi \cos^2 \varphi \end{pmatrix}.$$

The tangent plane at  $\mathbf{R}(\varphi, t)$  is:

$$\mathbf{n} \cdot (\mathbf{F} - \mathbf{R}(\varphi, t)) = 0,$$

where  $\mathbf{F} = (F_1, F_2, F_3)$  is a point on the plane.

This surface is developable (the tangent plane is constant along each generator  $l_\varphi$ ), so we can find the plane by taking  $t = 0$ :

$$\text{At } t = 0, \mathbf{R}(\varphi, 0) = (\cos \varphi, -\sin \varphi, 0), \mathbf{n} = (-\cos \varphi, \sin \varphi, \sin \varphi \cos \varphi).$$

The plane equation becomes:

$$-\cos \varphi (F_1 - \cos \varphi) + \sin \varphi (F_2 + \sin \varphi) + \sin \varphi \cos \varphi (F_3 - 0) = 0.$$

Simplify:

$$-\cos \varphi F_1 + \cos^2 \varphi + \sin \varphi F_2 + \sin^2 \varphi + \sin \varphi \cos \varphi F_3 = 0,$$

$$-\cos \varphi F_1 + \sin \varphi F_2 + \sin \varphi \cos \varphi F_3 + 1 = 0.$$

Renaming  $(F_1, F_2, F_3)$  back to  $(f_1, f_2, f_3)$  yields the result. One may verify that for any  $t$ , the points of  $l_\varphi$  satisfy this equation, confirming the plane contains the line. Thus

$$-\cos \varphi f_1 + \sin \varphi f_2 + \sin \varphi \cos \varphi f_3 + 1 = 0$$

is the equation of the tangent plane  $T_\varphi$  along the generator  $l_\varphi$ . ■

Substituting the equation of the tangent plane into the discriminant surface and then factoring, we isolate the double factor representing a straight line. The remaining part of the intersection of the plane and the surface is a quartic plane curve. Therefore, there is also a family of quartic curves on the discriminant surface, with the following equation.

$$\begin{aligned} & \Delta (\tan \varphi f_2 + \sin \varphi f_3 + \sec \varphi, f_2, f_3) \\ &= -\frac{1}{8} \sec^6 \varphi (f_2 + f_3 \cos^3 \varphi + \sin \varphi)^2 \\ & \times \left[ 32 (1 + 4f_2^2 + f_2^4) + 12 (15 + 4f_2^2) f_3^2 - 10f_3^4 + 24f_2f_3 (20 + 3f_3^2) \cos \varphi \right. \\ & + (-64f_2^2 + 176f_3^2 - 15f_3^4) \cos 2\varphi + 4f_2f_3 (-8 - 16f_2^2 + f_3^2) \cos 3\varphi \\ & - 2f_3^2 (2 + 24f_2^2 + 3f_3^2) \cos 4\varphi - 12f_2f_3^3 \cos 5\varphi - f_3^4 \cos 6\varphi \\ & \left. + 64f_2 (2 + 2f_2^2 + 3f_3^2) \sin \varphi + 32f_3 (3 + 9f_2^2 + 2f_3^2) \sin 2\varphi + 192f_2f_3^2 \sin 3\varphi + 32f_3^3 \sin 4\varphi \right] \end{aligned}$$

### 5.2.4 Ruledness Theorem under the Affine Diagonal Condition

**Theorem 5.3 (Ruledness):** The generic part of discriminant surface of characteristic polynomial of matrix

$$H = \begin{bmatrix} g_1(f_1, f_2, f_3) & f_1 & f_2 \\ -f_1 & g_2(f_1, f_2, f_3) & f_3 \\ -f_2 & f_3 & g_3(f_1, f_2, f_3) \end{bmatrix}$$

is ruled, where  $g_1, g_2, g_3$  are linear functions of parameters  $f_1, f_2, f_3$ , i.e.  $g_i = a_{i,1}f_1 + a_{i,2}f_2 + a_{i,3}f_3 + b_i$ . Furthermore, the discriminant surface of  $H$  is tangent developable and the associated directing curve is the curve of three multiplicity.

**Theorem 5.4:** When  $g_1, g_2, g_3$  are constants numbers, the ruledness theorem is right in this special case.

**Proof:** Without loss of generality, we may assume  $g_1 + g_2 + g_3 = 0$ . Let  $g_1, g_2, g_3$  be  $-2b, b - k, b + k$ , respectively. Here,  $b$  and  $k$  are variables. Thus, the characteristic polynomial is  $g(\lambda) = \lambda^3 + (-3b^2 + f_1^2 + f_2^2 - (f_3^2 + k^2))\lambda + 2b^3 - b(f_1^2 + f_2^2) - 2b(f_3^2 + k^2) - k(f_1^2 - f_2^2) + 2f_1f_2f_3$ . It leads that

$$\begin{cases} f_1^2 + f_2^2 = 3b^2 - 3t^2 + (f_3^2 + k^2) \\ k(f_1^2 - f_2^2) = -b^3 + 3bt^2 - 2t^3 - 3b(f_3^2 + k^2) + 2f_1f_2f_3, \end{cases} \quad (5-18)$$

where  $t$  is the repeated root in the characteristic polynomial. Thus, let  $f_3$  be  $k \sinh \eta$ , then for  $k > 0$  (for  $k = 0$ , it has been discussed as sw4),

$$\begin{cases} 2k(\cosh \frac{\eta}{2} f_1 - \sinh \frac{\eta}{2} f_2)^2 = (k \cosh \eta - (b - t))^2 (k \cosh \eta - (b + 2t)) \\ 2k(\sinh \frac{\eta}{2} f_1 + \cosh \frac{\eta}{2} f_2)^2 = (k \cosh \eta + (b - t))^2 (k \cosh \eta + (b + 2t)) \end{cases} \quad (5-19)$$

Consider special cases. when  $\cosh \frac{\eta}{2} f_1 - \sinh \frac{\eta}{2} f_2 = k \cosh \eta - (b - t) = 0$ , we have

$$2k((\tanh^2 \frac{\eta}{2} + 1) \cosh \frac{\eta}{2} f_2)^2 = 4k^2 \cosh^2 \eta (3b - k \cosh \eta). \quad (5-20)$$

Then,

$$f_2^2 = (3b - k \cosh \eta) \cdot (k + k \cosh \eta). \quad (5-21)$$

It implies that

$$\begin{cases} f_1 = \pm \sqrt{(3b - k \cosh \eta) \cdot (-k + k \cosh \eta)} \\ f_2 = \pm \sqrt{(3b - k \cosh \eta) \cdot (k + k \cosh \eta)} \\ f_3 = k \sinh \eta, \end{cases} \quad (5-22)$$

where  $3b \geq k$  and  $f_1 f_2 f_3 \geq 0$ .

In the same way, when  $\sinh \frac{\eta}{2} f_1 + \cosh \frac{\eta}{2} f_2 = k \cosh \eta + (b - t) = 0$ , we have

$$2k((\tanh^2 \frac{\eta}{2} + 1) \cosh \frac{\eta}{2} f_1)^2 = 4k^2 \cosh^2 \eta (-3b - k \cosh \eta). \quad (5-23)$$

Then,

$$f_1^2 = (-3b - k \cosh \eta) \cdot (k + k \cosh \eta). \quad (5-24)$$

It implies that

$$\begin{cases} f_2 = \pm \sqrt{(-3b - k \cosh \eta) \cdot (-k + k \cosh \eta)} \\ f_1 = \pm \sqrt{(-3b - k \cosh \eta) \cdot (k + k \cosh \eta)} \\ f_3 = k \sinh \eta, \end{cases} \quad (5-25)$$

where  $-3b \geq k$  and  $f_1 f_2 f_3 \leq 0$ .

For general cases, let  $k \cosh \eta - (b + 2t)$  be  $2k \cosh \eta \cdot \sin^2 \theta$ , then  $t = \frac{1}{2}(k \cosh \eta \cos(2\theta) - b)$ . Thus,

$$\begin{cases} \cosh \frac{\eta}{2} f_1 - \sinh \frac{\eta}{2} f_2 = (k \cosh \eta - (b - t)) \sqrt{\cosh \eta} \cdot \sin \theta \\ \sinh \frac{\eta}{2} f_1 + \cosh \frac{\eta}{2} f_2 = (k \cosh \eta + (b - t)) \sqrt{\cosh \eta} \cdot \cos \theta \end{cases} \quad (5-26)$$

Thus, we give equations on  $f_1$ ,  $f_2$  and  $f_3$  as follows.

$$\begin{cases} f_1 = k \sqrt{\cosh \eta} (\sin \theta \cosh \frac{\eta}{2} + \cos \theta \sinh \frac{\eta}{2}) - \frac{3b - k \cosh \eta \cos(2\theta)}{2\sqrt{\cosh \eta}} (\sin \theta \cosh \frac{\eta}{2} - \cos \theta \sinh \frac{\eta}{2}) \\ f_2 = k \sqrt{\cosh \eta} (\cos \theta \cosh \frac{\eta}{2} - \sin \theta \sinh \frac{\eta}{2}) + \frac{3b - k \cosh \eta \cos(2\theta)}{2\sqrt{\cosh \eta}} (\cos \theta \cosh \frac{\eta}{2} + \sin \theta \sinh \frac{\eta}{2}) \\ f_3 = k \sinh \eta \end{cases} \quad (5-27)$$

Let  $\frac{\cosh \frac{\eta}{2}}{\sqrt{\cosh \eta}} = \cos \alpha$  and  $\frac{\sinh \frac{\eta}{2}}{\sqrt{\cosh \eta}} = \sin \alpha$ , then

$$\begin{cases} f_1 = k \sec(2\alpha) \sin(\theta + \alpha) - \frac{3b - k \sec(2\alpha) \cos(2\theta)}{2} \sin(\theta - \alpha) \\ f_2 = k \sec(2\alpha) \cos(\theta + \alpha) + \frac{3b - k \sec(2\alpha) \cos(2\theta)}{2} \cos(\theta - \alpha) \\ f_3 = k \tan(2\alpha) \end{cases} \quad (5-28)$$

Let  $\theta$  be  $\theta + \alpha$ , then

$$\begin{cases} f_1 = k \sec(2\alpha) \sin(\theta + 2\alpha) - \frac{3b - k \sec(2\alpha) \cos(2\theta + 2\alpha)}{2} \sin \theta \\ f_2 = k \sec(2\alpha) \cos(\theta + 2\alpha) + \frac{3b - k \sec(2\alpha) \cos(2\theta + 2\alpha)}{2} \cos \theta \\ f_3 = k \tan(2\alpha) \end{cases} \quad (5-29)$$

From the trigonometric addition formulas, we obtain

$$\begin{cases} f_1 = k \sin \theta + f_3 \cos \theta - \frac{3b - k \cos(2\theta) + f_3 \sin(2\theta)}{2} \sin \theta \\ f_2 = k \cos \theta - f_3 \sin \theta + \frac{3b - k \cos(2\theta) + f_3 \sin(2\theta)}{2} \cos \theta \\ f_3 = f_3 \end{cases} \quad (5-30)$$

Rearranging, we obtain

$$\begin{cases} f_1 = \frac{3b}{2} + \frac{k \sin \theta}{2} (2 + \cos(2\theta)) + f_3 \cos^3 \theta \\ f_2 = \frac{3b}{2} + \frac{k \cos \theta}{2} (2 - \cos(2\theta)) - f_3 \sin^3 \theta \\ f_3 = f_3 \end{cases} \quad (5-31)$$

Note that for  $k < 0$ ,

$$\begin{cases} -2k(\sinh \frac{\eta}{2} f_1 + \cosh \frac{\eta}{2} f_2)^2 = (-k \cosh \eta - (b - t))^2 (-k \cosh \eta - (b + 2t)) \\ -2k(\cosh \frac{\eta}{2} f_1 - \sinh \frac{\eta}{2} f_2)^2 = (-k \cosh \eta + (b - t))^2 (-k \cosh \eta + (b + 2t)) \end{cases} \quad (5-32)$$

Thus, for general cases, it is equivalent to the case  $k > 0$ . For special cases,  $k \cosh \eta$  changes its sign, it leads that

$$\begin{cases} f_1 = \pm \sqrt{(-3b + k \cosh \eta) \cdot (k - k \cosh \eta)} \\ f_2 = \pm \sqrt{(-3b + k \cosh \eta) \cdot (-k - k \cosh \eta)} \\ f_3 = k \sinh \eta, \end{cases} \quad (5-33)$$

where  $-3b \geq -k$  and  $f_1 f_2 f_3 \geq 0$  and that

$$\begin{cases} f_2 = \pm \sqrt{(3b + k \cosh \eta) \cdot (k - k \cosh \eta)} \\ f_1 = \pm \sqrt{(3b + k \cosh \eta) \cdot (-k - k \cosh \eta)} \\ f_3 = k \sinh \eta, \end{cases} \quad (5-34)$$

where  $3b \geq -k$  and  $f_1 f_2 f_3 \leq 0$ .

Define the sign function

$$\text{sign}(x) = \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{if } x = 0, \\ -1 & \text{if } x < 0, \end{cases}$$

then we have

$$\begin{cases} f_{\frac{3-\text{sign}(kb)}{2}} = \pm \text{sign}(b) \sqrt{\frac{\text{sign}(kb) \cdot 3b - k \cosh \eta}{k + k \cosh \eta}} \cdot k \sinh \eta \\ f_{\frac{3+\text{sign}(kb)}{2}} = \pm \sqrt{(k + k \cosh \eta)(\text{sign}(kb) \cdot 3b - k \cosh \eta)} \\ f_3 = k \sinh \eta \end{cases} \quad (5-35)$$

for  $k, b \neq 0$  and  $|k| \leq 3|b|$ .

Note that for special cases and for distinct  $b$  and  $k$ , the curve always can be represented by one of the above equations.

When  $f_3 = 0$ ,  $f_{\frac{3-\text{sign}(kb)}{2}} = 0$  and  $f_{\frac{3+\text{sign}(kb)}{2}} = \pm \sqrt{2k(\text{sign}(kb) \cdot 3b - k)}$ . ■

After settling the case where the Hamiltonian matrix has constant diagonal entries, we can now tackle the more general situation with linear functions on the diagonal using the method of variation of constants.

**Proof of Theorem 5.3:** Notice in formula (A) that each component  $f_i$  can be expressed as a linear combination of  $k$  and  $b$  with trigonometric coefficients, namely

$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} \sec(2\alpha)(3 \sin(\theta + \alpha) + \sin(3\theta - \alpha)) & -\frac{3}{2} \sin(\theta - \alpha) \\ \frac{1}{4} \sec(2\alpha)(3 \cos(\theta + \alpha) - \cos(3\theta - \alpha)) & \frac{3}{2} \cos(\theta - \alpha) \\ \tan(2\alpha) & 0 \end{bmatrix} \begin{bmatrix} k \\ b \end{bmatrix}. \quad (5-36)$$

Denote the above  $3 \times 2$  matrix of trigonometric functions by  $A$ .

We now introduce a variation of  $k, b$  that also contains a linear part in  $f_1, f_2, f_3$ :

$$\begin{cases} k = c_{11}f_1 + c_{12}f_2 + c_{13}f_3 + c_k, \\ b = c_{21}f_1 + c_{22}f_2 + c_{23}f_3 + c_b, \end{cases} \quad \text{or equivalently} \quad \begin{bmatrix} k \\ b \end{bmatrix} = C \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} + \begin{bmatrix} c_k \\ c_b \end{bmatrix},$$

where

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \end{bmatrix}.$$

The Hamiltonian is given in the form

$$H = \begin{bmatrix} g_1(f_1, f_2, f_3) & f_1 & f_2 \\ -f_1 & g_2(f_1, f_2, f_3) & f_3 \\ -f_2 & f_3 & g_3(f_1, f_2, f_3) \end{bmatrix}.$$

Since adding a constant multiple of the identity to  $H$  does not alter the geometry of its degeneracy surfaces, we may subtract  $\left(\frac{g_1 + g_2 + g_3}{3}\right) \cdot I$  to make the trace zero. Setting the trace to zero forces new diagonal entries  $g'_i$  satisfying  $g'_1 + g'_2 + g'_3 = 0$ . A convenient choice satisfying this condition is

$$g'_1 = -2b, \quad g'_2 = b - k, \quad g'_3 = b + k.$$

Each  $g_i$  is originally an affine function of  $f_1, f_2, f_3$ , say

$$g_i = a_{i1}f_1 + a_{i2}f_2 + a_{i3}f_3 + b_i.$$

From the relations  $g'_1 = -2b$ ,  $g'_2 = b - k$ ,  $g'_3 = b + k$  together with the expression of  $k, b$  in terms of  $f_1, f_2, f_3$ , we can determine the coefficients  $c_{ij}$ ,  $c_k$ ,  $c_b$ .

$$\begin{aligned} c_{11} &= \frac{a_{31} - a_{21}}{2}, & c_{12} &= \frac{a_{32} - a_{22}}{2}, & c_{13} &= \frac{a_{33} - a_{23}}{2}, \\ c_{21} &= -\frac{1}{2} \left( a_{11} - \frac{a_{11} + a_{21} + a_{31}}{3} \right), \\ c_{22} &= -\frac{1}{2} \left( a_{12} - \frac{a_{12} + a_{22} + a_{32}}{3} \right), \\ c_{23} &= -\frac{1}{2} \left( a_{13} - \frac{a_{13} + a_{23} + a_{33}}{3} \right), \\ c_k &= \frac{b_3 - b_2}{2}, & c_b &= -\frac{1}{2} \left( b_1 - \frac{b_1 + b_2 + b_3}{3} \right). \end{aligned}$$

Substituting the expression of  $k, b$  into (A) gives

$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = A \left( C \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} + \begin{bmatrix} c_k \\ c_b \end{bmatrix} \right) = AC \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} + A \begin{bmatrix} c_k \\ c_b \end{bmatrix}.$$

Solving this linear system yields

$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = (I - AC)^{-1} A \begin{bmatrix} c_k \\ c_b \end{bmatrix}.$$

Using the matrix identity  $(I - AC)^{-1}A = A(I - CA)^{-1}$  (valid when the inverses exist), the right-hand side can be rewritten as

$$A(I - CA)^{-1} \begin{bmatrix} c_k \\ c_b \end{bmatrix},$$

which involves inverting only the  $2 \times 2$  matrix  $I - CA$  and thereby simplifies the compu-

tation. ■

### 5.3 Conics on the Sextic Surface

Enumerative geometry is characterized by its fundamental question: “How many” geometric objects of a specific type satisfy given conditions? Any valid answer must be a natural number.

**Theorem 5.5 (27 Lines on a Smooth Cubic Surface):** Let  $S$  be a smooth cubic surface in the complex projective space  $\mathbb{P}^3$ . The surface  $S$  contains exactly 27 distinct lines. Each of these lines intersects exactly 10 of the others. The configuration of these 27 lines is independent of the specific smooth cubic surface  $S$ ; it is determined up to isomorphism by the following incidence relations: 30 points where three lines meet (trihedral points); 45 planes, each containing exactly three of the lines (trihedral planes).

**Theorem 5.6 (3264 Conics Tangent to Five Conics):** Let  $C_1, C_2, C_3, C_4, C_5$  be five smooth conics in general position in the complex projective plane  $\mathbb{P}^2$ . The number of distinct smooth conics in  $\mathbb{P}^2$  that are tangent to all five given conics is precisely 3264.

**Remark 5.3:** The restriction condition “general position” is a technical condition meaning the conics are chosen to avoid degenerate situations where the number of tangent conics would be infinite or where the intersection calculation would be incorrect due to non-transversality. And tangent means the conic meets the given conic at a point with intersection multiplicity at least 2.

**Remark 5.4:** Historical context: This result corrects an earlier, naive calculation by J. Steiner (1848), who argued that since the condition of tangency to a conic is of degree 6, the number of solutions should be  $6^5 = 7776$ . The correct number, **3264**, accounts for the contribution of degenerate conics (like double lines) in a compactified moduli space of conics, a resolution achieved in the 19th century by geometers including Chasles, de Jonquières, and Zeuthen.

From the example above, we can see that degeneracy is a crucial factor when counting conics. The properties of smooth conics and degenerate conics are often fundamentally different and must be treated separately. In the sextic algebraic surface we are studying, since it is a ruled surface, it contains infinitely many lines. Moreover, two intersecting lines can form a plane, and the pair of intersecting lines in that plane constitutes a degenerate conic. Consequently, on our discriminant surface, there are infinitely many degenerate conics, but smooth conics are extremely rare, in a special cases, we can find three.

**Proposition 5.3:** For sw4 surface give by the discriminant of characteristic polynomial of matrix

$$\begin{bmatrix} 1 & f_1 & f_2 \\ -f_1 & -1 & f_3 \\ -f_2 & f_3 & -1 \end{bmatrix},$$

there are at least 3 smooth conics on it.

**Proof:** There are three ellipses on the surface parametrized by:

$$\begin{cases} f_1 = \cos \psi \\ f_2 = -\cos \psi \\ f_3 = 1 + \sin \psi \end{cases} \quad \begin{cases} f_1 = \cos \psi \\ f_2 = \cos \psi \\ f_3 = -1 - \sin \psi \end{cases} \quad \begin{cases} f_1 = \cos \varphi \\ f_2 = \sin \varphi \\ f_3 = 0 \end{cases}$$

Among these, the first two are oblate ellipses, while the third is a perfect circle. ■

## 5.4 Generalizations and Future Work

In this chapter, we have explored the geometric properties of the discriminant surface associated with a family of Hamiltonian matrices. We established that under the affine diagonal condition, the generic part of the discriminant surface is ruled and tangent developable, with the directing curve corresponding to points of triple multiplicity. Additionally, we identified specific families of conics on the surface, including infinitely many degenerate conics formed by intersecting lines and at least three smooth conics in special cases.

Quantization of surfaces generally refers to transforming the classical geometry and symplectic structure of the moduli space of surfaces into noncommutative algebraic structures and operator representations through quantization methods, thereby constructing the corresponding Hilbert spaces and quantum invariants. In this process, the combinatorial structure of the surface provides a foundation for coordinatization and discretization, while algebraic tools such as quantum groups and representation theory are used to characterize its quantum symmetries, particularly the representation of the mapping class group on the quantum state space.

Non-Hermitian quantum topology studies topological structures in open, dissipative, gain-loss, or nonreciprocal quantum systems where the Hamiltonian is no longer Hermitian. Core issues include the classification of complex spectra via point and line gaps, Berry phases and quantum geometry defined with biorthogonal eigenstates, exceptional

points and their branch-point topology, the non-Hermitian skin effect and the resulting breakdown and reconstruction of bulk-boundary correspondence, as well as many-body non-Hermitian topology and topological responses in stochastic dynamics. From the perspective of mathematical physics, it connects to K-theory, homotopy classification, spectral theory of non-normal operators, complex analytic geometry, singularity theory, and non-Bloch band theory. Experimentally, relevant platforms include photonic lattices, cold atoms, electrical circuits, and mechanical/acoustic metamaterials. A key recent trend is to treat non-Hermiticity itself as a source of topology—rather than merely a perturbation of Hermitian topology—focusing in particular on unified frameworks linking exceptional points, the skin effect, and point-gap topology, together with their experimental controllability.

## CONCLUSION

The core innovations of this dissertation are as follows:

(1) We investigate the relationship between the closed geodesic system and the homology groups of the real moment-angle complex surfaces  $Z(C_n, (D^1, S^0))$ . By restricting the Hurewicz map to the free abelian group generated by the systoles (shortest closed geodesics), we construct a standard basis for the kernel of this map by algebraic normal form methods. The vectors in this basis are in one-to-one correspondence with the nonempty independent sets of the cycle graph  $C_n$ .

(2) Using combinatorial enumeration methods, we derive explicit formulas for the ranks of the kernel and the image of the Hurewicz map. These formulas are precisely expressed in terms of the Lucas numbers  $L(n)$ . Furthermore, by analyzing orbit counts under the action of cyclic groups, we obtain number-theoretic identities involving the Möbius function, the Euler totient function, and the Lucas numbers.

(3) We extend the problem of studying the homological relations of closed geodesics on closed surfaces to the broader context of regular maps. For the regular embeddings of the hypercube graph  $Q_n$ , we prove a formula for the kernel of Hurewicz map of closed geodesics to homology group. This formula is expressed in terms of the Pell–Lucas number  $Q(n)$ .

(4) We reveal a connection between the Morse index of the systole function and the combinatorial structure of the surface. Endowing the surface with the standard hyperbolic metric, we prove the conjecture that the Morse index of the systole function is of the form  $n \cdot 2^{n-3} - (n+3)$ . The proof proceeds by first obtaining the local parameter space and constraints from the combinatorial structure, then applying the Walsh–Hadamard transform to decompose into different frequencies, and finally counting the gradient directions that contribute.

(5) We study the geometric structures that arise as critical surfaces in non-Hermitian quantum mechanics. We prove that, under an affine parameter condition, these surfaces are algebraic ruled surfaces equipped with a flat metric.

Future directions for further research in this area include the following:

(1) It is natural to ask whether the subgroup generated by systolic geodesics has any homotopy-theoretic significance. After suspension, the systoles become spheres, and

it would be worth understanding how this phenomenon can be incorporated into the language and framework of homotopy theory.

(2) How to generalize the notion of independent sets in the context of polyhedral products. In particular, one may ask: what should the analogue of geodesics be in the case of complex moment-angle complexes?

(3) It is also natural to ask whether there exist suitable analogues of Fenchel–Nielsen coordinates in our setting. Although maximal independent sets cut the surface into certain basic pieces, they do not yield a canonical pants decomposition. Moreover, if one forcibly decomposes the surface into pairs of pants, the existing cell structure is inevitably destroyed. It would therefore be desirable to find a system of coordinates on Teichmüller space that is more natural for our setting and uses fewer parameters than the current description in terms of the edge lengths and angles of each piece.

(4) More generally, our current work only studies the shortest geodesics and their length functions. In fact, one may consider geodesics of arbitrary length, that is, the full length spectrum. Because of the nature of hyperbolic geometry, the multiplicity of geodesics of a given length is unbounded as the length increases. This suggests introducing a filtration by geodesic length and studying how the size of the kernel of the Hurewicz map changes as the length bound grows.

(5) There are also some intriguing numerical coincidences. The genus of the surface  $Z(C_n, (D^1, S^0))$  studied here coincides with the genus of the thickened surface associated with an embedding of the cube graph  $Q_{n-3}$  in three-dimensional space. It is natural to ask whether there is any deeper connection between them. Likewise, the regular embeddings corresponding to the N–S solutions  $M(n+1, -1)$  and  $M(n, 1)$  for the cube graph  $Q_{n-3}$  have the same genus. Is there a natural way to relate them? What is the distance between these two points in Teichmüller space? Is there a natural path connecting them? If such a path exists, do the surfaces along it possess any special geometric properties? Finally, how are the geodesic systems on these surfaces related to the geodesic system on the surface studied in this thesis?

(6) We can study the symmetric groups of the surfaces and the action of the tangent space of Teichmüller space. This may provide a new perspective on the relationship between the geometry of the surface and its combinatorial structure.

(7) Future research can be further developed from the perspective of algorithms and computational topology. For infinite-type surfaces with ends and their moduli spaces,

one may consider using combinatorial decompositions of surfaces, such as ideal regular polygon decompositions, pants decompositions, or graph-model representations, to transform the complex topology of surfaces into computable discrete data. At the same time, by combining such decompositions with parametrized coordinates, such as Fenchel-Nielsen coordinates, length-spectrum coordinates, or other coordinate systems, one may construct a discrete representation framework for moduli spaces, thereby enabling numerical sampling, simulation of geometric structures, and computational classification of different types of surfaces. This direction may not only help reveal the local and global structures of moduli spaces of infinite-type surfaces, but also provide more systematic numerical evidence for related conjectures.

(8) While retaining the existing numerical verification results, future work may also further explore possible routes toward rigorous proof. In particular, two perspectives seem especially promising: the action of symmetry groups and the extremal principle in hyperbolic geometry. On the one hand, one may analyze the action of the infinite cyclic subgroup contained in the automorphism group of the limiting surface, and study its constraints on the end space, geodesic structure, and moduli parameters. On the other hand, one may use extremal objects in hyperbolic geometry, such as shortest closed geodesics, extremal lengths, maximal embedded regions, or geodesic rays tending toward ends, to characterize the geometric complexity and topological invariants of the surface. These methods may help transform current computational observations into deeper structural results.

(9) The current numerical results are still limited by computational range, sampling scale, and algorithmic complexity. In future work, it would be possible to further enlarge the data scale and improve simulation accuracy by introducing high-performance computing resources, optimizing surface decomposition algorithms, improving sampling strategies in parameter spaces, and developing more efficient criteria for homeomorphism classification. With the continued improvement of computational power and algorithmic methods, this direction is expected to provide stronger numerical support for the study of infinite-type surfaces, big mapping class groups, and the corresponding moduli spaces.

(10) In addition, this research also has potential value for interdisciplinary applications. Combinatorial decompositions of surfaces and parametrization methods for moduli spaces may be connected with complex network structures and periodic or quasi-periodic surface models in materials science. Meanwhile, surface classification, geometric fea-

## CONCLUSION

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ture extraction, and moduli-space sampling may provide new mathematical models for geometric deep learning, topological data analysis, and graph neural networks. Therefore, this direction is not only theoretically significant in pure mathematics, but may also lead to further applications in computational topology, materials modeling, and artificial intelligence.

## APPENDIX A HYPERBOLIC GEOMETRY TRIGONOMETRY

### Choice of Hyperbolic Geometry Model

The hyperbolic plane (curvature  $-1$ ) has several classical models: the Poincaré disk, the upper half-plane, the hyperboloid model, and the Klein model. When deriving trigonometric formulas, the **hyperboloid model** is the most concise because it transforms geometric problems into linear algebra problems. References can be found in<sup>[39,74]</sup>, and there are numerous textbooks that prove these theorems using other models, such as<sup>[18,27,38]</sup>.

Next, the hyperbolic plane is realized as one sheet of a two-sheeted hyperboloid in three-dimensional Minkowski space.

Consider the Lorentzian inner product on  $\mathbb{R}^{2,1}$  (Minkowski space):

$$\langle x, y \rangle_L = -x_0 y_0 + x_1 y_1 + x_2 y_2. \quad (\text{A-1})$$

The hyperbolic plane  $\mathbb{H}^2$  is defined as the upper sheet of the hyperboloid, or understood as a sphere with radius equal to the imaginary unit  $i$ :

$$\mathbb{H}^2 = \{x \in \mathbb{R}^{2,1} : \langle x, x \rangle_L = -1, x_0 > 0\}. \quad (\text{A-2})$$

The characteristic of this model is that geodesics are simple and geometrically intuitive: Geodesics are the intersections of the hyperboloid with two-dimensional planes passing through the origin (i.e., hyperbolas or straight lines). This description is very direct in its analogy to “great circles” in spherical geometry: great circles on a sphere are intersections of the sphere with planes through the sphere’s center; geodesics on the hyperboloid are intersections of the hyperboloid with timelike or lightlike planes through the origin.

The isometry group of the hyperboloid is the branch of the Lorentz group  $O(2, 1)$  that preserves time orientation,  $SO^+(2, 1)$ . The hyperbolic distance formula can be directly given by the Minkowski inner product:

**Lemma A.1 (Distance Formula):** For any  $u, v \in \mathbb{H}^2$ ,

$$\cosh d(u, v) = -\langle u, v \rangle_L. \quad (\text{A-3})$$

**Remark A.1:** This perfectly mirrors the distance formula in spherical geometry:

$\cos d(u, v) = \langle u, v \rangle$ ,  $\forall u, v \in S^2$ , where the inner product is the Euclidean inner product.

**Definition A.1 (Tangent Space and Angle):** The tangent space at a point  $A \in \mathbb{H}^2$  is

$$T_A \mathbb{H}^2 = \{w : \langle w, A \rangle_L = 0\}. \quad (\text{A-4})$$

The angle between two geodesics at  $A$  is defined as the angle between their tangent vectors in the Riemannian metric induced on this tangent space. Note that in the hyperboloid model, when drawing two intersecting geodesics (hyperbolas), the hyperbolic angle at their intersection is defined by the Minkowski metric and does not coincide with the visual angle defined by the Euclidean metric in three-dimensional Euclidean space.

## Law of Cosines for Sides (SSS)

**Theorem A.1 (Law of Cosines for Sides):** In a hyperbolic triangle  $\triangle ABC$ , let  $a = d(B, C)$ ,  $b = d(C, A)$ ,  $c = d(A, B)$ , and  $\alpha = \angle A$ . Then

$$\cosh a = \cosh b \cosh c - \sinh b \sinh c \cos \alpha. \quad (\text{SSS})$$

**Proof:** Using  $O(2, 1)$  isometries, place the triangle in a standard position:

(1) First, set  $A = (1, 0, 0)$ .

(2) Place  $B$  in the plane  $x_2 = 0$ . From  $d(A, B) = c$  and the distance formula (A-3), we can take

$$B = (\cosh c, \sinh c, 0).$$

(3) Place  $C$  at an angle  $\alpha$  from  $AB$ . From  $d(A, C) = b$  and rotational symmetry, we can take

$$C = (\cosh b, \sinh b \cos \alpha, \sinh b \sin \alpha).$$

Verification:  $\langle A, A \rangle_L = \langle B, B \rangle_L = \langle C, C \rangle_L = -1$ , and  $\langle A, B \rangle_L = -\cosh c$ ,  $\langle A, C \rangle_L = -\cosh b$ . From the distance formula (A-3):

$$\cosh a = -\langle B, C \rangle_L. \quad (\text{A-5})$$

Compute the inner product:

$$\begin{aligned} \langle B, C \rangle_L &= -(\cosh c)(\cosh b) + (\sinh c)(\sinh b \cos \alpha) + 0 \cdot (\sinh b \sin \alpha) \\ &= -\cosh b \cosh c + \sinh b \sinh c \cos \alpha. \end{aligned} \quad (\text{A-6})$$

Therefore,

$$\cosh a = -\langle B, C \rangle_L = \cosh b \cosh c - \sinh b \sinh c \cos \alpha. \quad (\text{A-7})$$

Cyclic permutation yields the other two formulas. ■

## Law of Sines

**Theorem A.2 (Law of Sines):** In a hyperbolic triangle  $\triangle ABC$ ,

$$\boxed{\frac{\sin \alpha}{\sinh a} = \frac{\sin \beta}{\sinh b} = \frac{\sin \gamma}{\sinh c}}. \quad (\text{Sine})$$

**Proof:** Let  $A, B, C \in \mathbb{H}^2 \subset \mathbb{R}^{2,1}$  be the vertices of the hyperbolic triangle, satisfying  $\langle A, A \rangle = \langle B, B \rangle = \langle C, C \rangle = -1$ . Denote the side lengths as  $a = d(B, C)$ ,  $b = d(C, A)$ ,  $c = d(A, B)$ , and the opposite angles as  $\alpha = \angle A$ ,  $\beta = \angle B$ ,  $\gamma = \angle C$ .

In  $\mathbb{R}^{2,1}$ , define the generalized cross product  $X \times Y$  as the unique vector satisfying:

- $\langle X \times Y, X \rangle = \langle X \times Y, Y \rangle = 0$ ,
- $\langle X \times Y, X \times Y \rangle = \sinh^2 d(X, Y)$ ,

Define the scalar triple product

$$\Delta = \det(A, B, C) = \langle A, B \times C \rangle,$$

where the determinant is computed with respect to an orthonormal basis.

Using determinant identities,  $\Delta^2$  can be computed:

$$\Delta^2 = \det \begin{pmatrix} \langle A, A \rangle & \langle A, B \rangle & \langle A, C \rangle \\ \langle B, A \rangle & \langle B, B \rangle & \langle B, C \rangle \\ \langle C, A \rangle & \langle C, B \rangle & \langle C, C \rangle \end{pmatrix}.$$

Substitute  $\langle A, A \rangle = \langle B, B \rangle = \langle C, C \rangle = -1$ ,  $\langle A, B \rangle = -\cosh c$ ,  $\langle A, C \rangle = -\cosh b$ ,  $\langle B, C \rangle = -\cosh a$ . Factor out a negative sign from each row (three rows give a factor  $(-1)^3 = -1$ ):

$$\Delta^2 = -\det \begin{pmatrix} 1 & \cosh c & \cosh b \\ \cosh c & 1 & \cosh a \\ \cosh b & \cosh a & 1 \end{pmatrix}.$$

This determinant can be simplified to

$$\det = 1 + 2 \cosh a \cosh b \cosh c - \cosh^2 a - \cosh^2 b - \cosh^2 c = \sinh^2 b \sinh^2 c \sin^2 \alpha.$$

(The last equality can be verified using the hyperbolic law of cosines  $\cosh a = \cosh b \cosh c - \sinh b \sinh c \cos \alpha$ .) Therefore,

$$\Delta^2 = \sinh^2 b \sinh^2 c \sin^2 \alpha,$$

i.e.,

$$|\Delta| = \sinh b \sinh c \sin \alpha.$$

Taking a specific coordinate representation also verifies the above formula. For example, set

$$A = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad B = \begin{pmatrix} \cosh c \\ \sinh c \\ 0 \end{pmatrix}, \quad C = \begin{pmatrix} \cosh b \\ \sinh b \cos \alpha \\ \sinh b \sin \alpha \end{pmatrix},$$

and compute  $\det(A, B, C)$  directly to obtain

$$\Delta = \sinh b \sinh c \sin \alpha.$$

Since the determinant is antisymmetric, its absolute value remains unchanged under cyclic permutation of vertices:

$$|\langle A, B \times C \rangle| = |\langle B, C \times A \rangle| = |\langle C, A \times B \rangle|.$$

Compute the other two expressions:  $|\langle B, C \times A \rangle| = \sinh c \sinh a \sin \beta$  and  $|\langle C, A \times B \rangle| = \sinh a \sinh b \sin \gamma$ . Thus, we obtain:

$$\sinh b \sinh c \sin \alpha = \sinh c \sinh a \sin \beta = \sinh a \sinh b \sin \gamma.$$

Dividing all three expressions by  $\sinh a \sinh b \sinh c$  yields

$$\frac{\sin \alpha}{\sinh a} = \frac{\sin \beta}{\sinh b} = \frac{\sin \gamma}{\sinh c}.$$

■

## Law of Cosines for Angles (AAA)

**Theorem A.3 (Law of Cosines for Angles):** In a hyperbolic triangle  $\triangle ABC$ ,

$$\boxed{\cos \alpha = -\cos \beta \cos \gamma + \sin \beta \sin \gamma \cosh a}. \quad (\text{AAA})$$

**Proof:** The following uses the polar triangle duality method to directly transform the law of cosines for sides into the law of cosines for angles.

For triangle  $ABC$ , define:  $n_a$  as the unit normal vector to side  $BC$  (also called the polar vector of the side),  $n_b$  as the unit normal vector to side  $CA$ , and  $n_c$  as the unit normal vector to side  $AB$ . Treat these normal vectors as points  $A^* = n_a$ ,  $B^* = n_b$ ,  $C^* = n_c$ , called the polar triangle  $A^*B^*C^*$  of the original triangle. Although these points lie on the de Sitter surface, they algebraically serve as dual objects, where lengths and angles

naturally become complex, interchanging the relationships between sides and angles.

The angles of the original triangle correspond to the sides of the polar triangle. Specifically, the angle  $A$  at vertex  $A$  equals the angle between the normal vectors  $n_b$  and  $n_c$ , i.e.,  $\langle n_b, n_c \rangle_L = \cos A$ . If  $B^*$  and  $C^*$  are treated as points, their hyperbolic distance satisfies  $\cosh(B^*C^*) = -\langle n_b, n_c \rangle_L = -\cos A$ . Using the identity  $\cosh(i(\pi - A)) = \cos(\pi - A) = -\cos A$ , we obtain:

$$B^*C^* = i(\pi - A), \quad C^*A^* = i(\pi - B), \quad A^*B^* = i(\pi - C). \quad (D1)$$

On the other hand, the sides of the original triangle correspond to the angles of the polar triangle. Consider the angle  $\angle C^*$  at vertex  $C^*$  of the polar triangle. Its relationship with the original side  $c = |AB|$  is  $\cos(\angle C^*) = -\cosh c$ . Since  $\cos(\pi - ic) = -\cos(ic) = -\cosh c$ , we have:

$$\angle C^* = \pi - ic, \quad \angle A^* = \pi - ia, \quad \angle B^* = \pi - ib. \quad (D2)$$

Now, apply the law of cosines for sides to the polar triangle  $A^*B^*C^*$ :

$$\cosh(A^*B^*) = \cosh(B^*C^*) \cosh(C^*A^*) - \sinh(B^*C^*) \sinh(C^*A^*) \cos(\angle C^*). \quad (*)$$

Substitute the dual relationships term by term. First, from (D1), the left-hand side is  $\cosh(A^*B^*) = \cosh(i(\pi - C)) = \cos(\pi - C) = -\cos C$ . In the first term on the right,  $\cosh(B^*C^*) = \cosh(i(\pi - A)) = \cos(\pi - A) = -\cos A$ , and similarly  $\cosh(C^*A^*) = -\cos B$ , so their product is  $\cos A \cos B$ . In the second term on the right,  $\sinh(B^*C^*) = \sinh(i(\pi - A)) = i \sin(\pi - A) = i \sin A$ , and similarly  $\sinh(C^*A^*) = i \sin B$ , while  $\cos(\angle C^*) = \cos(\pi - ic) = -\cosh c$ . After simplification, this yields the law of cosines for angles:

$$\cos C = -\cos A \cos B + \sin A \sin B \cosh c. \quad (\text{HAL})$$

Similarly, by cyclic permutation, the corresponding formulas for angles  $A$  and  $B$  can be obtained. ■

## Special Case: Right Triangles

**Corollary A.1 (Hyperbolic Pythagorean Theorem):** If  $\gamma = \pi/2$ , then

$$\boxed{\cosh c = \cosh a \cosh b}. \quad (\text{A-8})$$

**Proof:** Set  $\gamma = \pi/2$  in the law of cosines for sides (SSS) to obtain  $\cosh c = \cosh a \cosh b - \sinh a \sinh b \cdot 0 = \cosh a \cosh b$ . ■

**Corollary A.2 (Trigonometric Functions for Acute Angles):** In a right triangle with  $\gamma = \pi/2$ ,

$$\sin \alpha = \frac{\sinh a}{\sinh c}, \quad \sin \beta = \frac{\sinh b}{\sinh c}. \quad (\text{A-9})$$

**Proof:** Directly obtained from the law of sines (Sine) with  $\sin \gamma = 1$ . ■

**Corollary A.3 (Tangent Relationships):**

$$\cos \alpha = \tanh b \coth c, \quad \cos \beta = \tanh a \coth c. \quad (\text{A-10})$$

**Proof:** Obtained by combining the law of cosines for sides (SSS) and the Pythagorean theorem and eliminating terms. ■

## Comparison Table of Three Types of Geometry

Table A-1 Comparison of trigonometric formulas in hyperbolic, spherical, and Euclidean geometry

| Theorem                 | Hyperbolic ( $K = -1$ )                                                                  | Spherical ( $K = 1$ )                                                                 | Euclidean ( $K = 0$ )                                                  |
|-------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Law of cosines (sides)  | $\cosh a = \cosh b \cosh c - \sinh b \sinh c \cos \alpha$                                | $\cos a = \cos b \cos c + \sin b \sin c \cos \alpha$                                  | $a^2 = b^2 + c^2 - 2bc \cos \alpha$                                    |
| Law of sines            | $\frac{\sin \alpha}{\sinh a} = \frac{\sin \beta}{\sinh b} = \frac{\sin \gamma}{\sinh c}$ | $\frac{\sin \alpha}{\sin a} = \frac{\sin \beta}{\sin b} = \frac{\sin \gamma}{\sin c}$ | $\frac{\sin \alpha}{a} = \frac{\sin \beta}{b} = \frac{\sin \gamma}{c}$ |
| Law of cosines (angles) | $\cos \alpha = -\cos \beta \cos \gamma + \sin \beta \sin \gamma \cosh a$                 | $\cos \alpha = -\cos \beta \cos \gamma + \sin \beta \sin \gamma \cos a$               | $\alpha + \beta + \gamma = \pi$                                        |
| Pythagorean theorem     | $\cosh c = \cosh a \cosh b$                                                              | $\cos c = \cos a \cos b$                                                              | $c^2 = a^2 + b^2$                                                      |

## APPENDIX B ALGORITHM AND COMPUTATIONAL IMPLEMENTATION

### B.1 Recursive construction and rendering of the decorated planar diagram for the dual map of the hypercube graph embedding

We now describe the recursive construction of the decorated planar diagram  $\mathcal{D}_n$ . The construction is based on a family of axis-aligned rectangular loops generated recursively from a basic square. For each integer  $n \geq 4$ , let  $\mathcal{F}_n$  denote the family of closed rectangular loops produced at level  $n$ . Let  $\text{RectVertexSet}$  be the set of all rectangle vertices appearing in  $\mathcal{F}_n$ . Among all intersections of horizontal and vertical segments in  $\mathcal{F}_n$ , we distinguish two types: the set  $\text{NodeSet}$  of midpoint-type intersections, and the set  $\text{CrossingSet}$  of all remaining non-corner intersections. The final rendered planar diagram is denoted by  $\mathcal{D}_n$ .

For clarity, the construction of  $\mathcal{D}_n$  is divided into three stages. First, Algorithm B-1 recursively generates the rectangular loop family  $\mathcal{F}_n$  from the initial square at level  $n = 4$  by horizontal translation, vertical scaling, and quarter-turn rotation. Next, Algorithm B-2 extracts all horizontal and vertical segments in  $\mathcal{F}_n$  and classifies their intersections into rectangle corners, nodes, and ordinary crossings. Finally, Algorithm B-4 renders the planar diagram by drawing the rectangular skeleton, assigning the prescribed local decorations to distinguished points, and adding the auxiliary axis-centered marks.

In this way, the recursive procedure first constructs the global rectangular framework  $\mathcal{F}_n$ , then identifies its distinguished intersection sets, and finally converts this geometric-combinatorial data into the decorated planar diagram  $\mathcal{D}_n$ . The global shape of  $\mathcal{D}_n$  is therefore determined recursively, whereas its local structure is encoded by the corner, node, and crossing rules used in the rendering stage.

---

 Algorithm B-1 Recursive construction of the rectangular loop family  $\mathcal{F}_n$ 


---

**Data:** integer  $n \geq 4$ , base half-side length  $a > 0$   
**Result:** a family  $\mathcal{F}_n$  of closed axis-aligned rectangular loops

```

1 Function Build( $n, a$ )
2   if  $n = 4$  then
3     return the square centered at the origin with vertices
         $(-a, -a), (a, -a), (a, a), (-a, a)$  and then returning to  $(-a, -a)$ ;
4   else
5      $\mathcal{F}_{n-1} \leftarrow \text{Build}(n-1, a)$ ;
6      $\delta \leftarrow 2^{n-4}a$ ;
7     initialize  $\mathcal{U} \leftarrow \emptyset$ ;
8     foreach  $C \in \mathcal{F}_{n-1}$  do
9       | add Translate( $C, (-\delta, 0)$ ) and Translate( $C, (\delta, 0)$ ) to  $\mathcal{U}$ ;
10    end
11    foreach  $C \in \mathcal{U}$  do
12      | replace every point  $(x, y) \in C$  by  $(x, 2y)$ ;
13    end
14    initialize  $\mathcal{V} \leftarrow \emptyset$ ;
15    foreach  $C \in \mathcal{U}$  do
16      | add Rotate90( $C$ ) to  $\mathcal{V}$ ;                                //  $(x, y) \mapsto (-y, x)$ 
17    end
18    return  $\mathcal{U} \cup \mathcal{V}$ ;
19  end
    
```

---

## B.2 Homological redundancy calculation in the dual maps of hypercube graph embeddings

```

1 import itertools
2 import numpy as np
3
4 class GeodesicAnalyzer:
5     def __init__(self, m):
6         self.m = m
7         self.num_edges = m * (1 << (m-1)) # m * 2^(m-1)
8         self.d2 = self.build_d2_sparse()
9
10    def edge_index(self, i, bits):
11        """Calculate edge index, i is the position of *, bits
12           is the binary combination of other positions"""
13        return i * (1 << (self.m-1)) + bits
14
15    def build_d2_sparse(self):
    
```

---

**Algorithm B-2** Intersection classification in the rectangular loop family

---

**Data:** rectangular loop family  $\mathcal{F}_n$   
**Result:** RectVertexSet, NodeSet, and CrossingSet

- 1 extract all horizontal segments into  $\mathcal{H}$  and all vertical segments into  $\mathcal{V}$ ;
- 2 initialize RectVertexSet  $\leftarrow \emptyset$ ;
- 3 initialize NodeSet  $\leftarrow \emptyset$  and CrossingSet  $\leftarrow \emptyset$ ;
- 4 **foreach** loop  $C \in \mathcal{F}_n$  **do**
- 5     add all vertices of  $C$ , except the repeated final point, to RectVertexSet;
- 6 **end**
- 7 **foreach** vertical segment  $(x_v, y_1, y_2) \in \mathcal{V}$  **do**
- 8     **foreach** horizontal segment  $(y_h, x_1, x_2) \in \mathcal{H}$  **do**
- 9         **if**  $x_v \in [x_1, x_2]$  and  $y_h \in [y_1, y_2]$  **then**
- 10             let  $p = (x_v, y_h)$ ;
- 11             **if**  $p$  is simultaneously an endpoint of the horizontal segment and an  
                    endpoint of the vertical segment **then**
- 12                 continue ;                                                             // rectangle corner
- 13             **else if**  $x_v = \frac{x_1 + x_2}{2}$  and  $y_h = \frac{y_1 + y_2}{2}$  **then**
- 14                 add  $p$  to NodeSet;
- 15             **else**
- 16                 add  $p$  to CrossingSet;
- 17             **end**
- 18         **end**
- 19     **end**
- 20 **end**
- 21 remove duplicate points from all three sets and sort them lexicographically;

---



---

**Algorithm B-3** Local decoration rules

---

**Data:** point type and local orientation  
**Result:** drawing instruction around the point

- 1 **Procedure** *Corner decoration at a point  $p$  with orientation in  $\{\text{LU}, \text{RU}, \text{LD}, \text{RD}\}$*
- 2     draw a white square neighborhood centered at  $p$ ;
- 3     draw three concentric quarter-circular arcs of radii 1, 2, 3;
- 4     use the color pattern red–black–red according to the corner orientation;
- 5 **Procedure** *Node decoration at a point  $p$*
- 6     draw a white square neighborhood centered at  $p$ ;
- 7     draw four red quarter-circular arcs around the local block;
- 8     draw one horizontal black segment and one vertical black segment crossing at  $p$ ;
- 9     draw a blue solid dot at  $p$ ;
- 10 **Procedure** *Crossing decoration at a point  $p$*
- 11     draw a white square neighborhood centered at  $p$ ;
- 12     declare the horizontal branch to pass over the vertical branch;
- 13     draw a central horizontal black segment through  $p$ ;
- 14     draw two short vertical black stubs above and below  $p$ ;
- 15     draw two parallel red horizontal segments and four short red vertical side stubs;

---

---

**Algorithm B-4** Rendering of the decorated planar diagram  $\mathcal{D}_n$ 


---

**Data:** integer  $n \geq 4$ , base half-side length  $a$ , optional output path  
**Result:** decorated planar diagram  $\mathcal{D}_n$

- 1  $\mathcal{F}_n \leftarrow$  Algorithm B-1 applied to  $(n, a)$ ;
- 2 compute RectVertexSet, NodeSet, and CrossingSet by Algorithm B-2;
- 3 extract from each loop its bounding rectangle and determine the orientation of each corner;
- 4 compute the line width
$$w_n = \max\{0.30, 2.4 \cdot 0.65^{n-5}\};$$
compute the auxiliary coordinate set
$$\mathcal{Y}_n = \{\pm 4(2k + 1) : 0 \leq k \leq 2^{n-4} - 1\};$$
- 5 initialize a square canvas with equal aspect ratio and no visible axes;
- 6 determine a square viewing window containing all rectangles, decorations, and auxiliary points;

// Step 1: draw the rectangular skeleton

- 7 **foreach** *bounding rectangle*  $(x_{\min}, x_{\max}, y_{\min}, y_{\max})$  **do**
- 8 | draw the outer red rectangle  $(x_{\min} - 1, x_{\max} + 1, y_{\min} - 1, y_{\max} + 1)$ ;
- 9 | **if**  $x_{\min} + 1 < x_{\max} - 1$  **and**  $y_{\min} + 1 < y_{\max} - 1$  **then**
- 10 | | draw the inner red rectangle  $(x_{\min} + 1, x_{\max} - 1, y_{\min} + 1, y_{\max} - 1)$ ;
- 11 | **end**
- 12 | draw the middle black rectangle  $(x_{\min}, x_{\max}, y_{\min}, y_{\max})$ ;
- 13 **end**

// Step 2: decorate all special points

- 14 **foreach** *corner point*  $p$  **do**
- 15 | apply the corner decoration rule of Algorithm B-3;
- 16 **end**
- 17 **foreach**  $p \in \text{CrossingSet}$  **do**
- 18 | apply the crossing decoration rule of Algorithm B-3;
- 19 **end**
- 20 **foreach**  $p \in \text{NodeSet}$  **do**
- 21 | apply the node decoration rule of Algorithm B-3;
- 22 **end**

// Step 3: add axis-centered auxiliary marks

- 23 **foreach**  $t \in \mathcal{Y}_n$  **do**
- 24 | draw a short vertical black segment from  $(0, t - 1)$  to  $(0, t + 1)$ ;
- 25 | draw a short horizontal black segment from  $(t - 1, 0)$  to  $(t + 1, 0)$ ;
- 26 | draw larger blue dots at  $(0, t)$  and  $(t, 0)$ ;
- 27 | draw smaller blue dots at  $(0, t \pm 1)$  and  $(t \pm 1, 0)$ ;
- 28 **end**
- 29 **if** *an output path is specified* **then**
- 30 | save the rendered figure in PDF format;
- 31 **end**

---

```
15     """Construct sparse matrix for d2 boundary operator,
16         each row corresponds to boundary edges of a face"""
17     d2 = []
18     for surface in itertools.product([0, 1], repeat=self.m)
19         :
18         row = []
19         for i in range(self.m):
20             # Generate edge: change the i-th bit of surface
21             # to *
22             edge_bits = 0
23             for pos in range(self.m):
24                 if pos != i:
25                     edge_bits = (edge_bits << 1) | surface[
26                         pos]
27                 row.append(self.edge_index(i, edge_bits))
28             d2.append(row)
29     return d2
30
31 def generate_geodesics(self):
32     """Generate all geodesics, each geodesic consists of
33         four edges"""
34     cycles = []
35     m = self.m
36     for i in range(m):
37         left = (i - 1) % m
38         right = (i + 1) % m
39         free_pos = [pos for pos in range(m) if pos not in {
40             i, left, right}]
41         num_free = len(free_pos)
42         for bits in range(1 << num_free):
43             # Set bits for free positions
44             base = {}
45             for idx, pos in enumerate(free_pos):
46                 base[pos] = (bits >> (num_free - 1 - idx))
47                 & 1
48             # Generate four edges, corresponding to four
49             # combinations of left and right positions
50             cycle = []
```

```
45         for lv, rv in itertools.product([0, 1], repeat
46             =2):
47             edge_bits = 0
48             for pos in range(m):
49                 if pos == i:
50                     continue # Position of *
51                 if pos == left:
52                     bit = lv
53                 elif pos == right:
54                     bit = rv
55                 else:
56                     bit = base[pos]
57                 edge_bits = (edge_bits << 1) | bit
58                 edge_idx = self.edge_index(i, edge_bits)
59                 cycle.append(edge_idx)
60             cycles.append(cycle)
61         return cycles
62
63     def sparse_gf2_rank(self, matrix):
64         """Compute rank of sparse matrix over GF(2)"""
65         rows = [set(row) for row in matrix]
66         pivot_cols = {}
67         rank = 0
68
69         for r in range(len(rows)):
70             row = rows[r].copy()
71             # Find pivot column
72             while row:
73                 pivot = min(row)
74                 if pivot in pivot_cols:
75                     row ^= pivot_cols[pivot]
76                 else:
77                     pivot_cols[pivot] = row
78                     rank += 1
79                     break
80             return rank
81
82     def curve_homology_rank(self, cycles):
```

```

82     """Compute homology rank of cycle list in H1"""
83     # Convert d2 to set form
84     d2_rows = [set(row) for row in self.d2]
85     B_rank = self.sparse_gf2_rank(d2_rows)
86
87     # Convert cycles to set form
88     S_rows = [set(cycle) for cycle in cycles]
89     S_rank = self.sparse_gf2_rank(S_rows)
90
91     # Compute rank of S + B
92     combined = S_rows + d2_rows
93     S_plus_B_rank = self.sparse_gf2_rank(combined)
94
95     # Compute dimension of intersection
96     intersection_dim = S_rank + B_rank - S_plus_B_rank
97
98     # Return homology rank
99     return S_rank - intersection_dim
100
101 if __name__ == "__main__":
102     for m in range(3,22):
103         print(f"\nAnalyzing for m={m}")
104         g=1+(m-4)*2**(m-3)
105         print(f"Genus: {g}")
106         print(f"Homology H1: {2*g}")
107         analyzer = GeodesicAnalyzer(m)
108         geodesics = analyzer.generate_geodesics()
109         print(f"Generated geodesic cycles: {len(geodesics)}")
110         rank = analyzer.curve_homology_rank(geodesics)
111         print(f"Homology rank of geodesics: {rank}")

```

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## RESUME AND ACADEMIC ACHIEVEMENTS

### Resume

Qingrui Qu was born in 1997 in Wuhan, Hubei, China.

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### Academic Achievements during the Study for an Academic Degree

#### Academic Articles

- [1] Feng P, Qu Q, Zhang H, Yi S, Yu Z, Ding Z, Zhu Y. Topology-enhanced machine learning for consonant recognition, submitted.
- [2] Yu Z, Feng P, Qu Q, Zhang H, Zhu Y. Topological Deep Learning for Speech Recognition, submitted.